



香港科技大學
THE HONG KONG
UNIVERSITY OF SCIENCE
AND TECHNOLOGY

COMP 5212 ML

Course Review

Yi R. (May) Fung
May 7, 2026

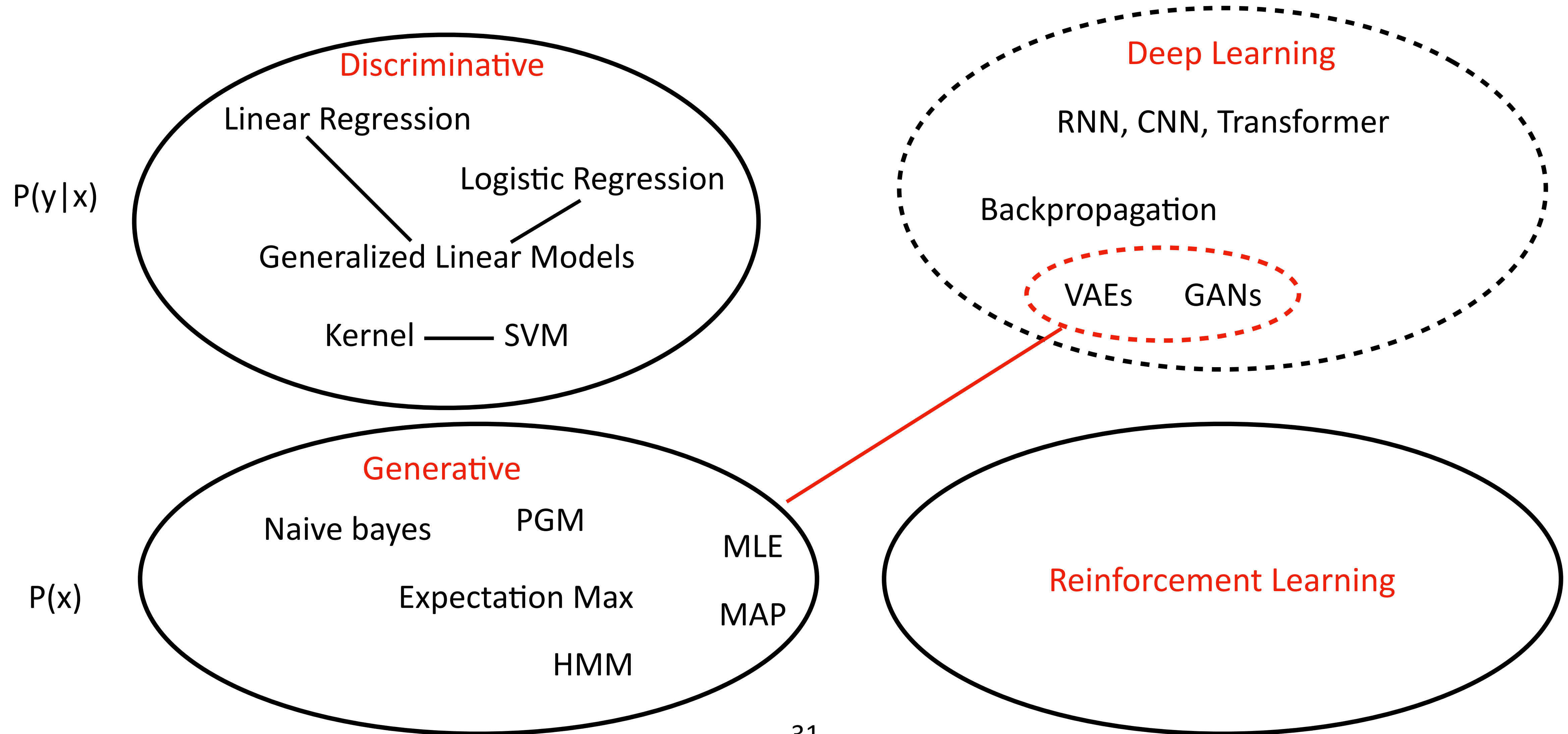
Final Exam

1. Two, double-sided A4-size cheatsheets
2. 2-hour duration
3. Contents cover both before mid-term and after mid-term, while emphasizing more on after mid-term
4. Format similar to mid-term exam, mixed multi-choice and open-ended questions

Will make more formal announcement on Canvas

May 21st 8:30-10:30AM, at location CYT Building LT-L.

Ending Remarks

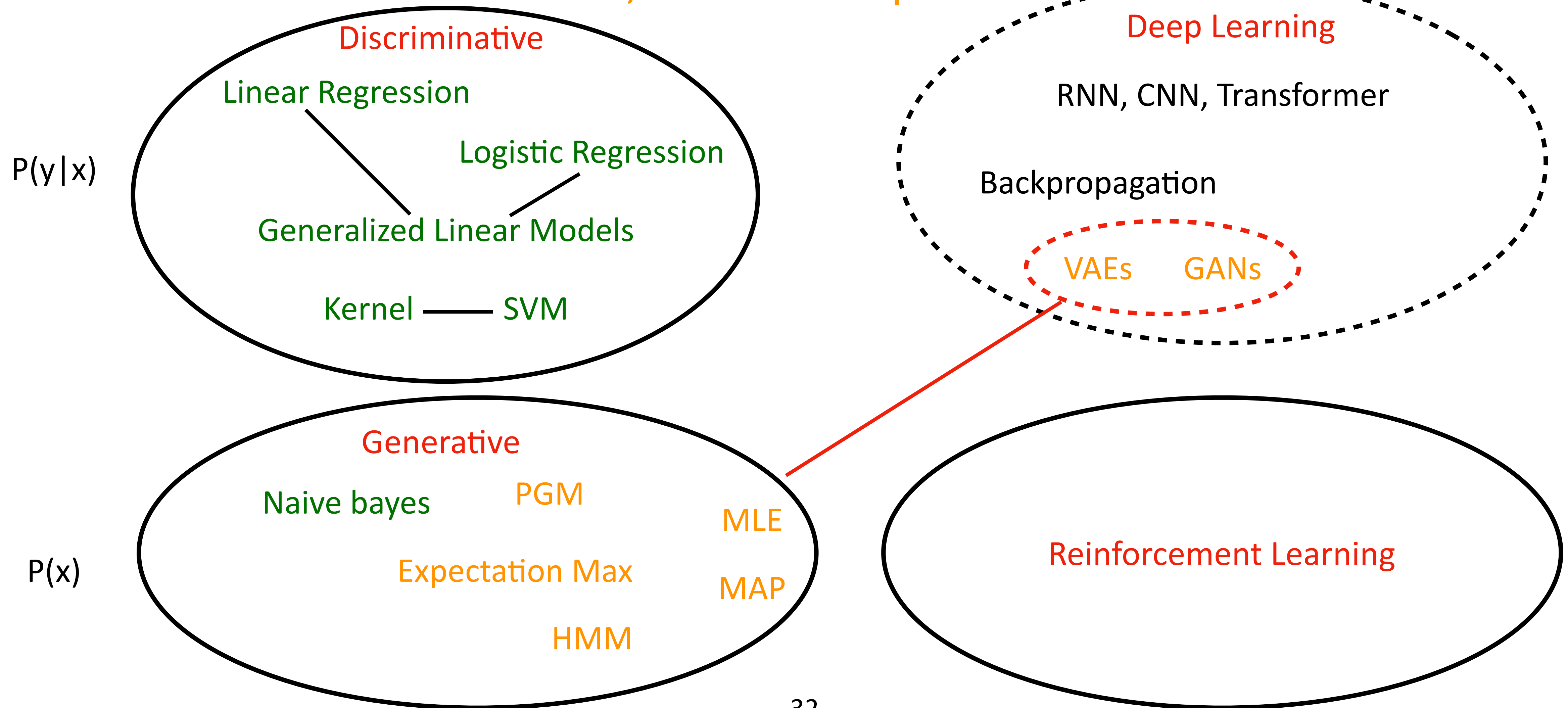


Ending Remarks

Supervised Learning

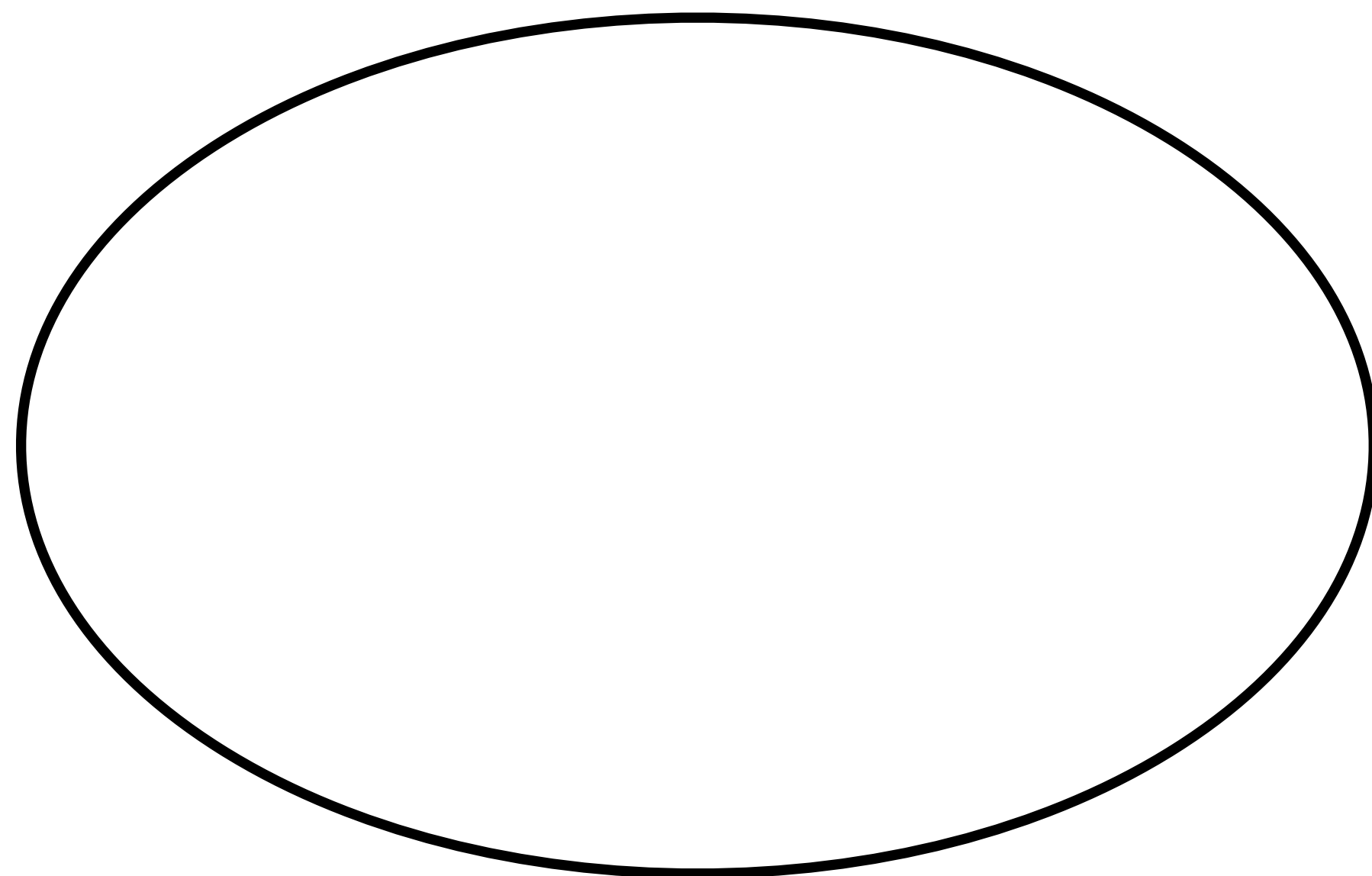
Unsupervised Learning

The unsupervised learning ones can actually do both, and semi-supervised

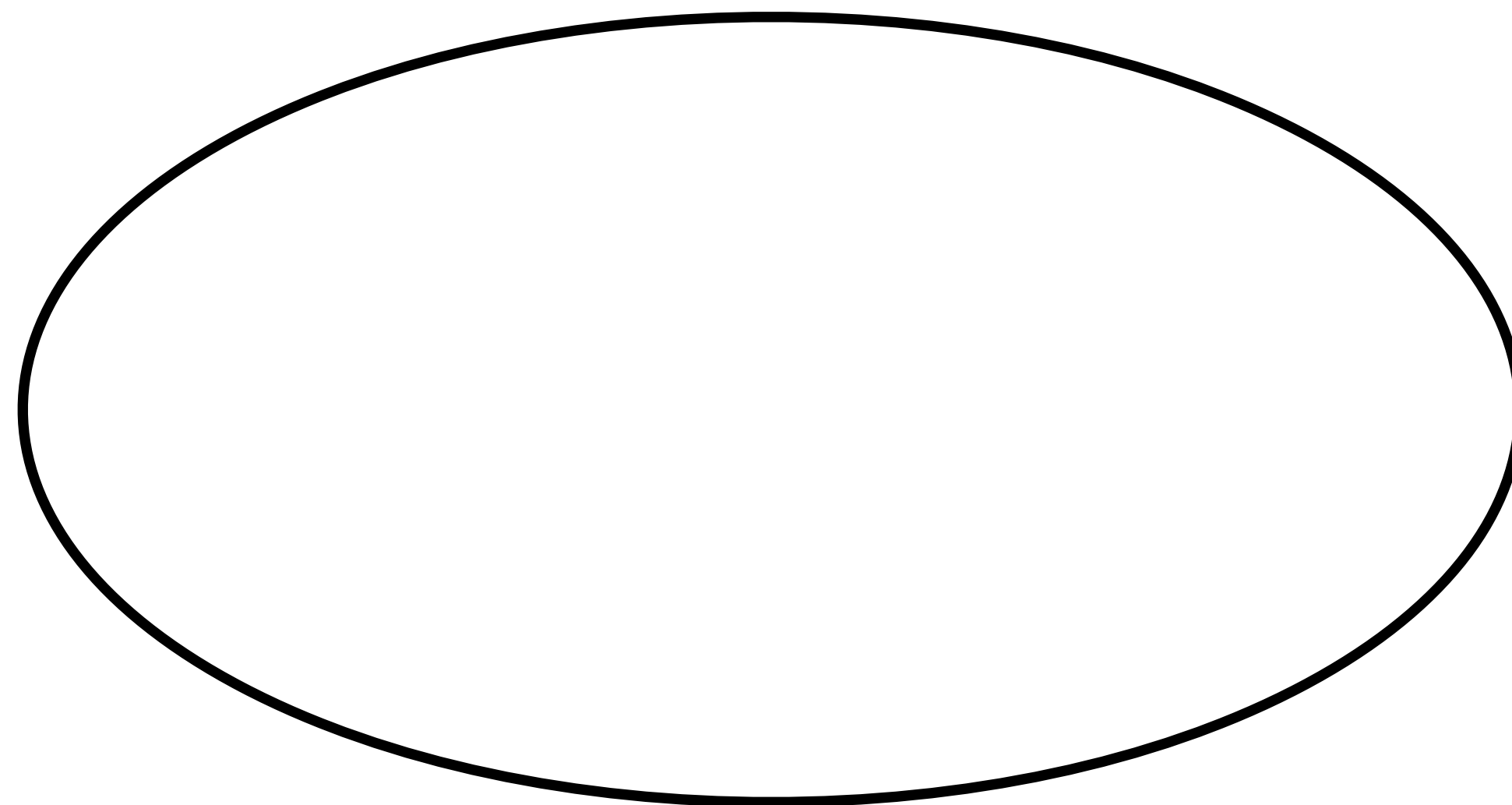


Review — Method

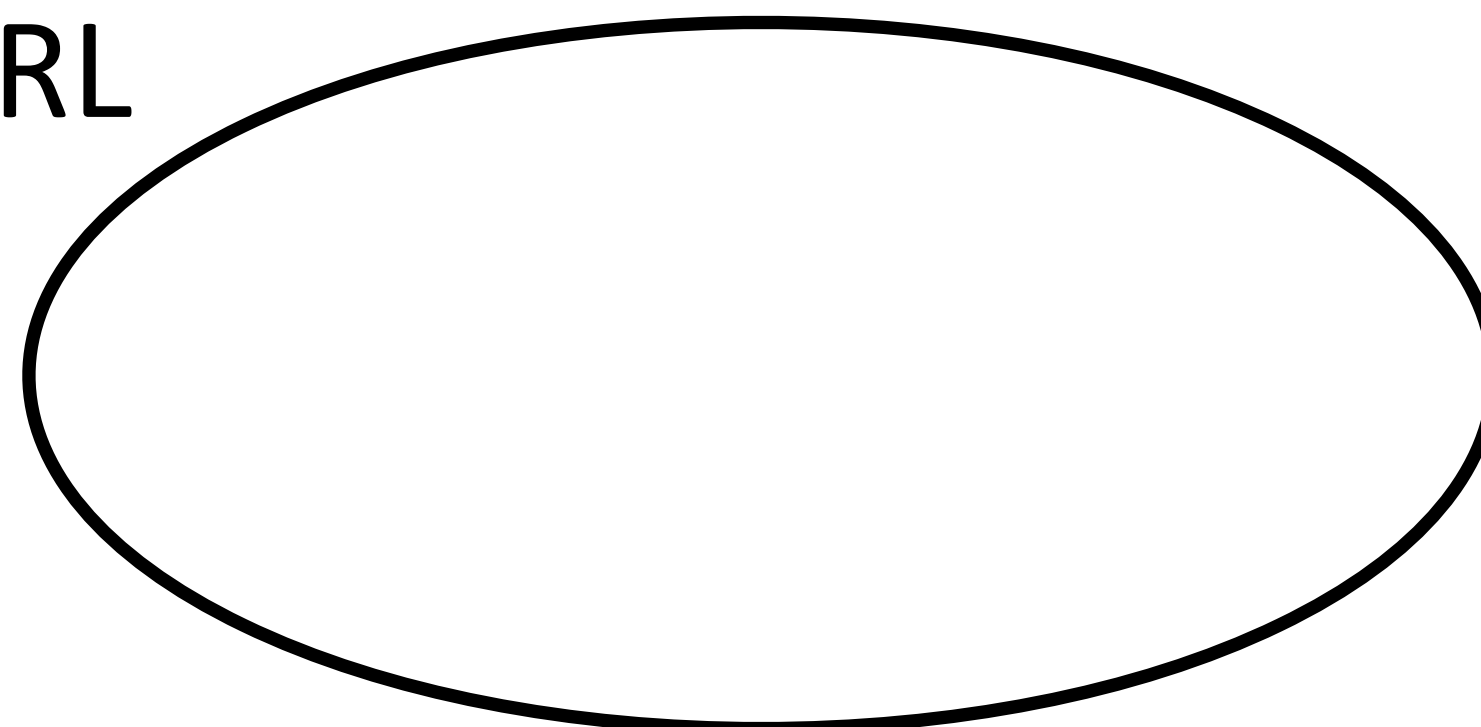
Pretraining



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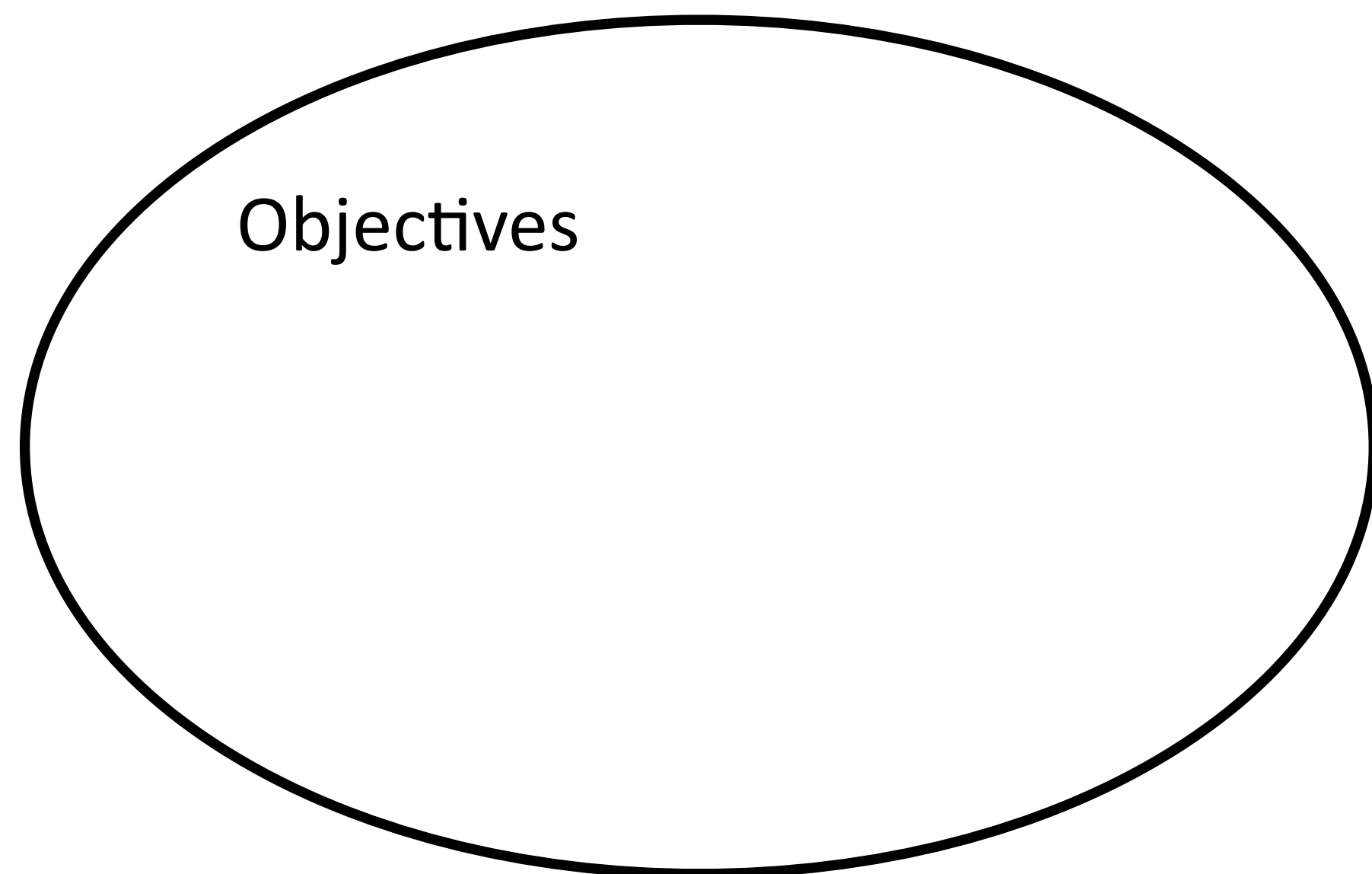


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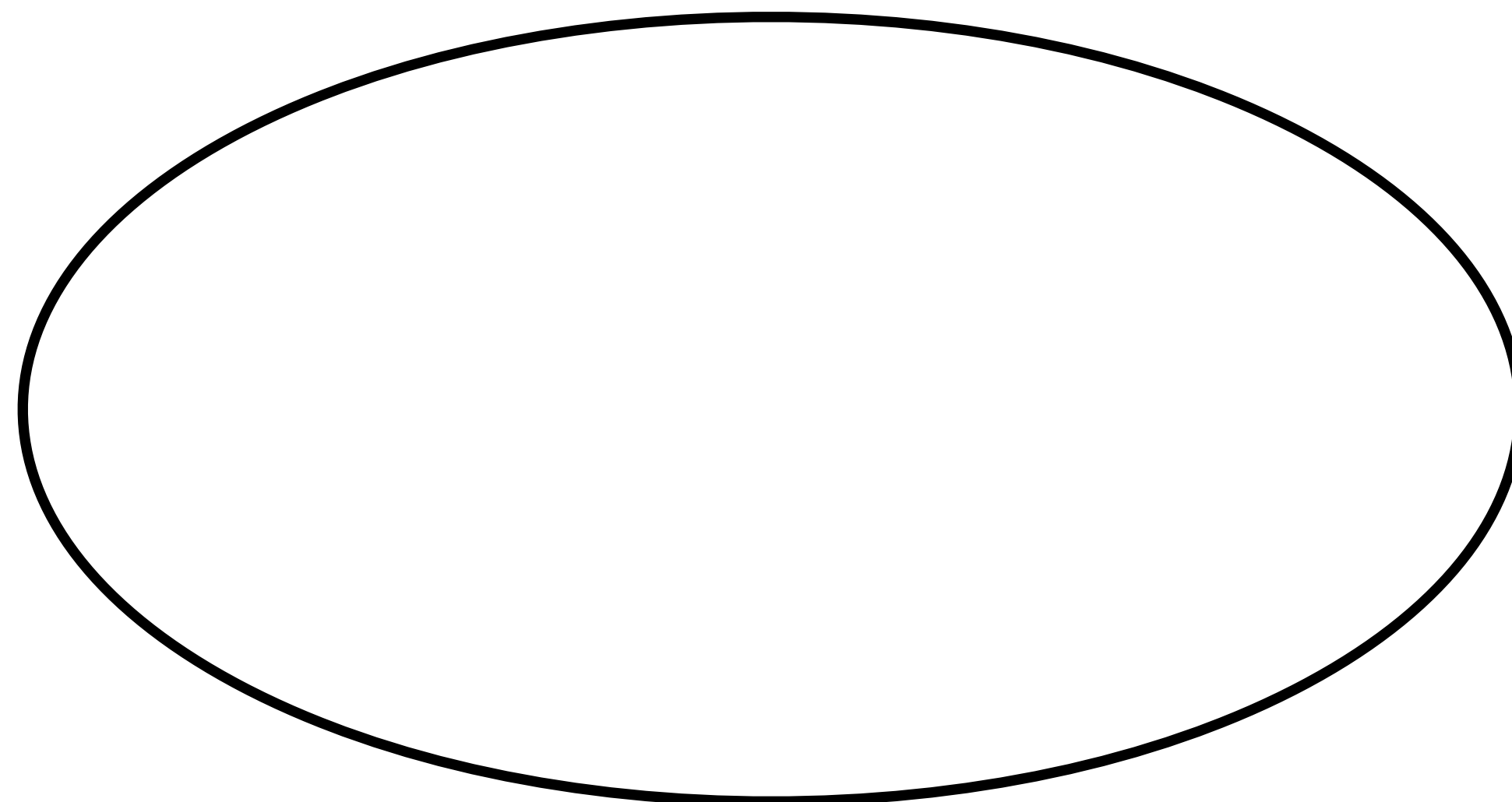


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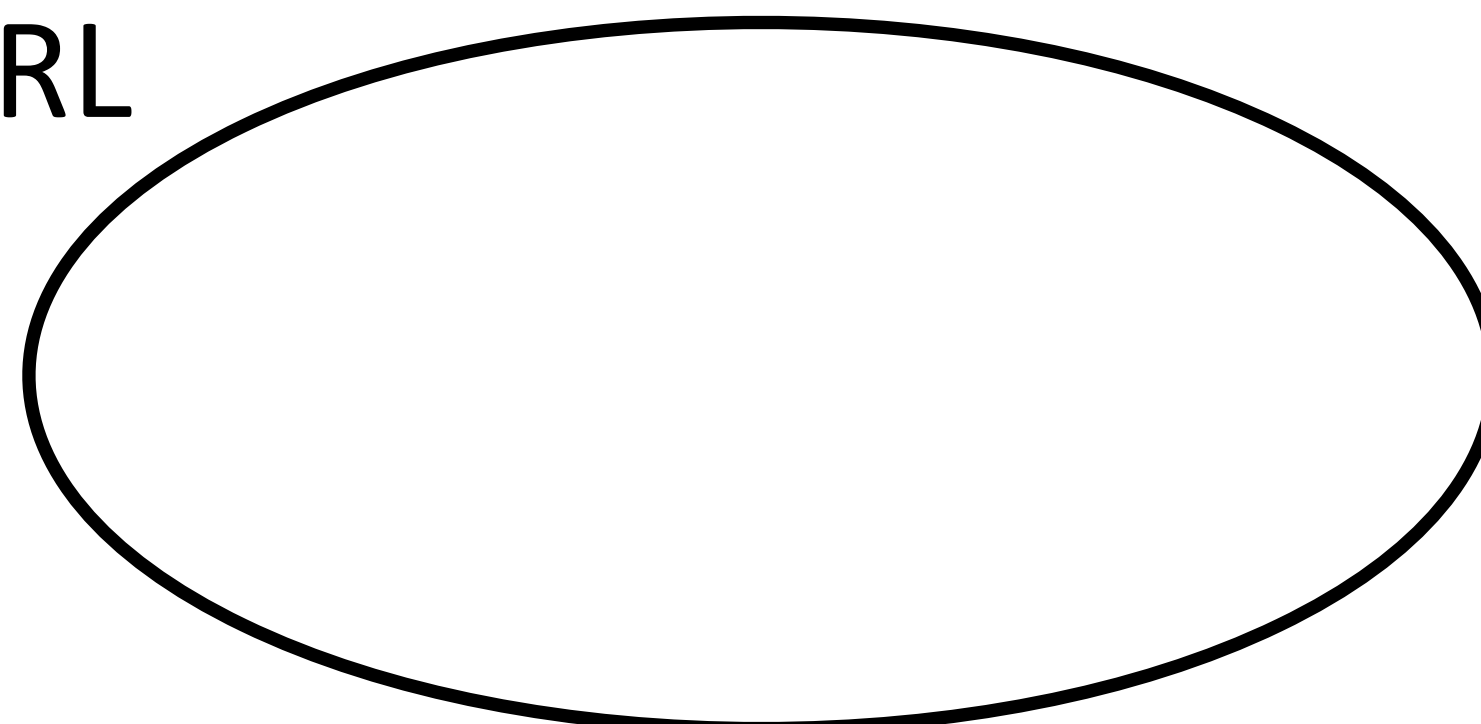
Pretraining



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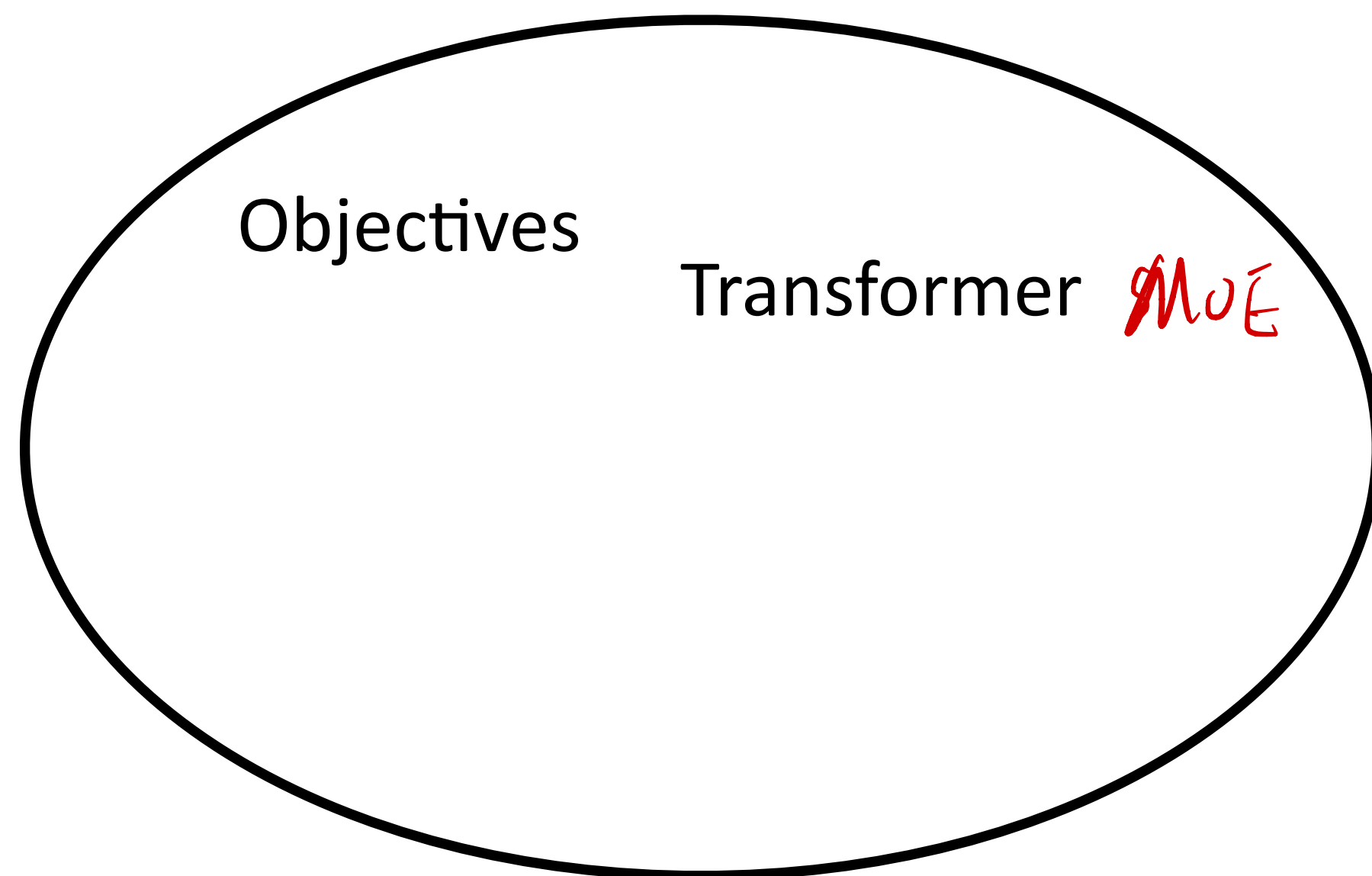


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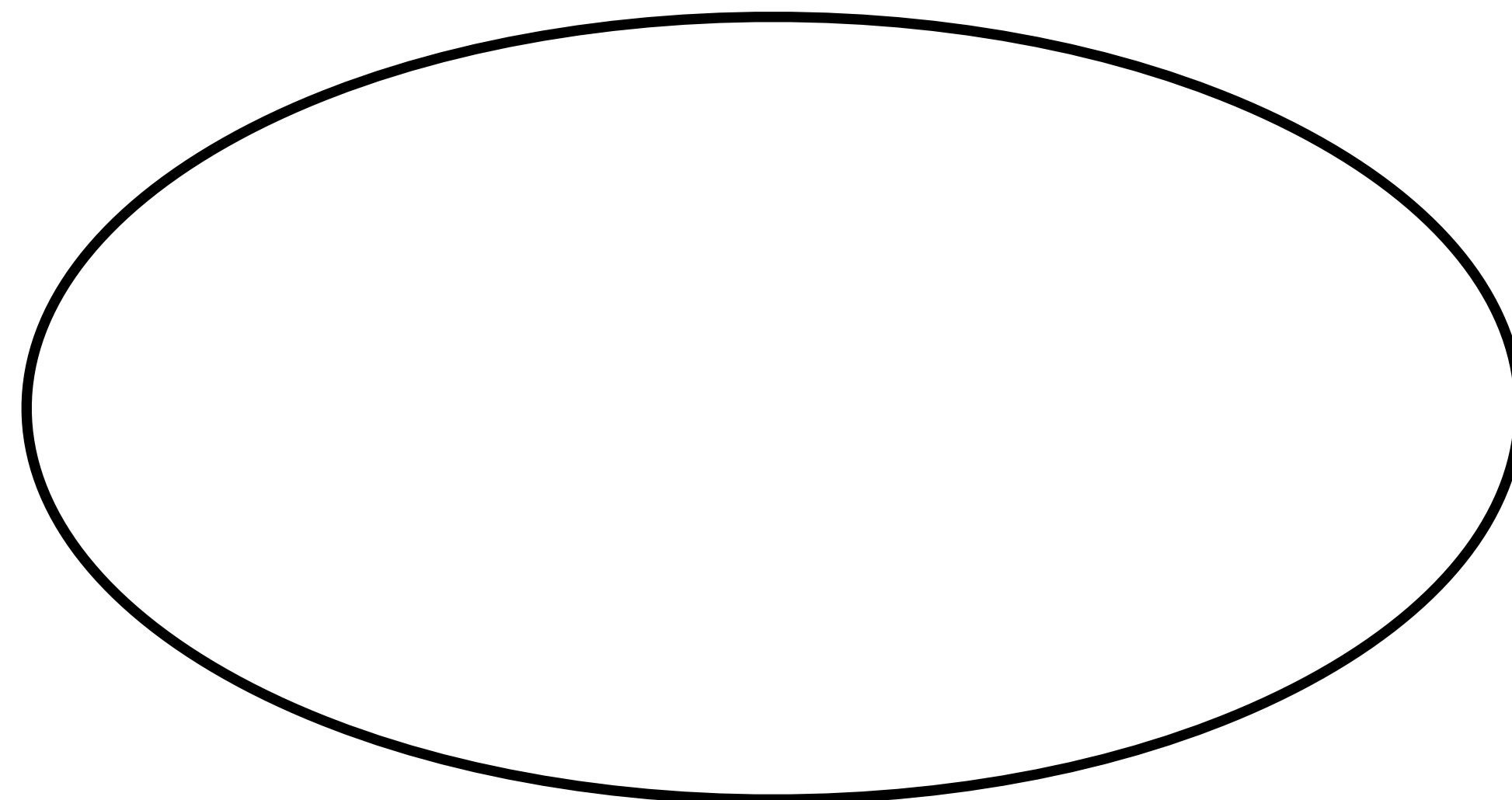


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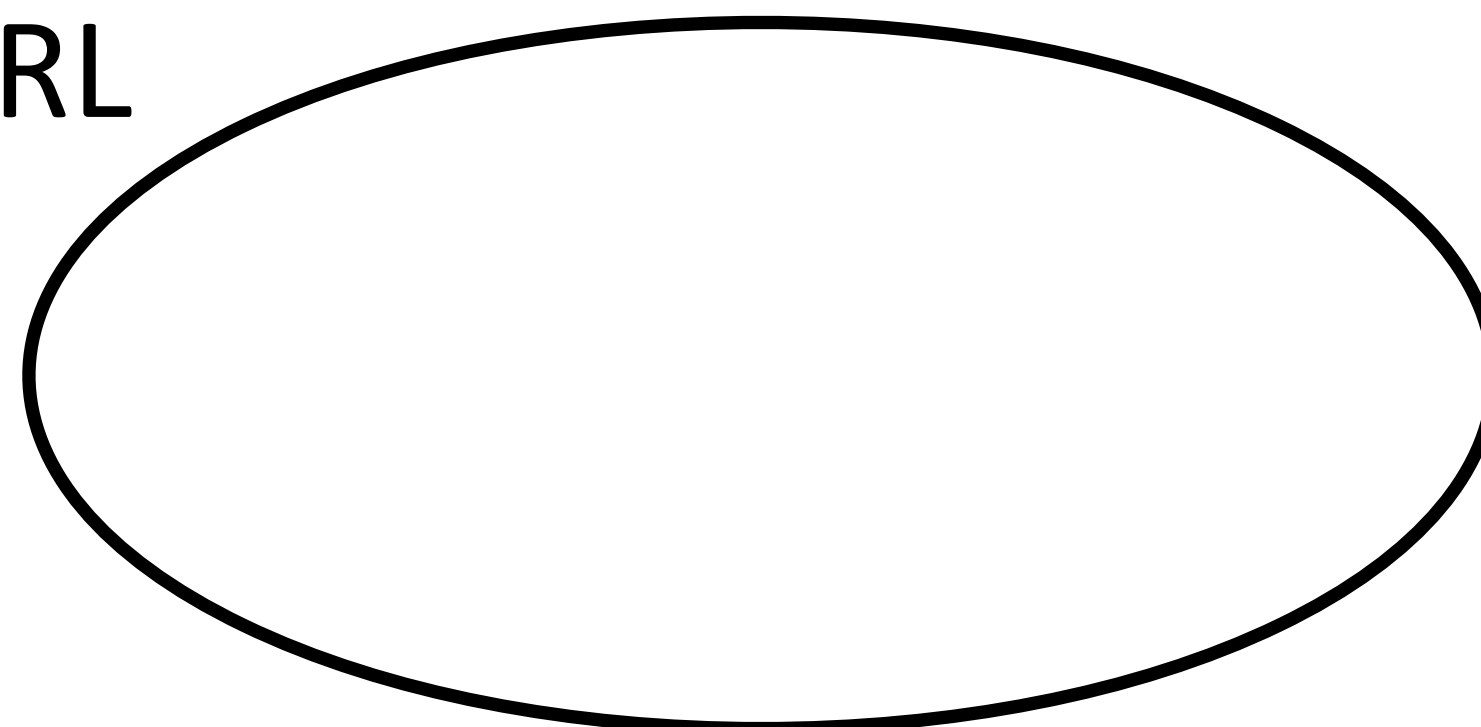
Pretraining



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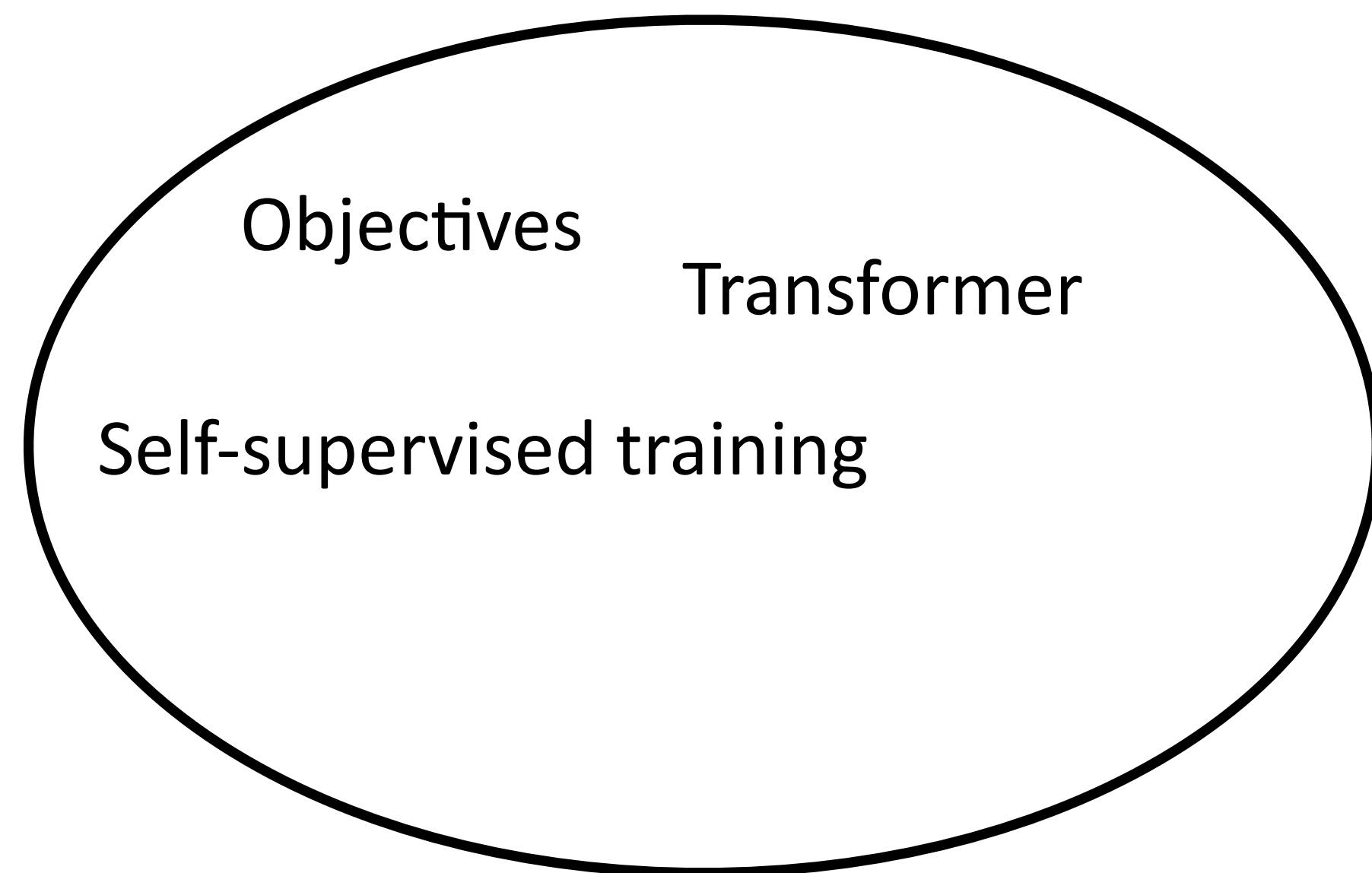


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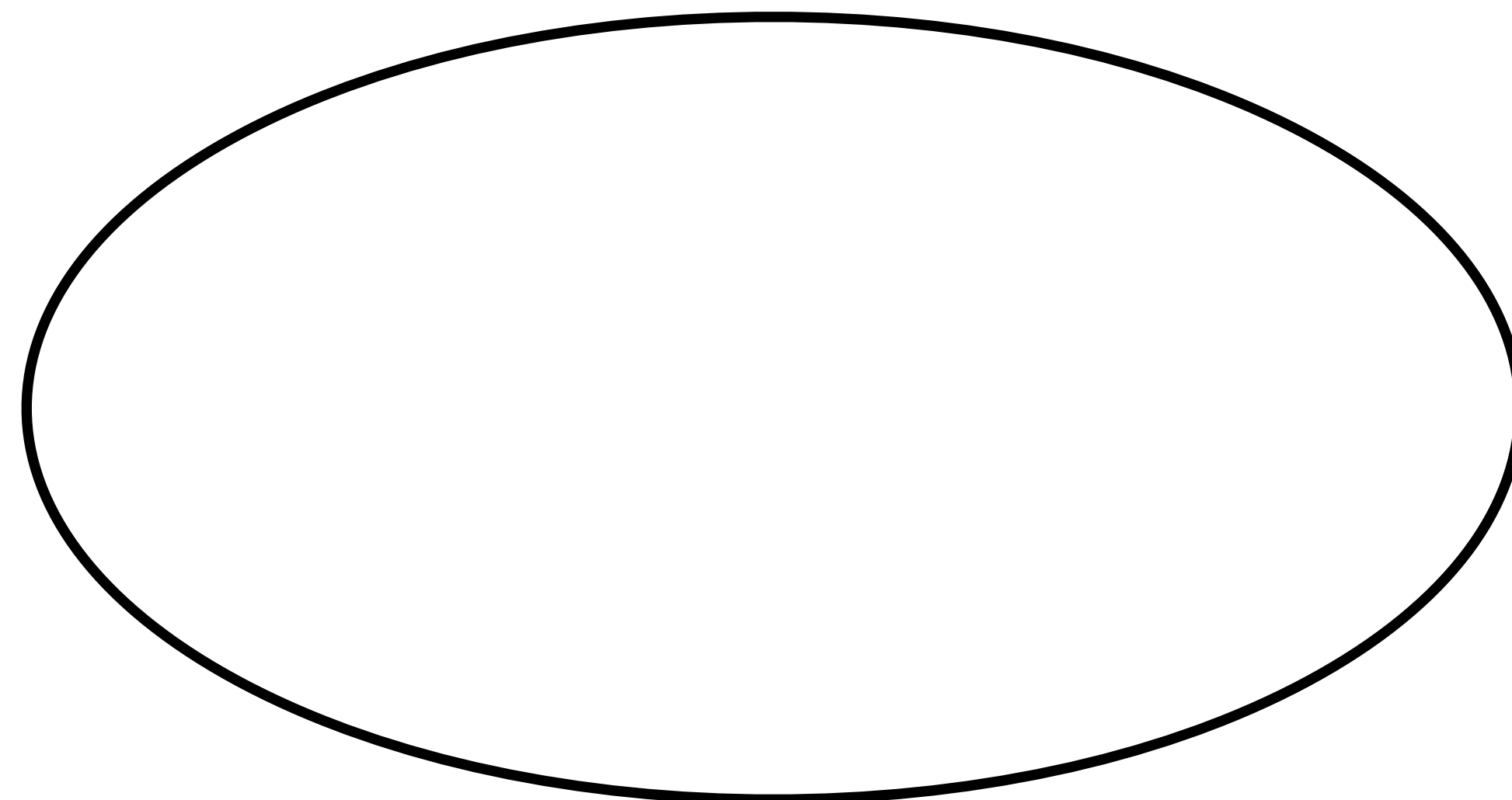


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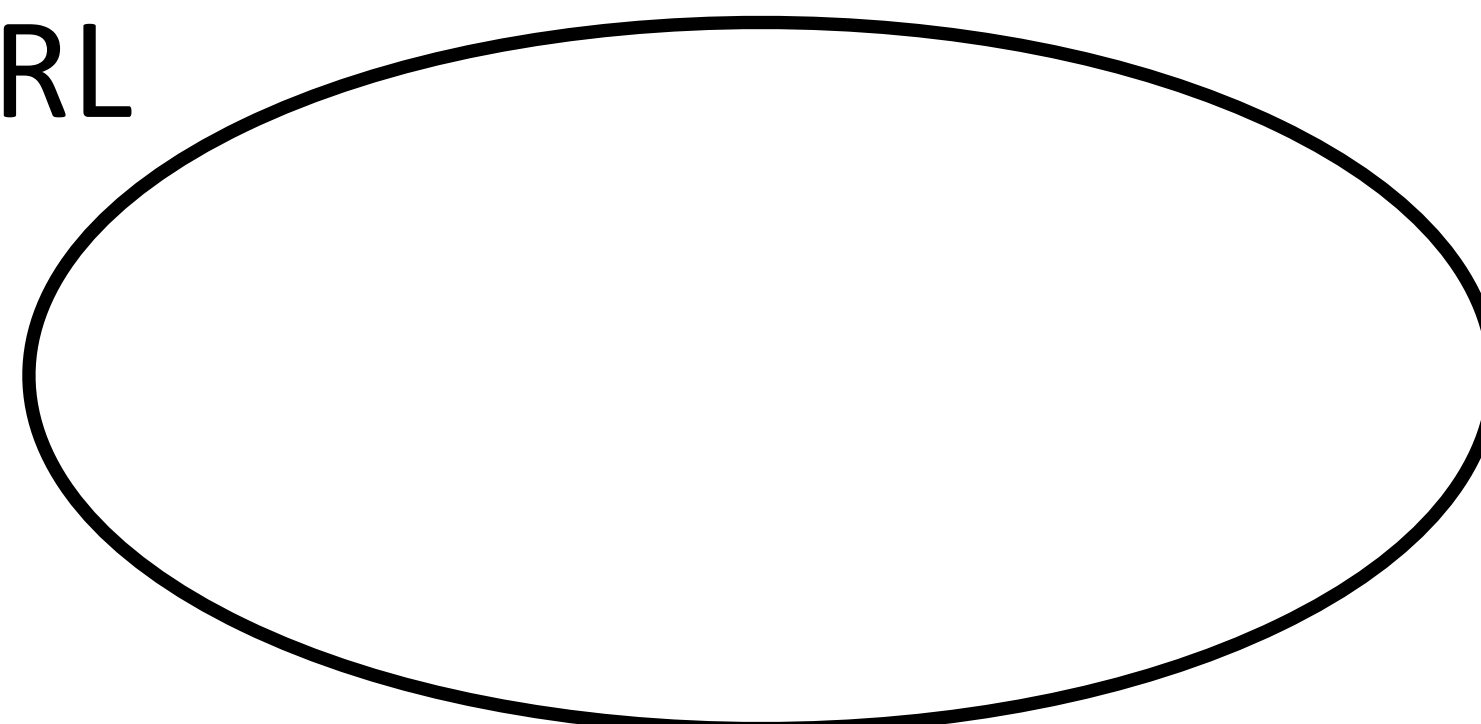
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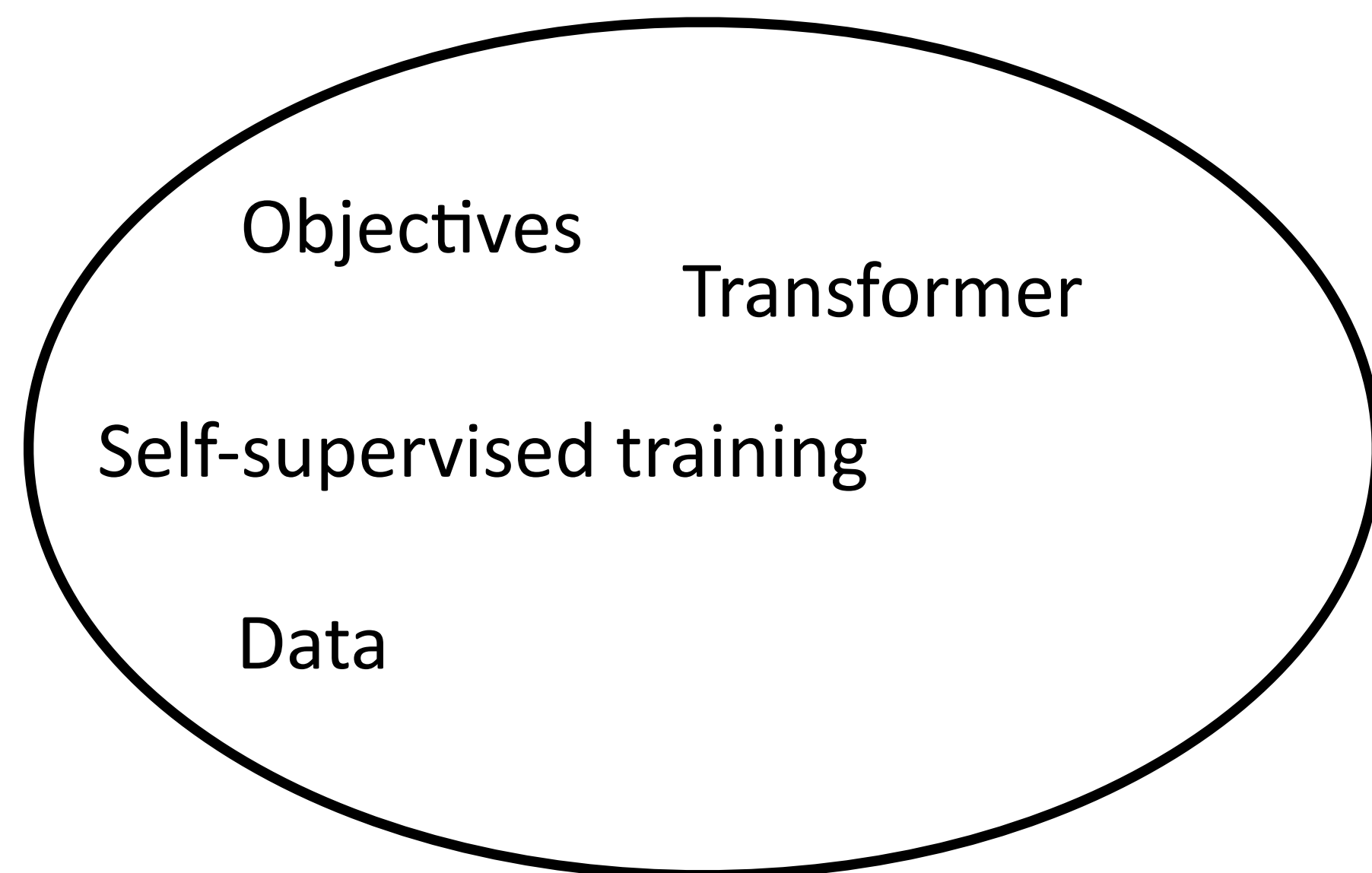


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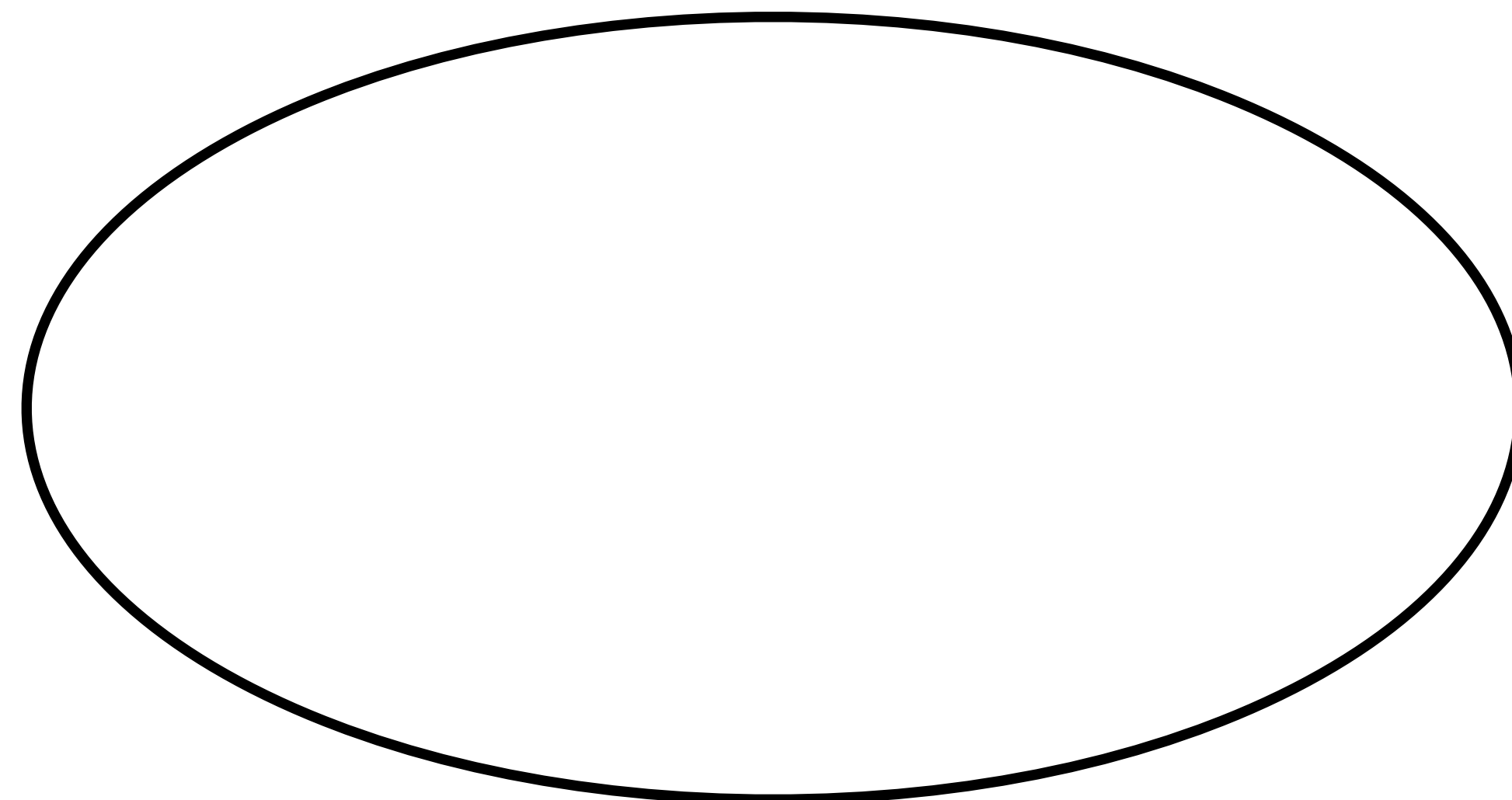


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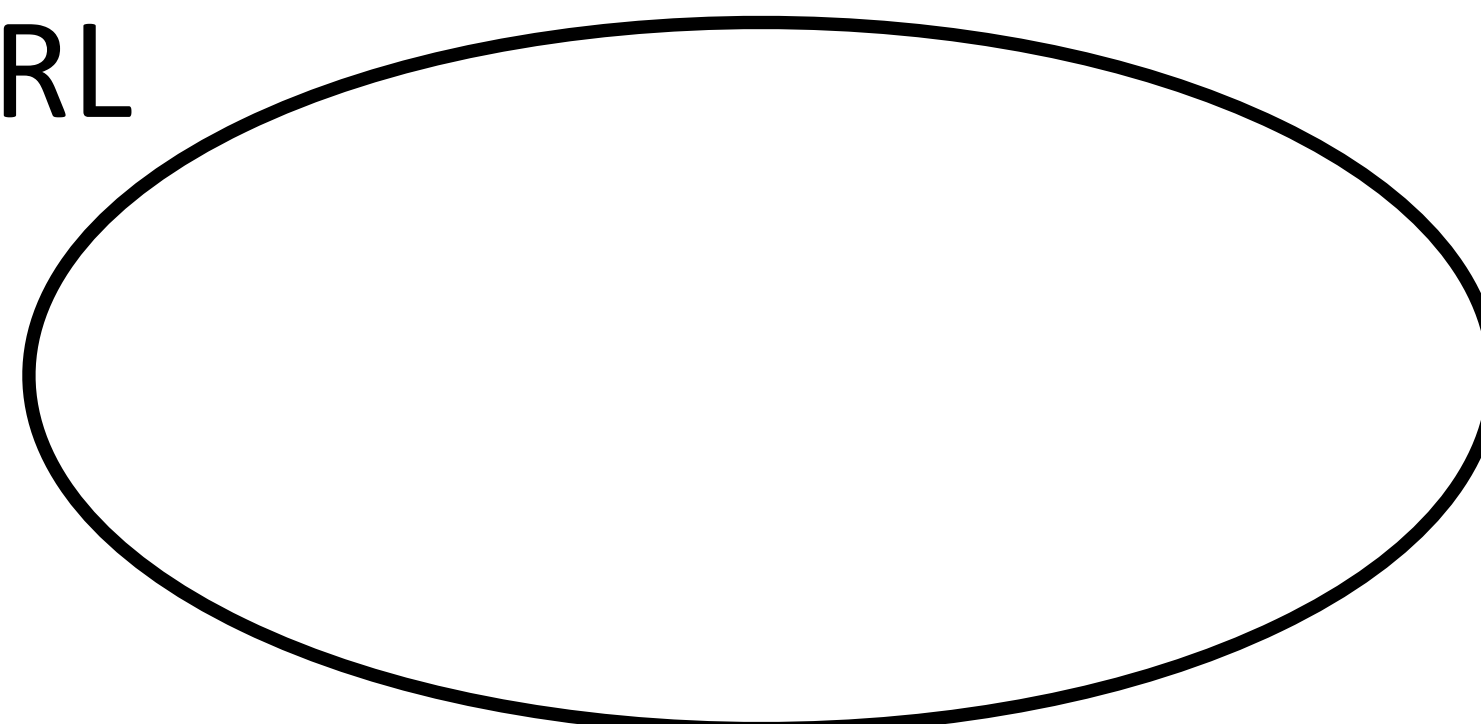
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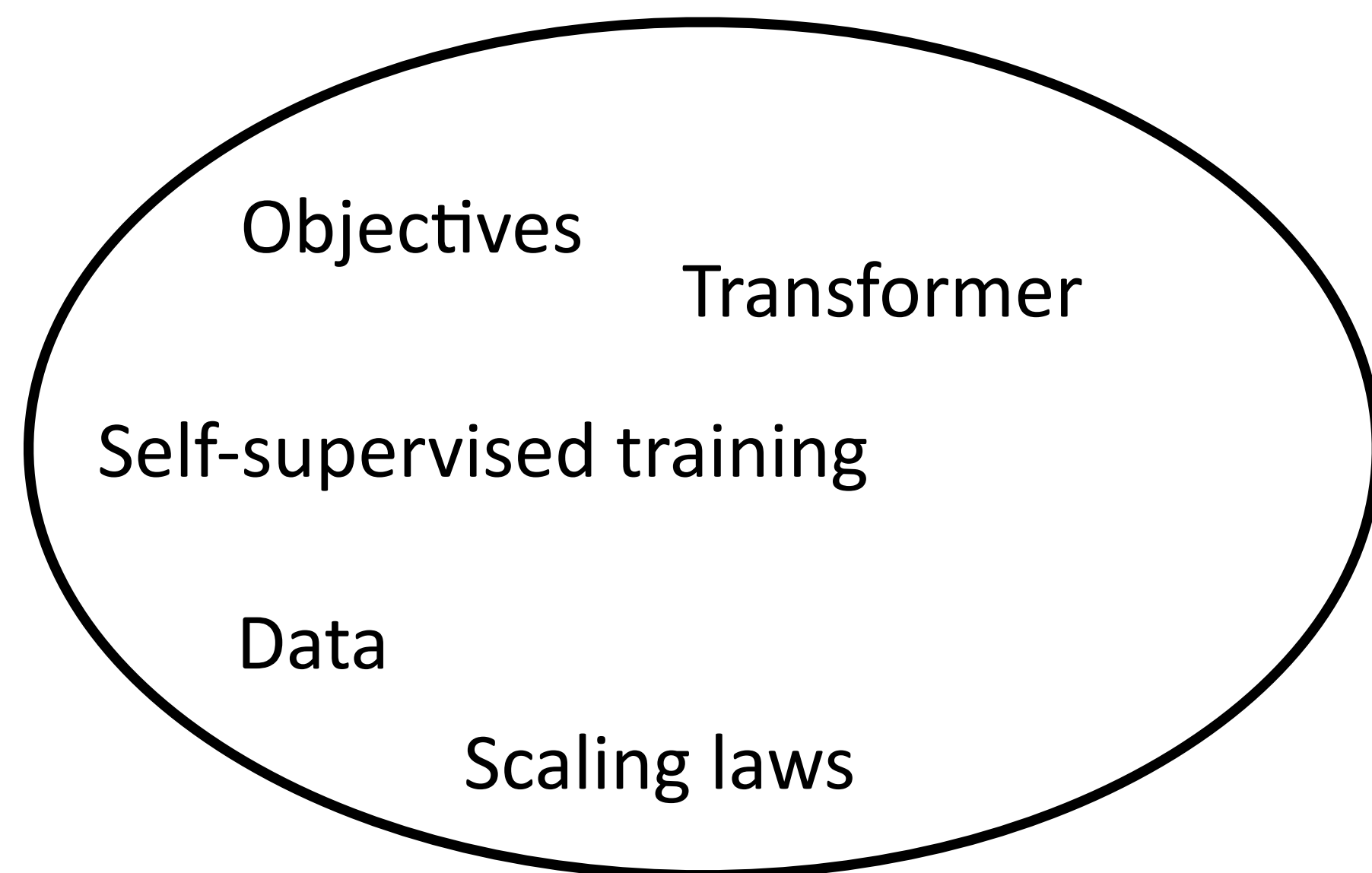


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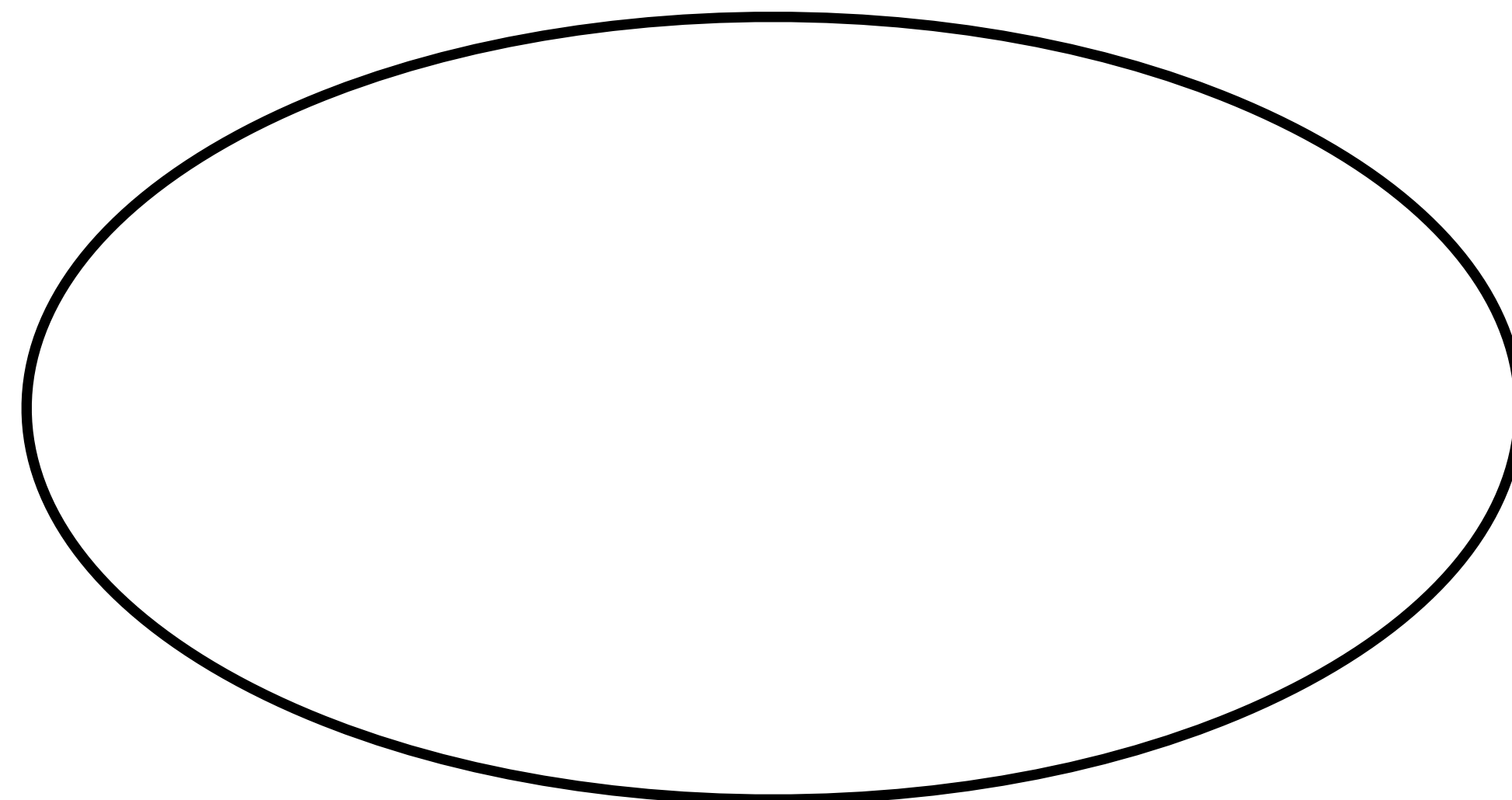


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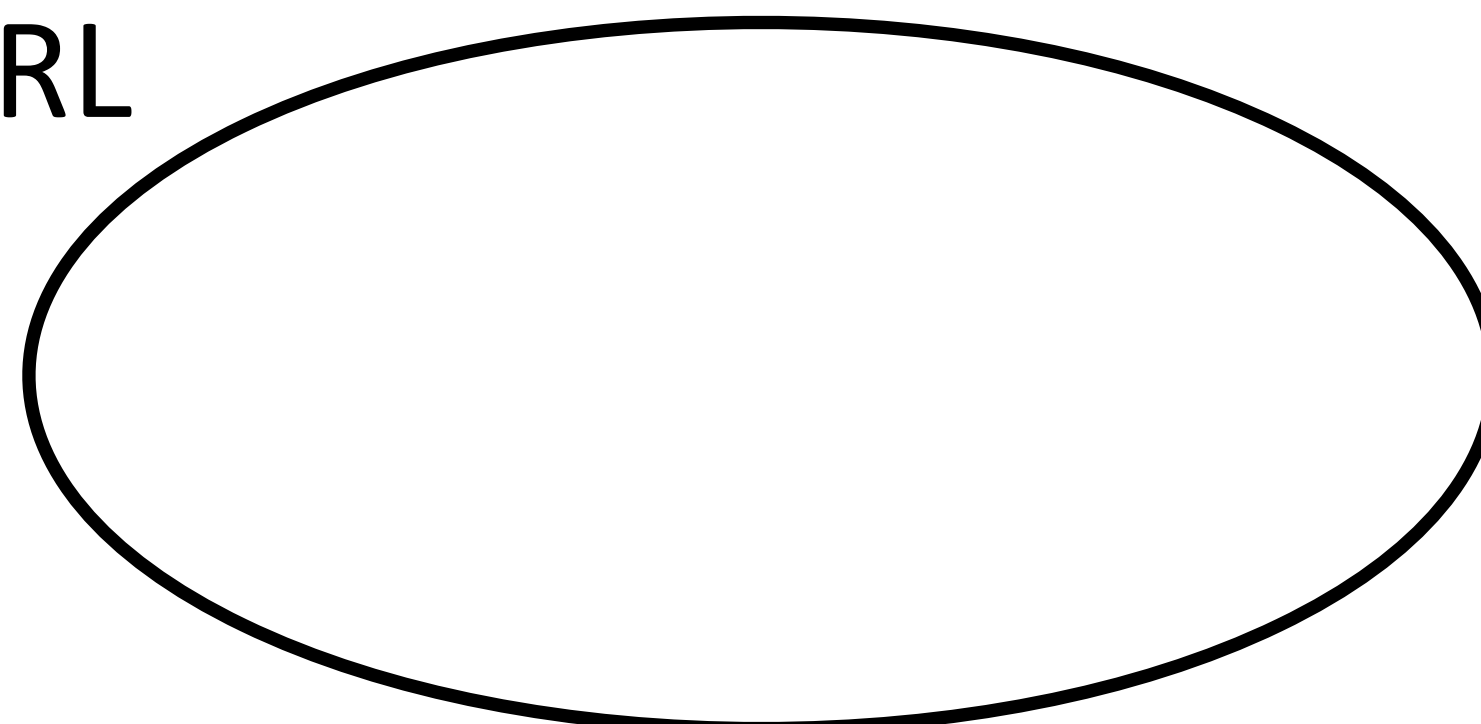
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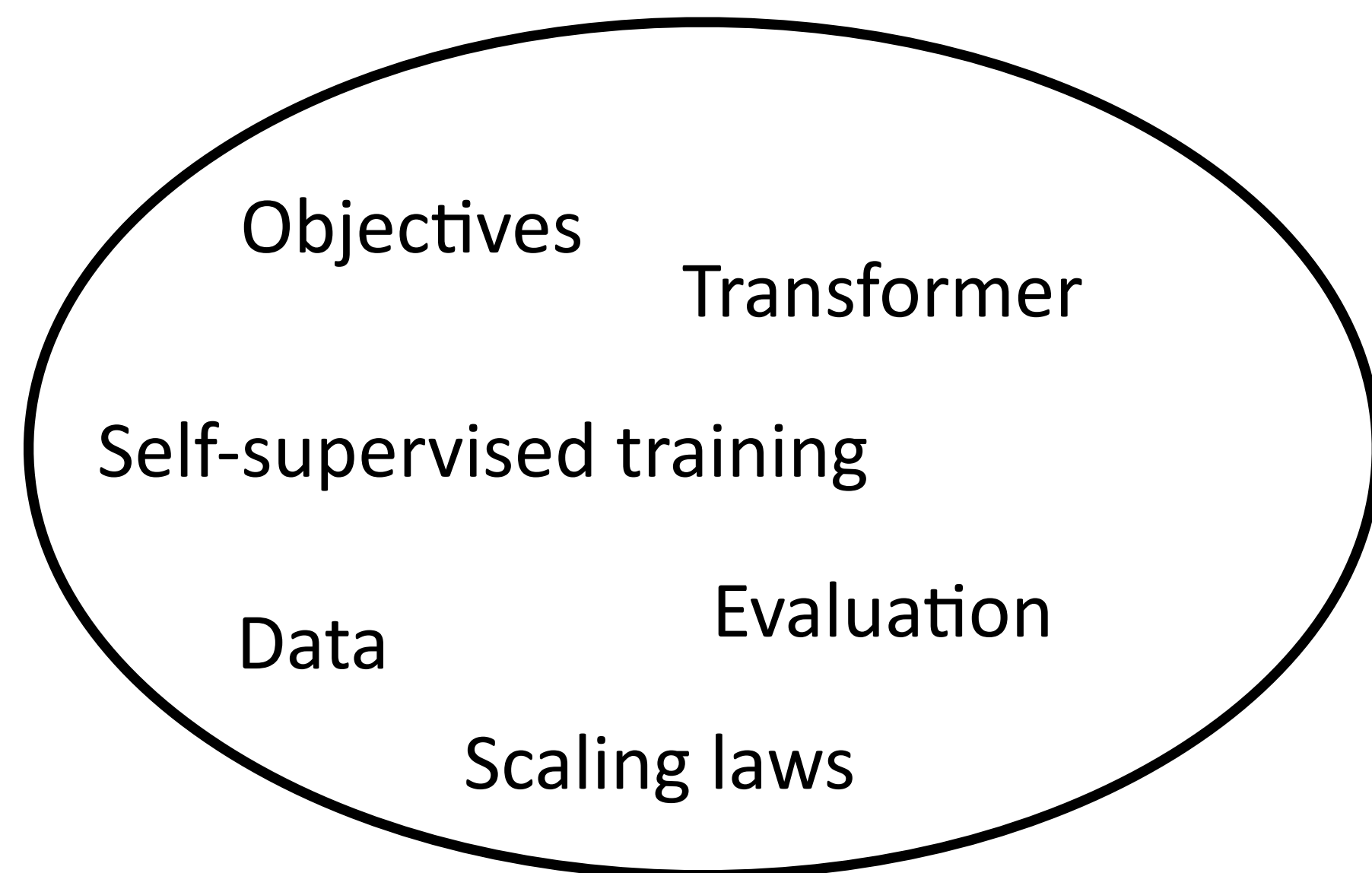


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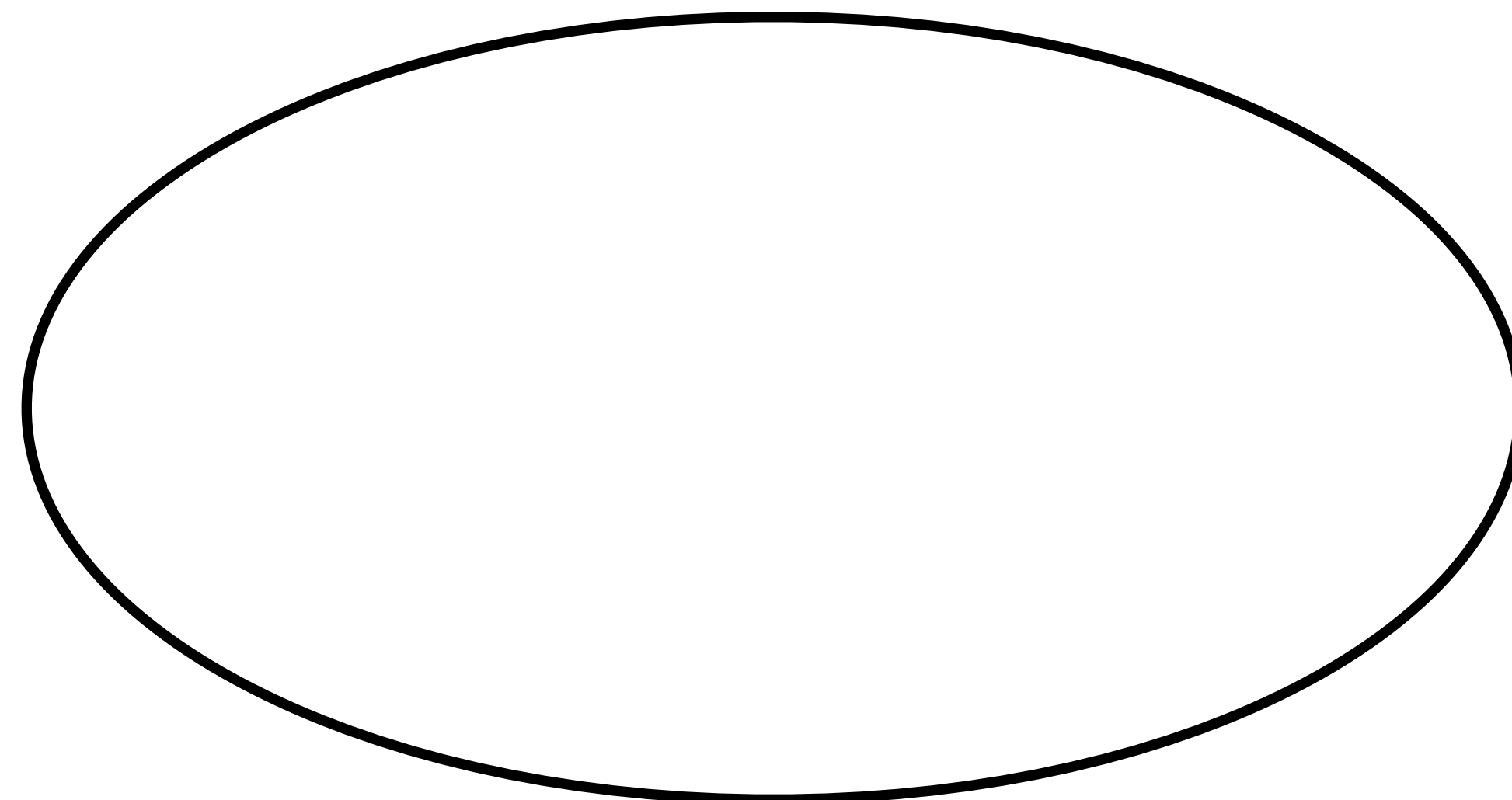


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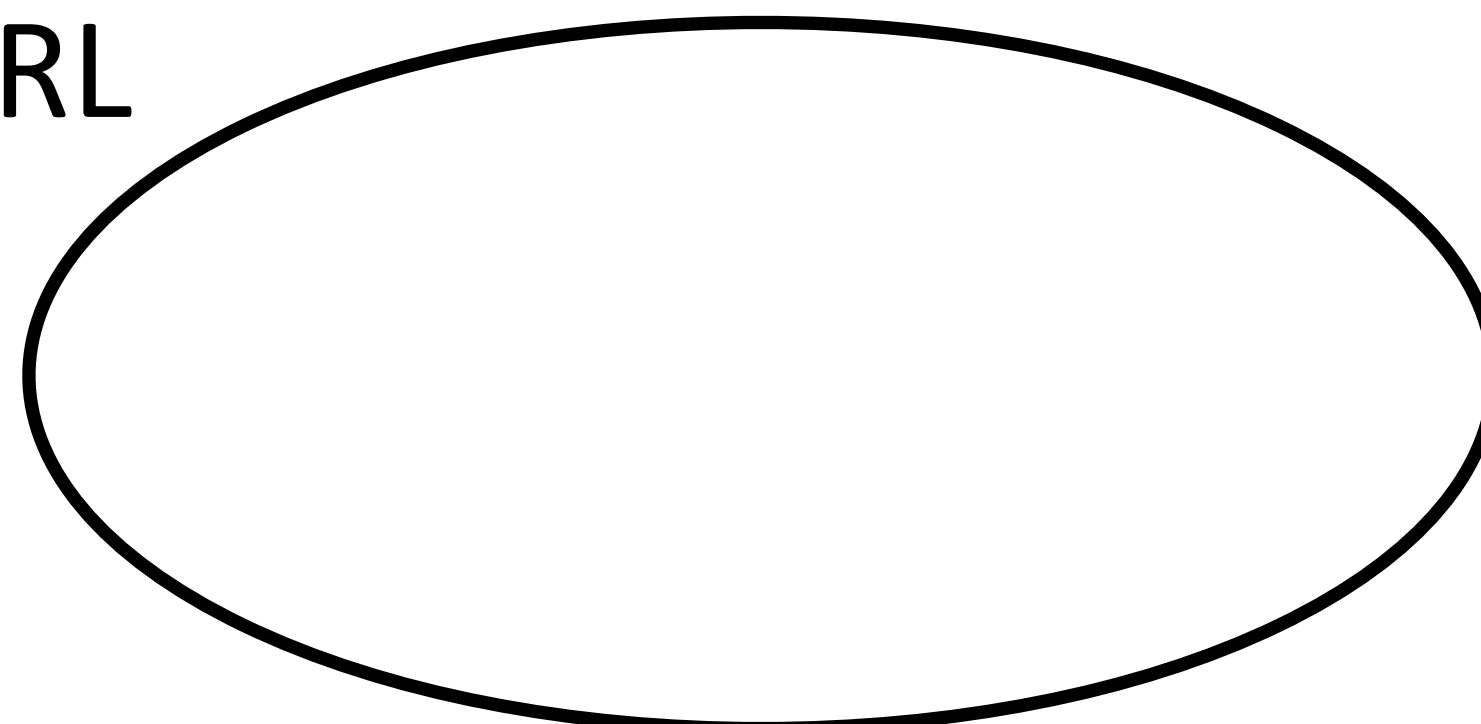
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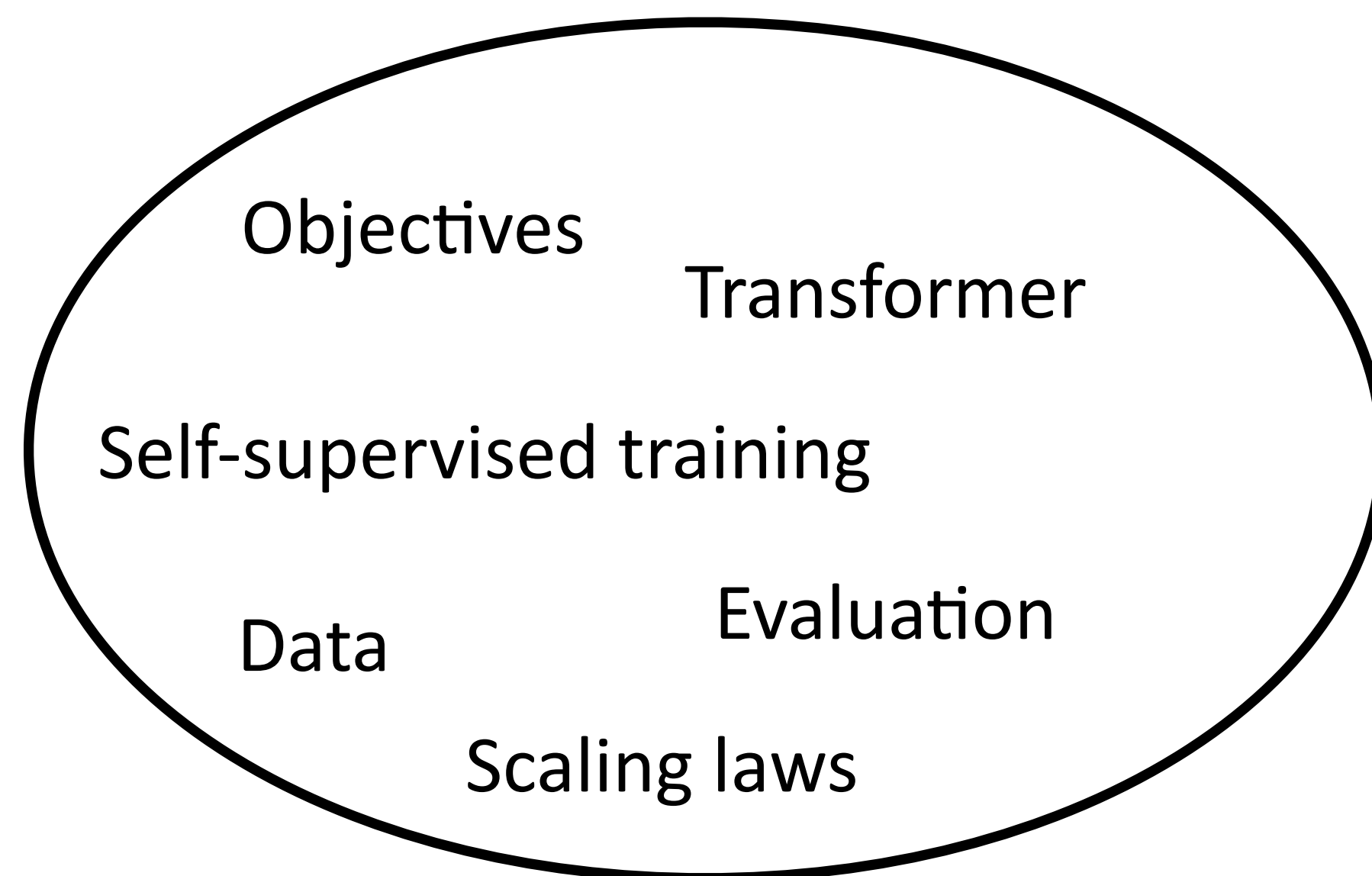


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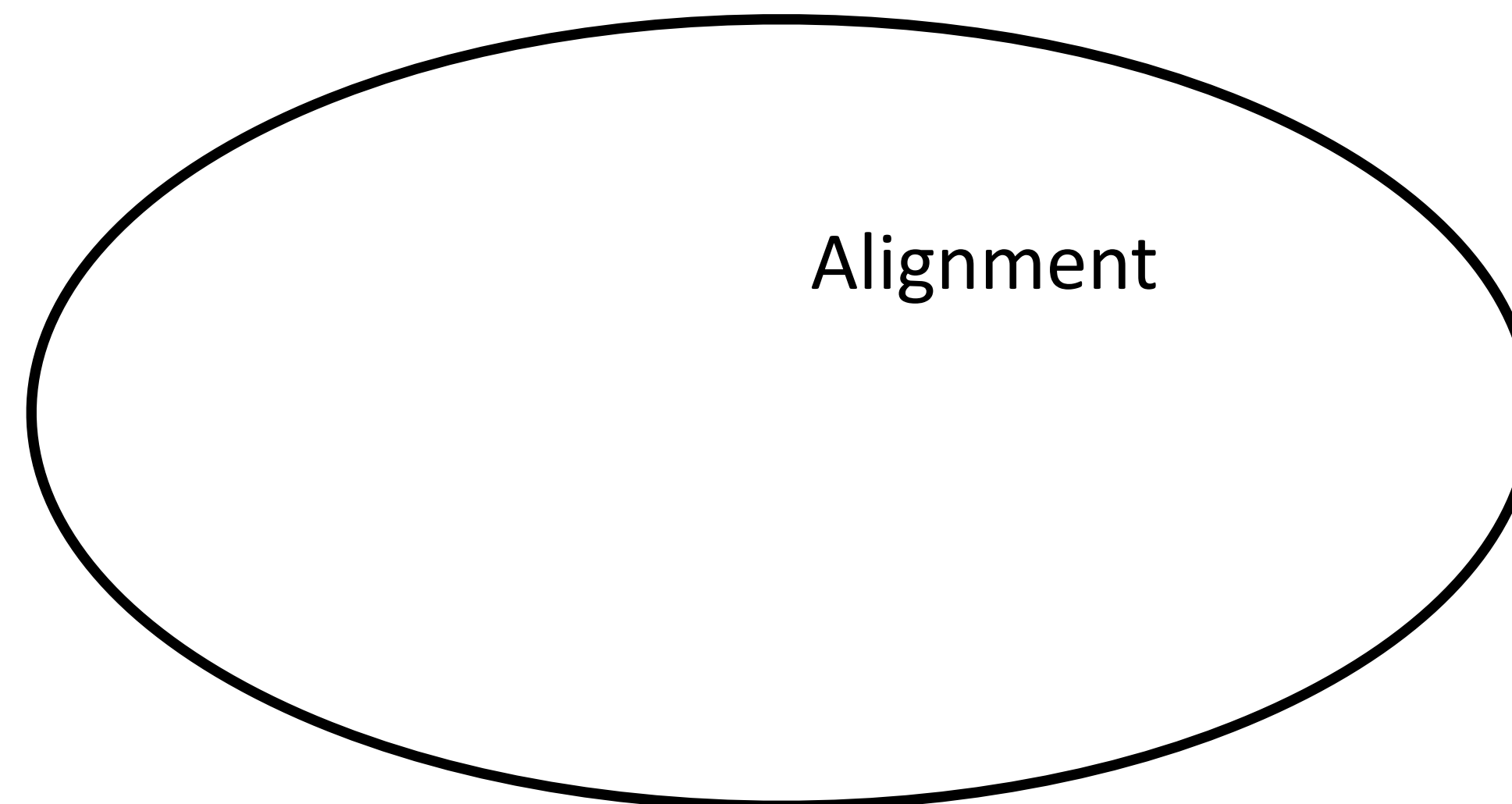


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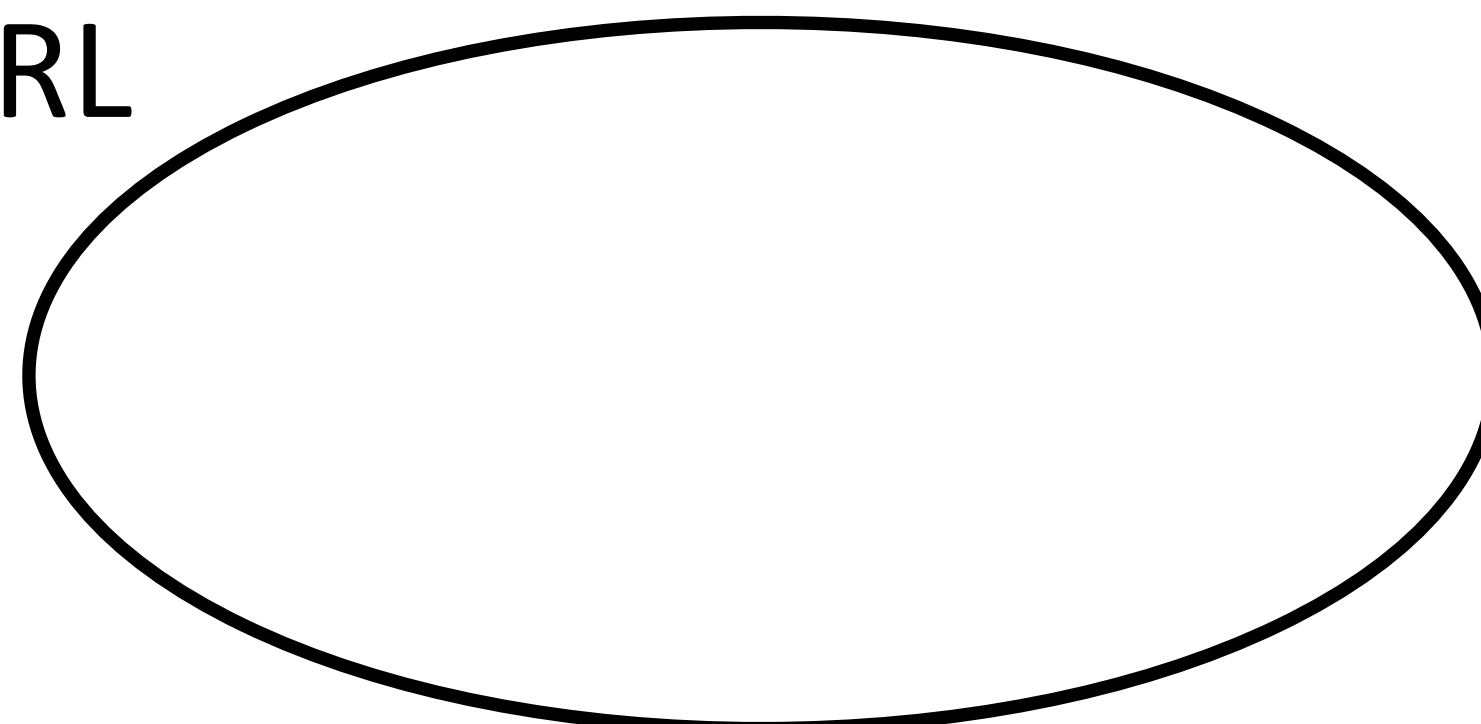
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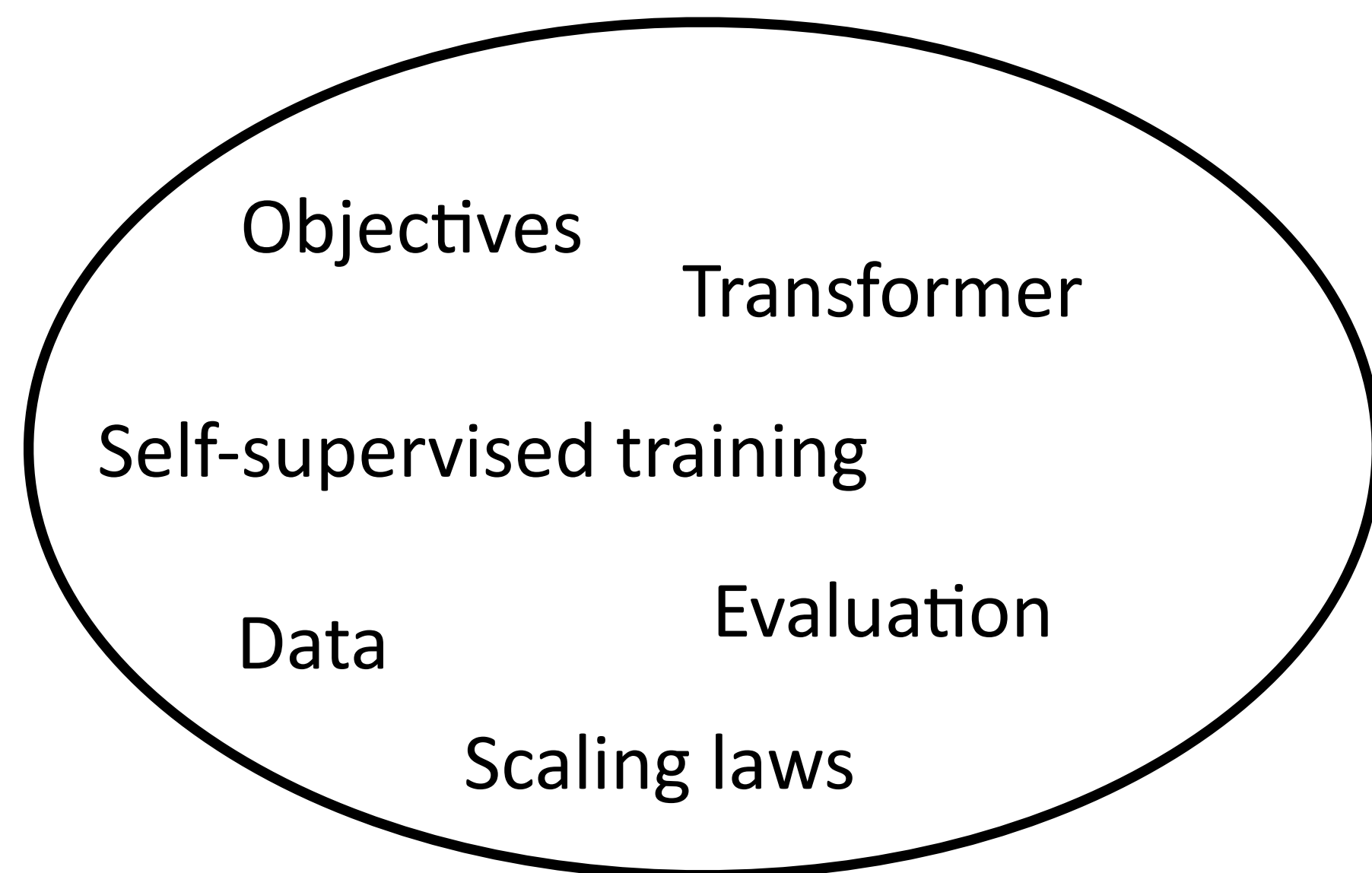


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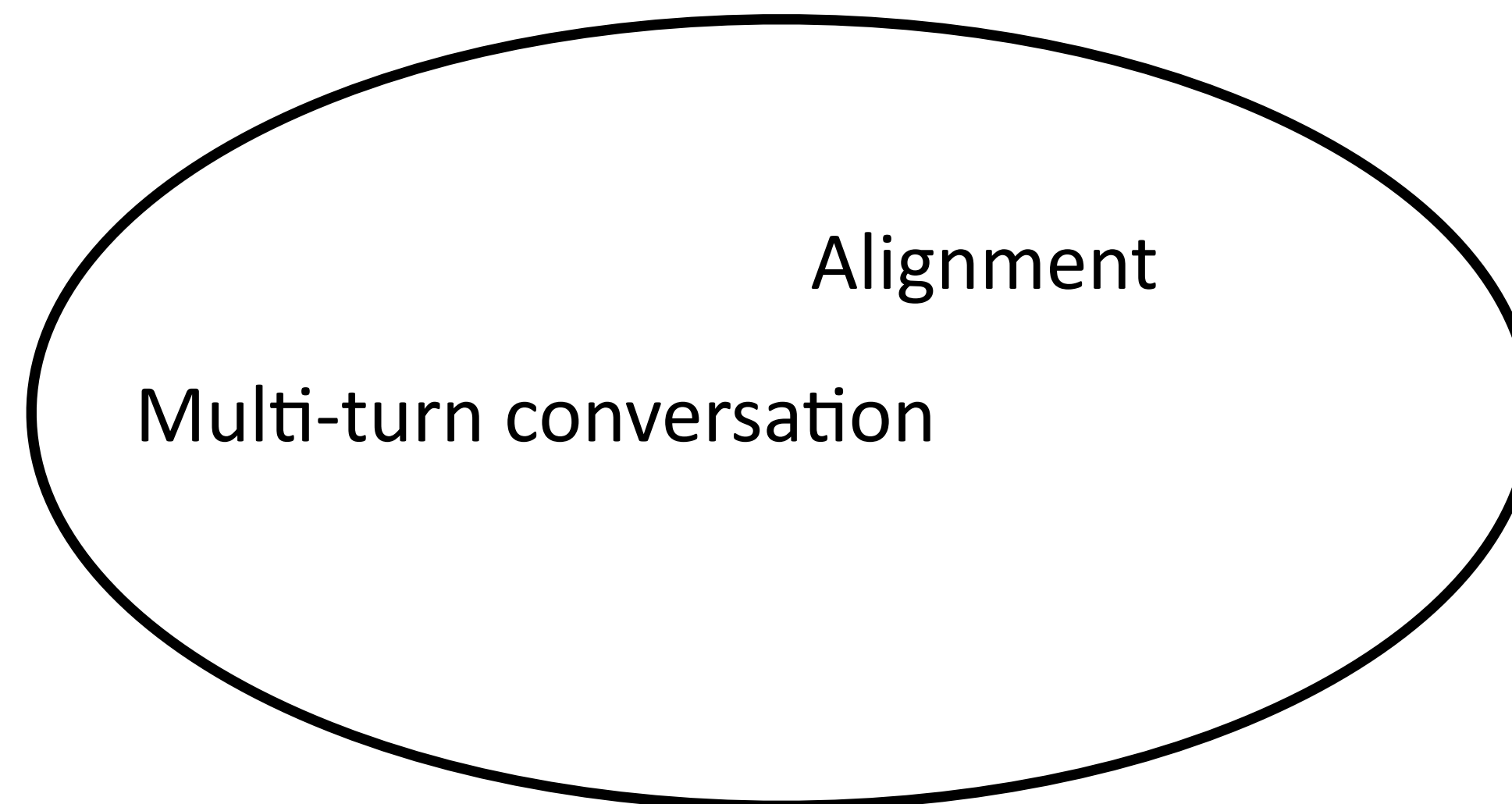


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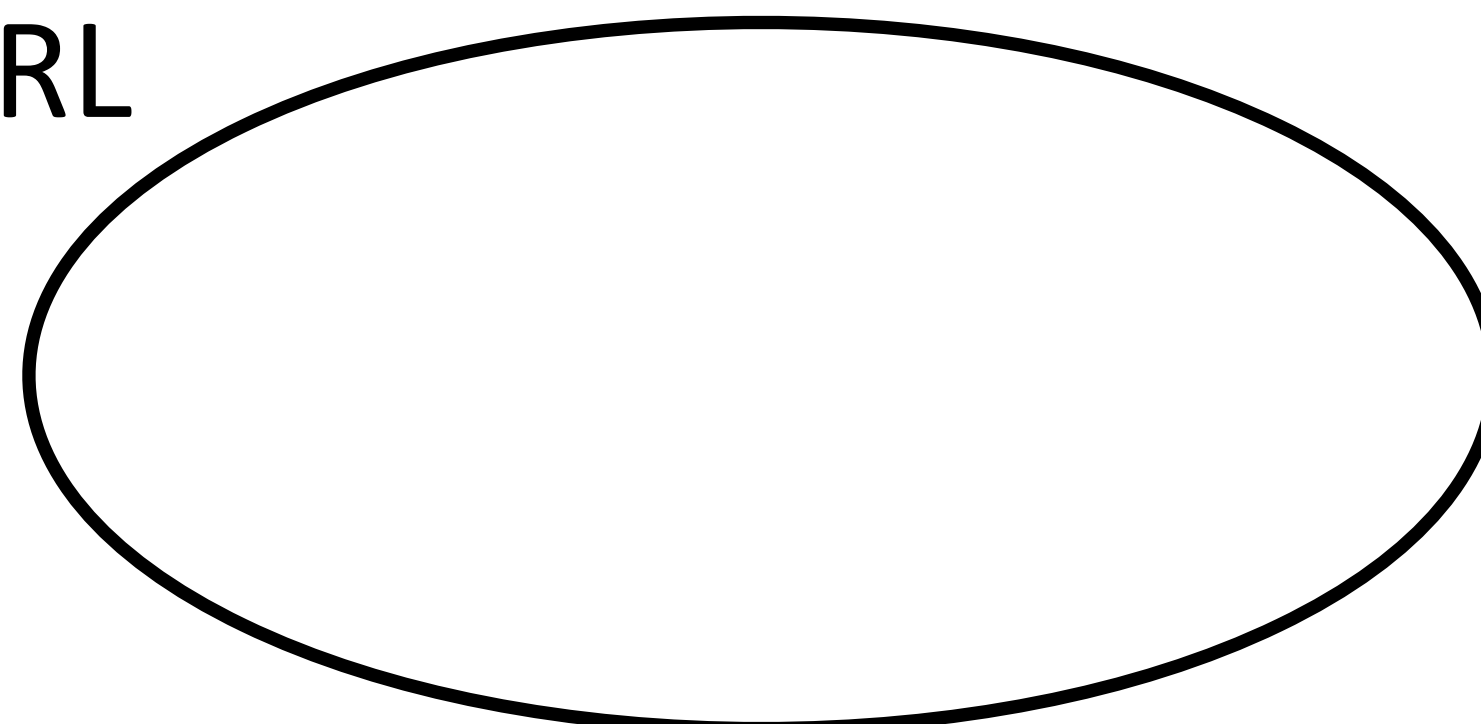
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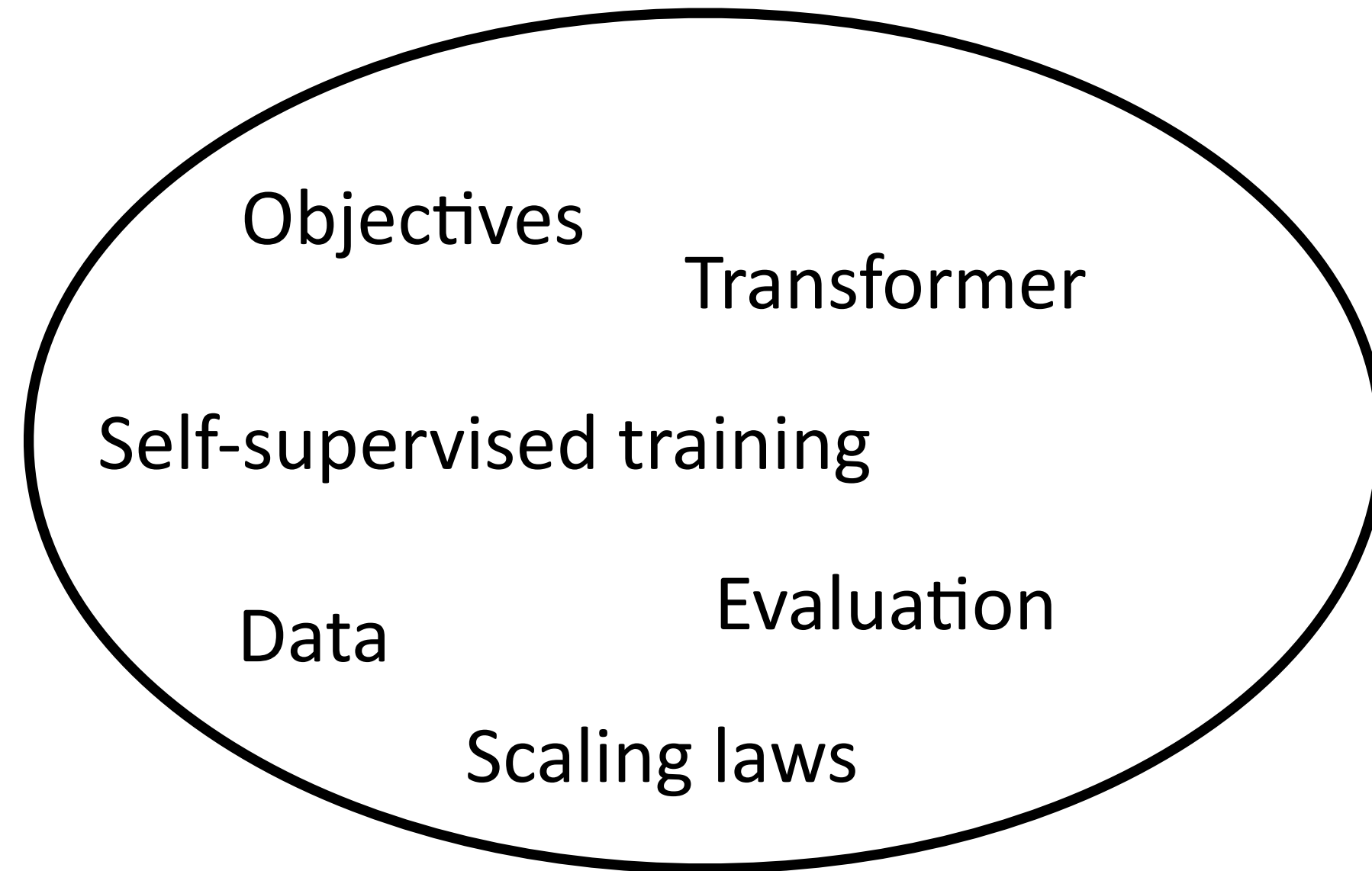


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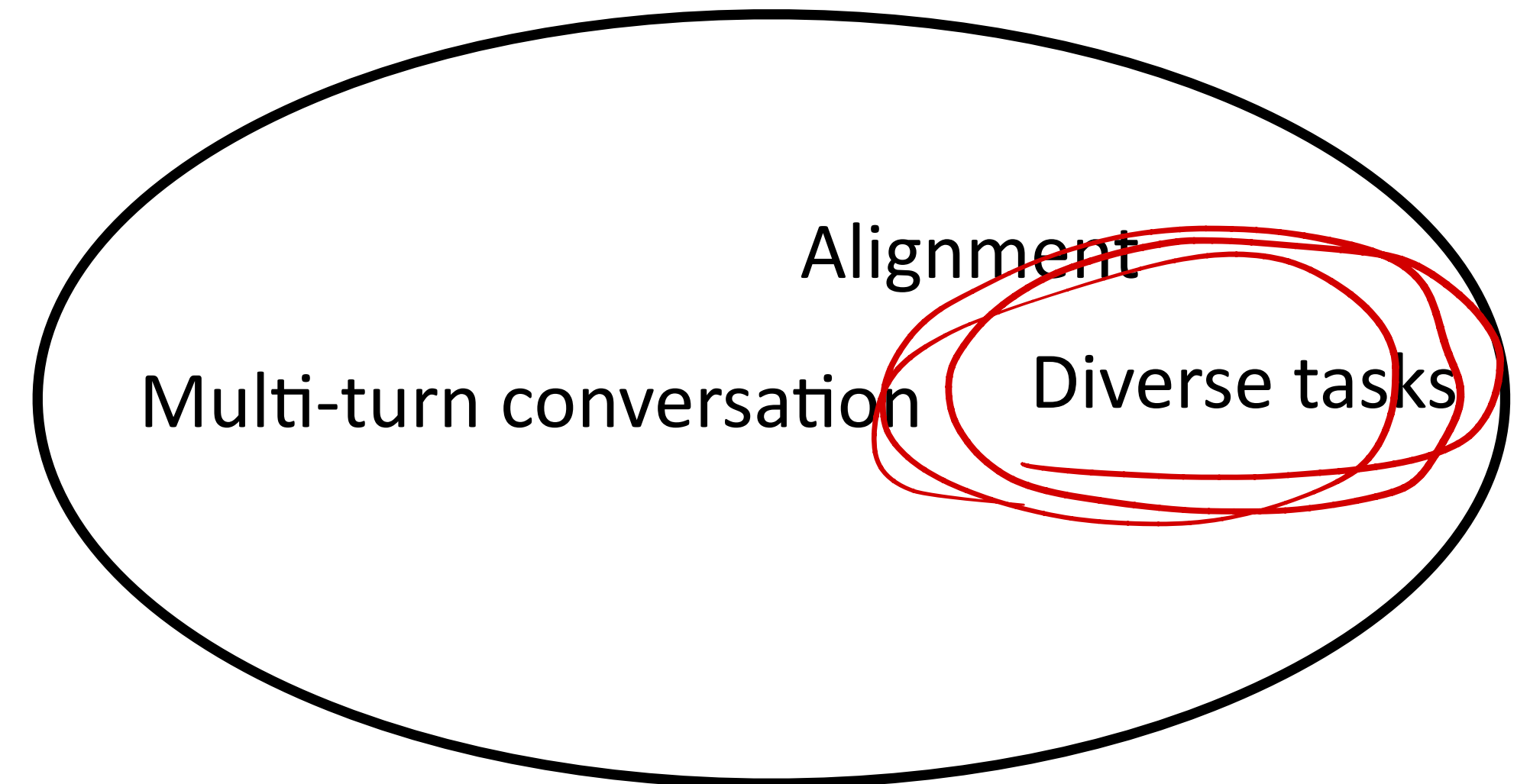


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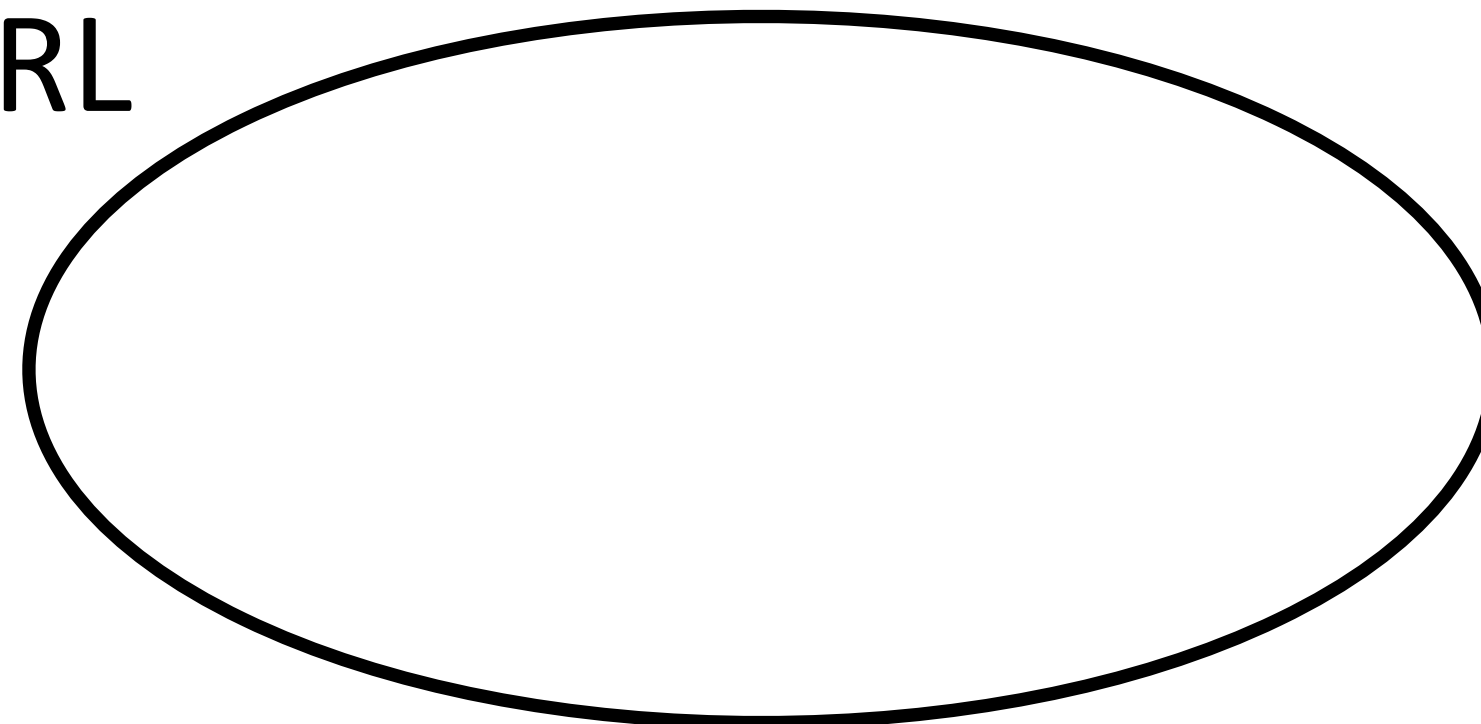
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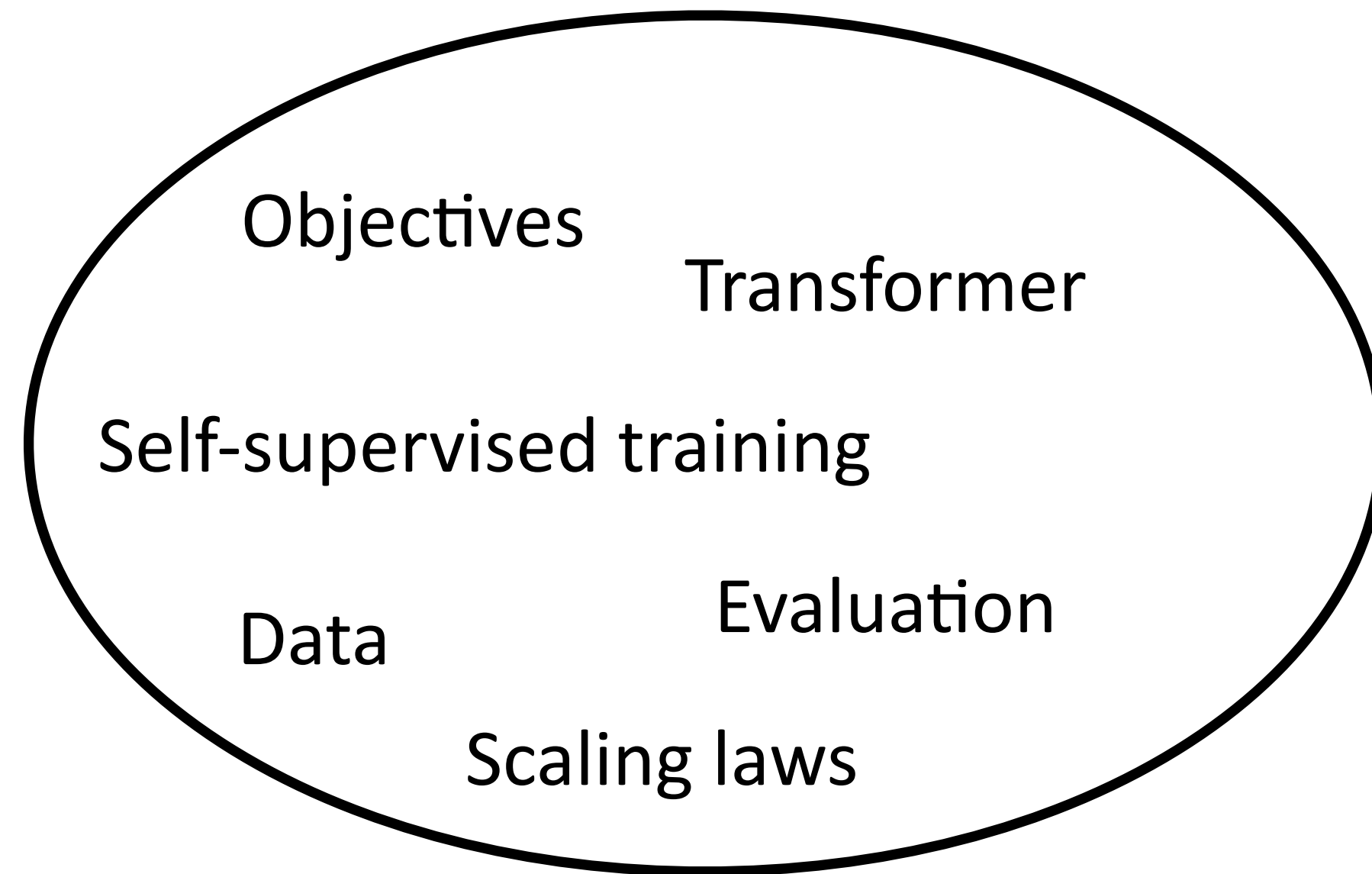


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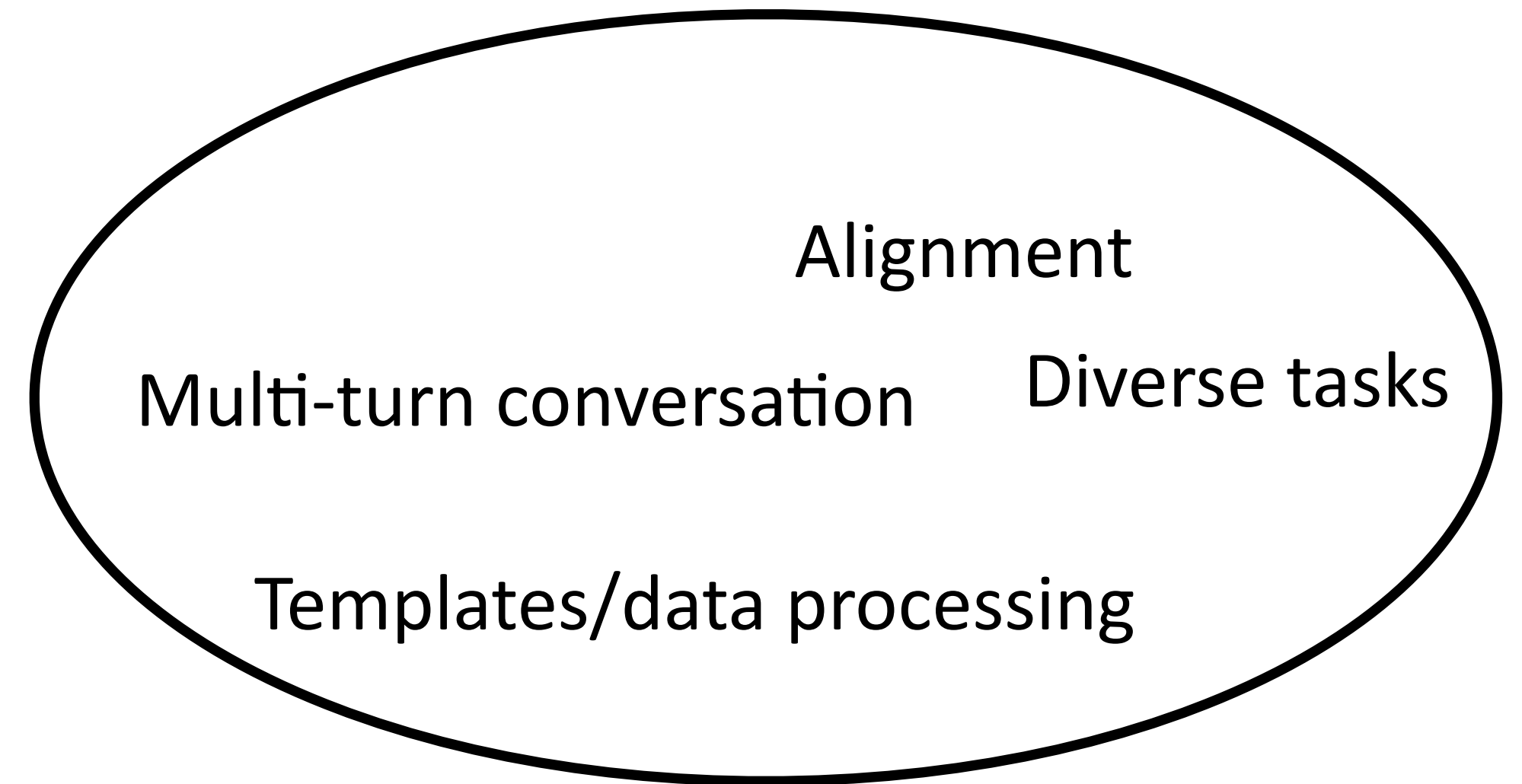
Review — Method

Pretraining

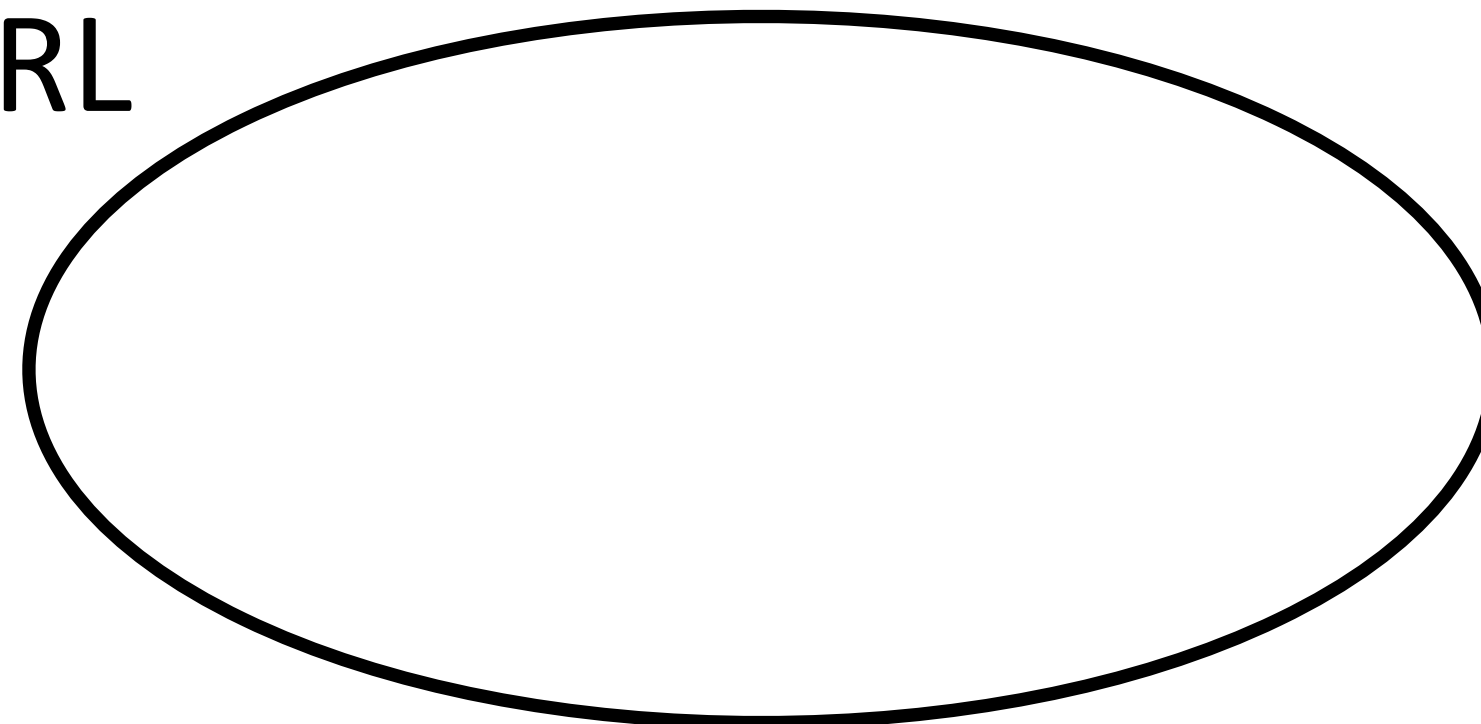


SFT

= instruction tuning

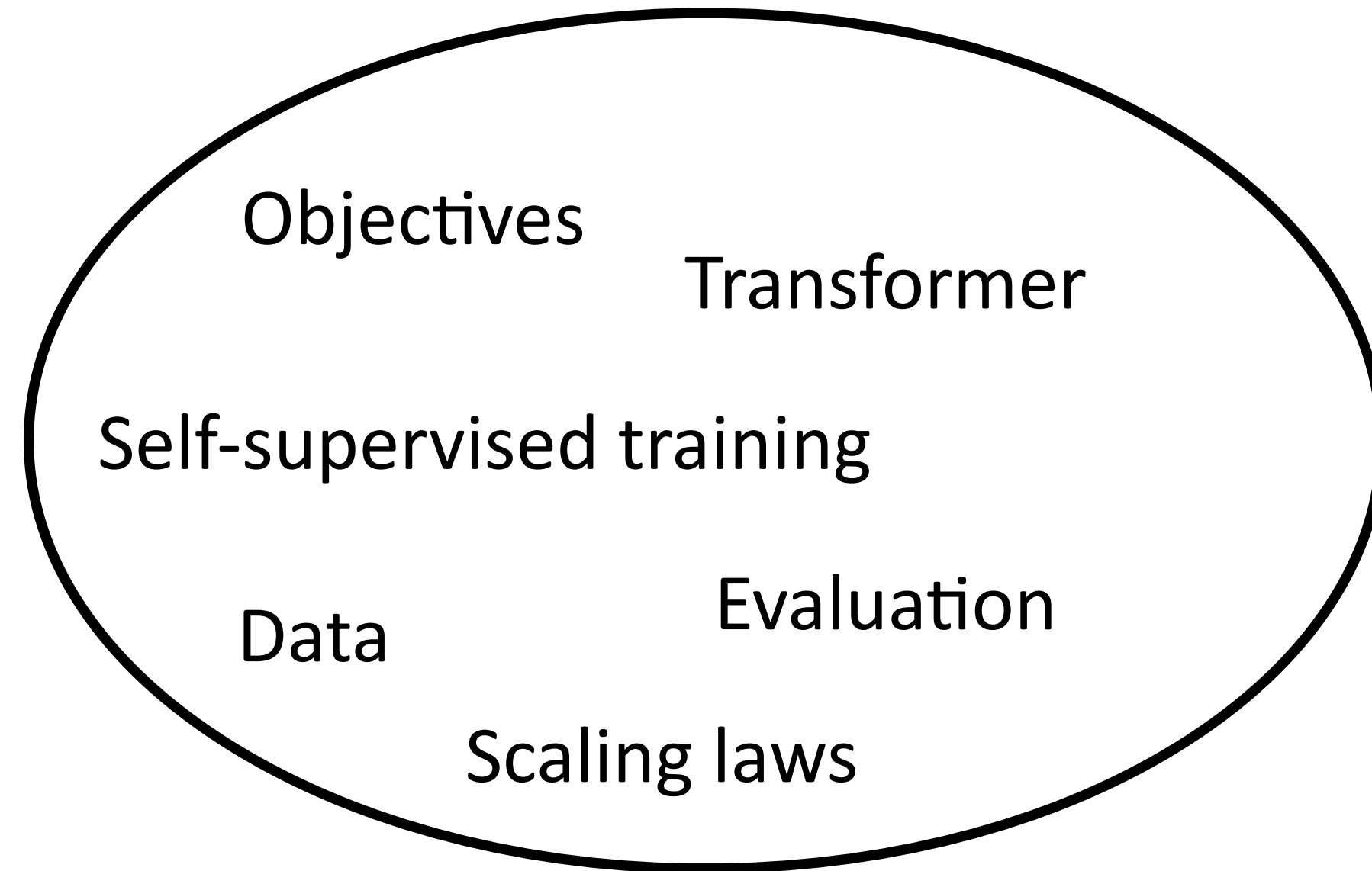


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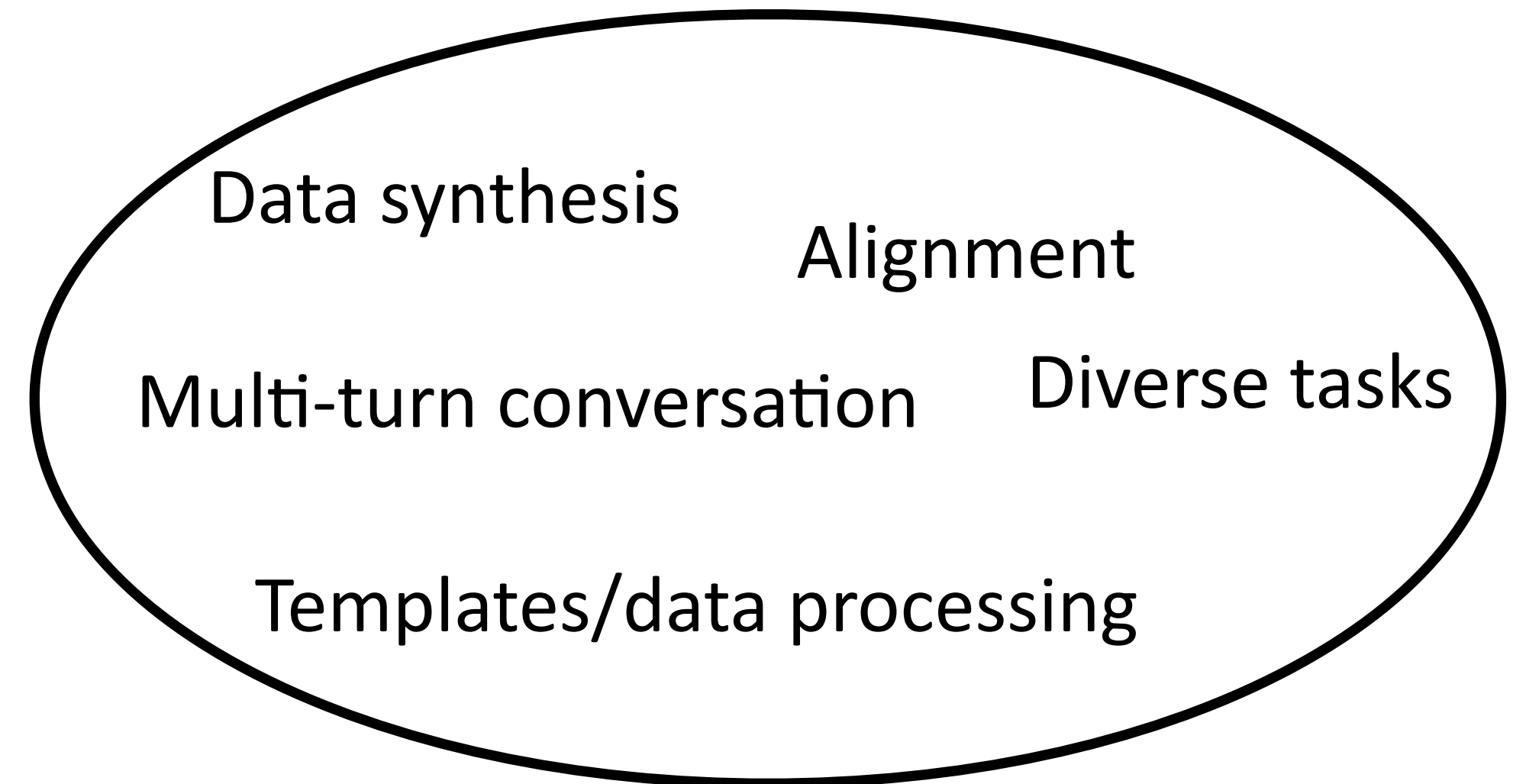


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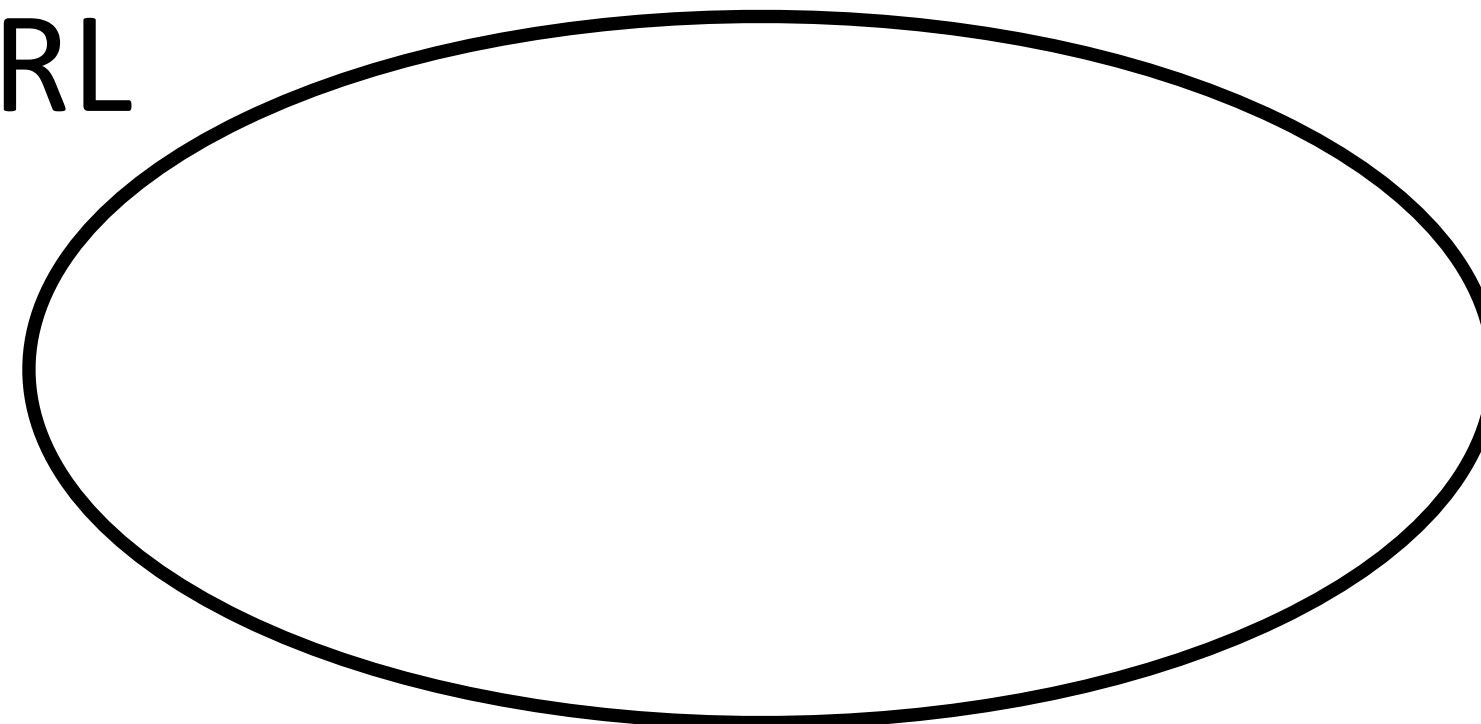
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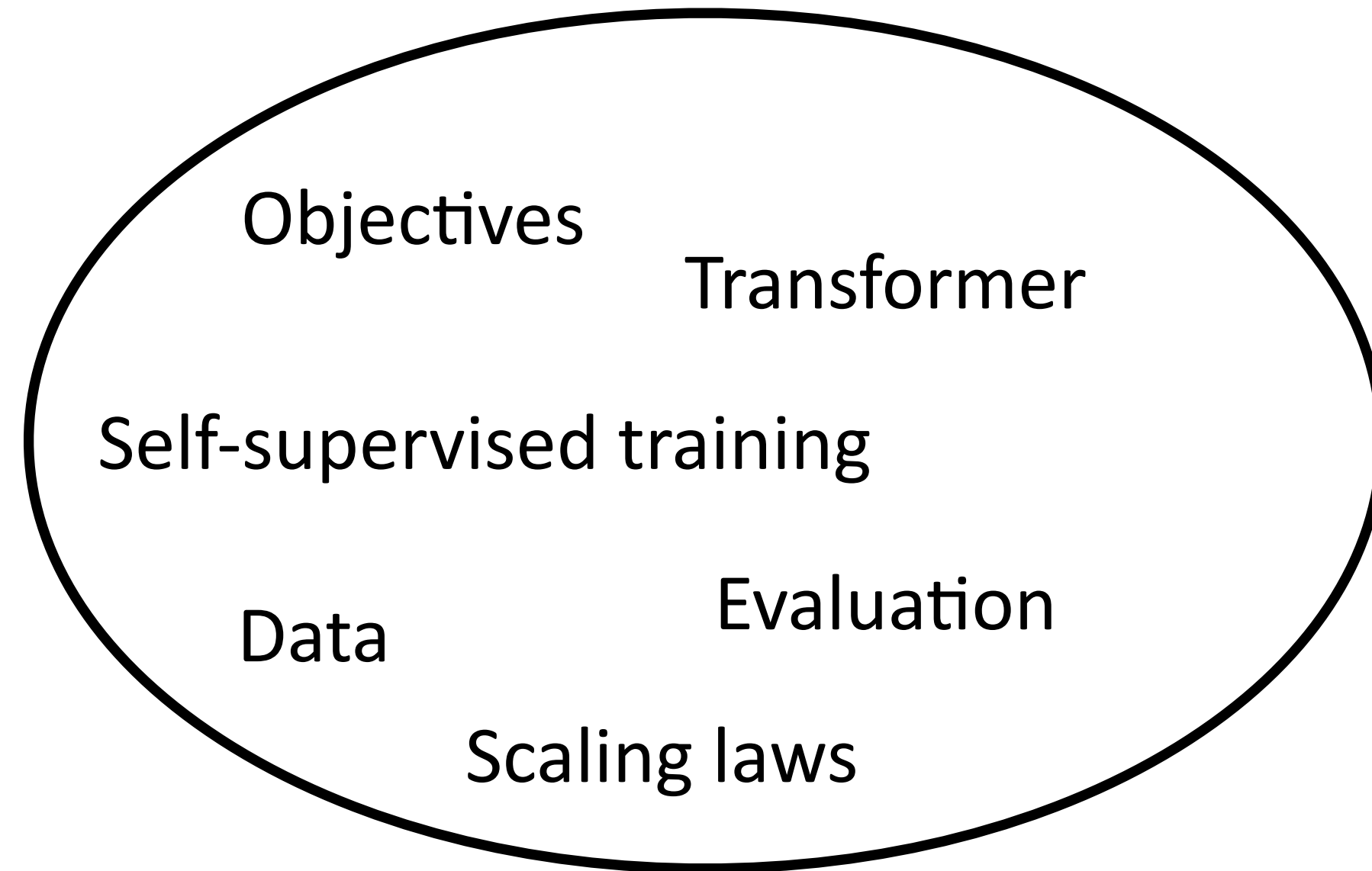


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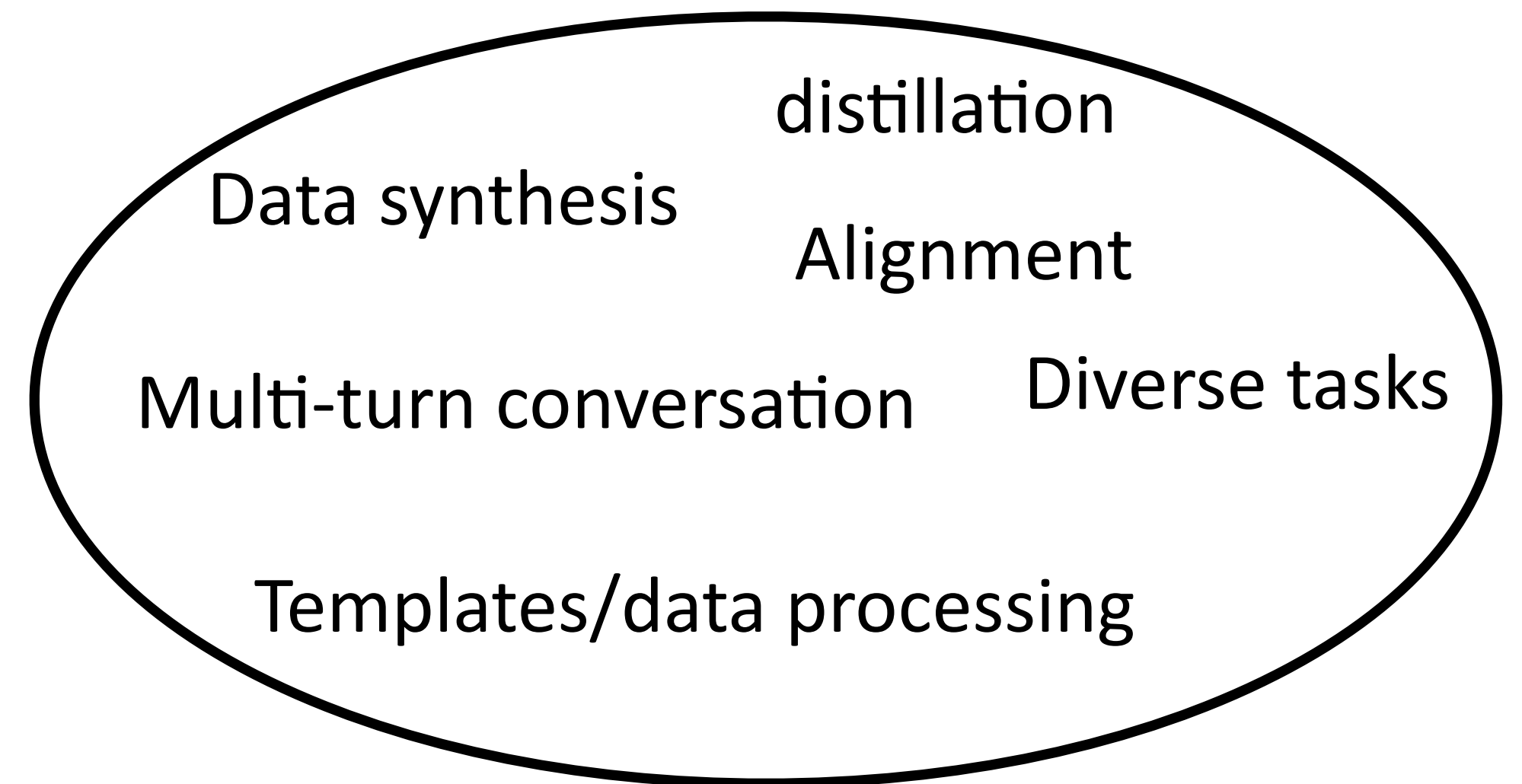


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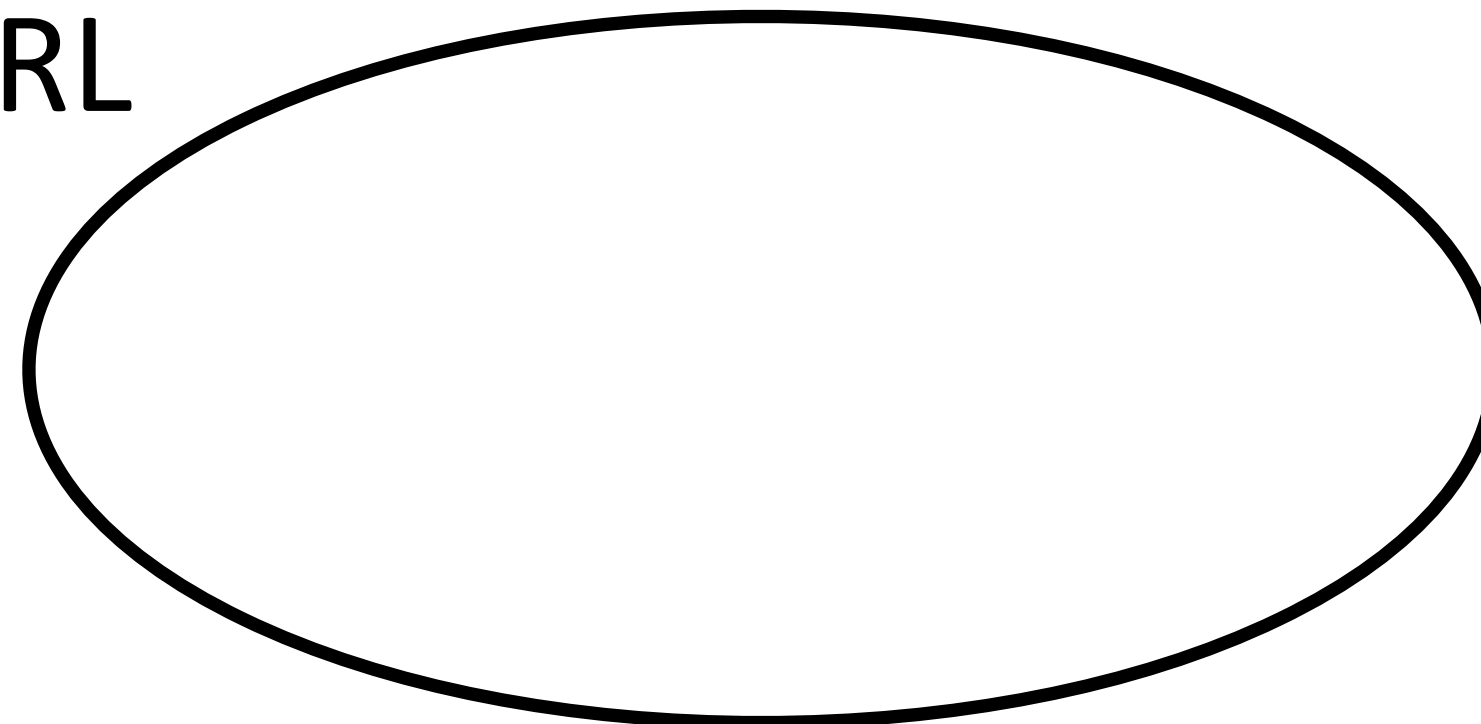
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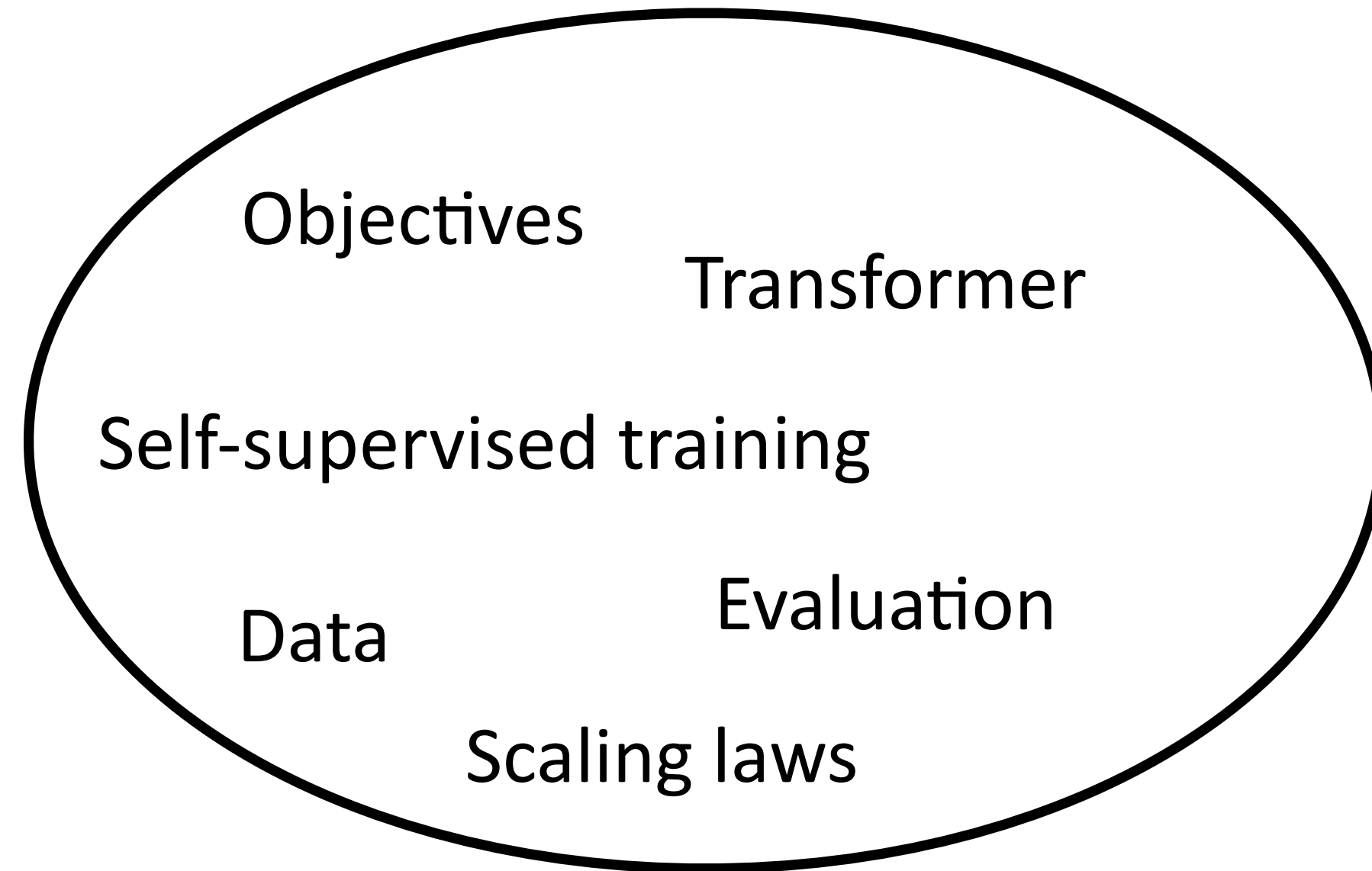


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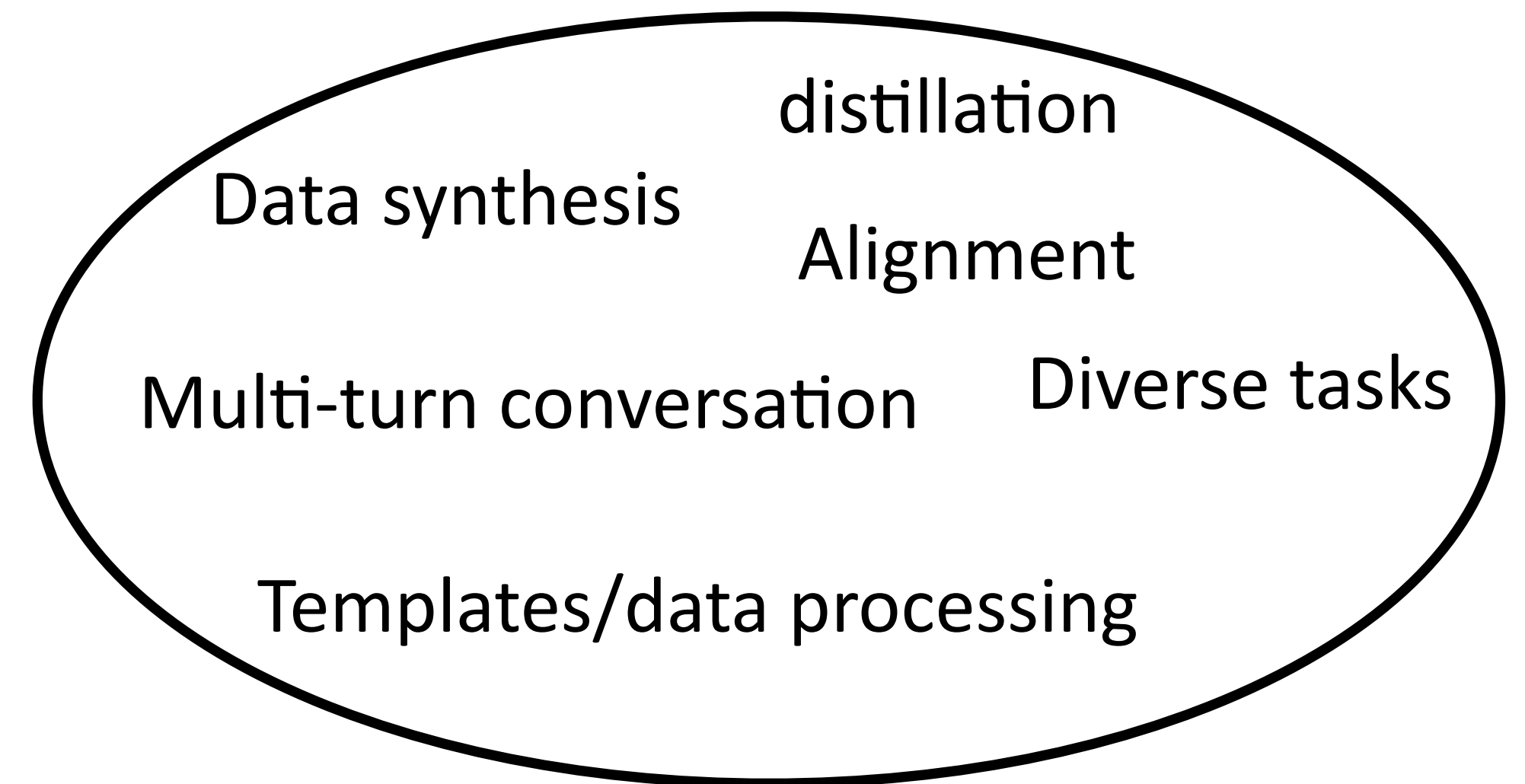


Review — Method

Pretraining



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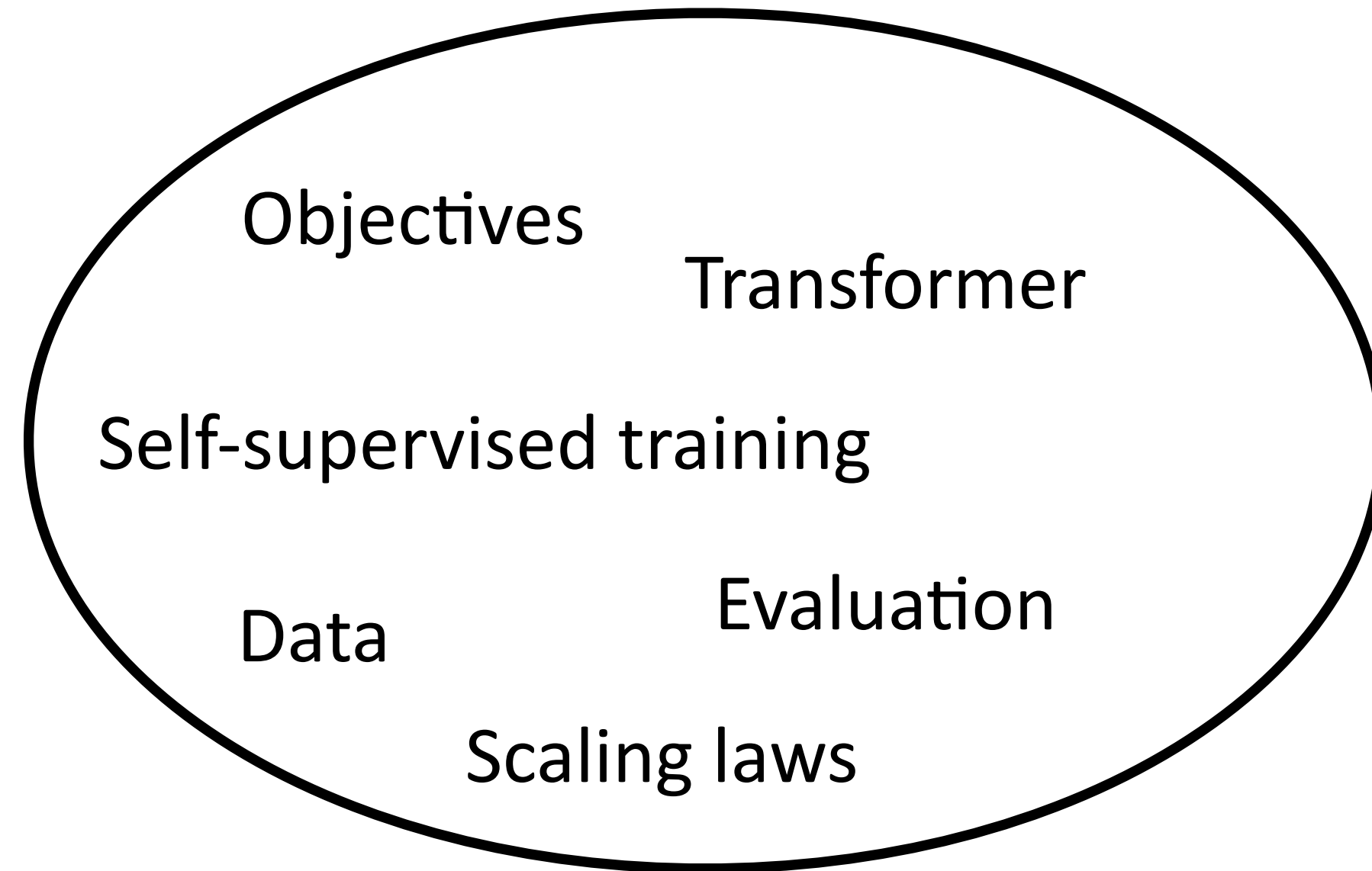


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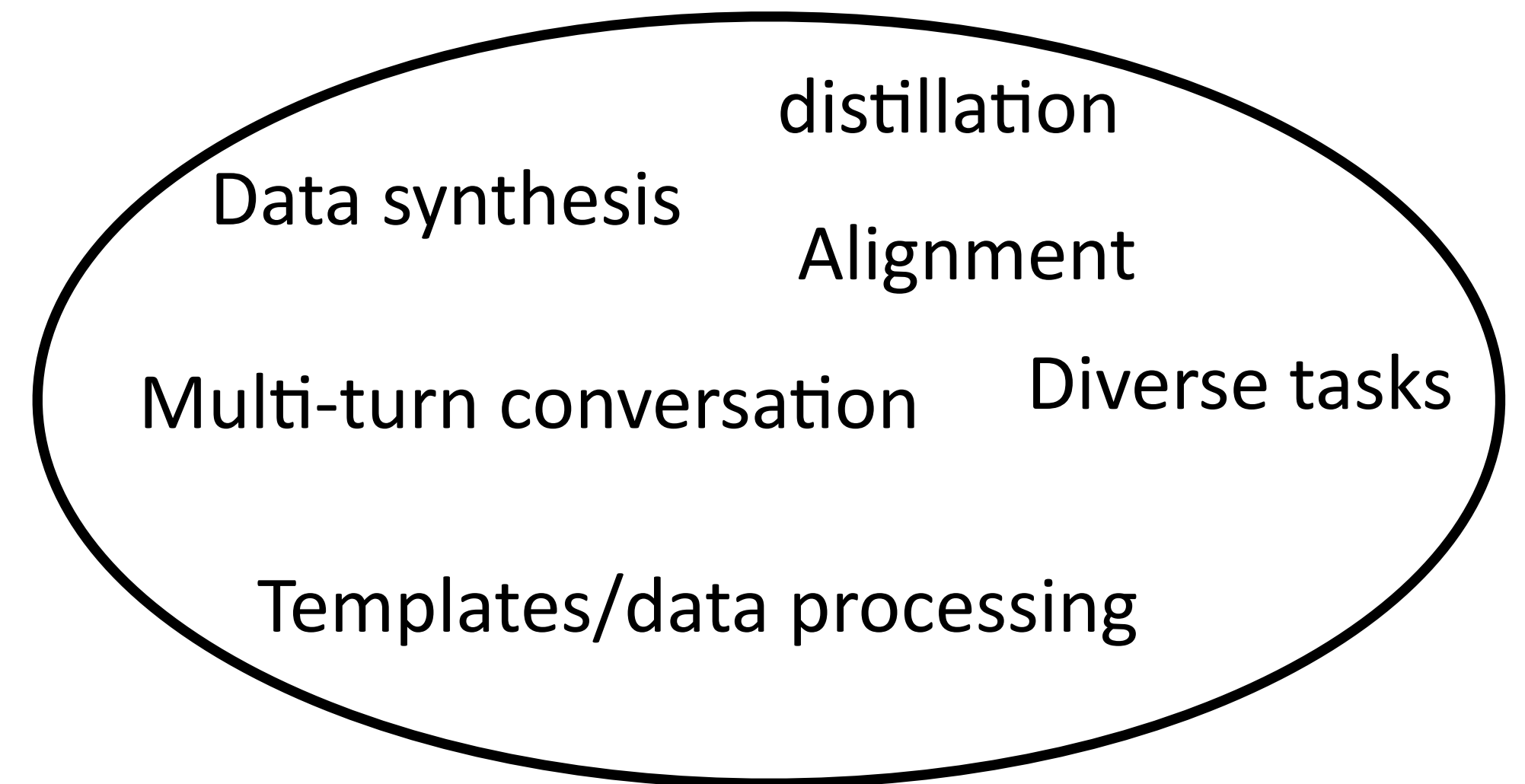


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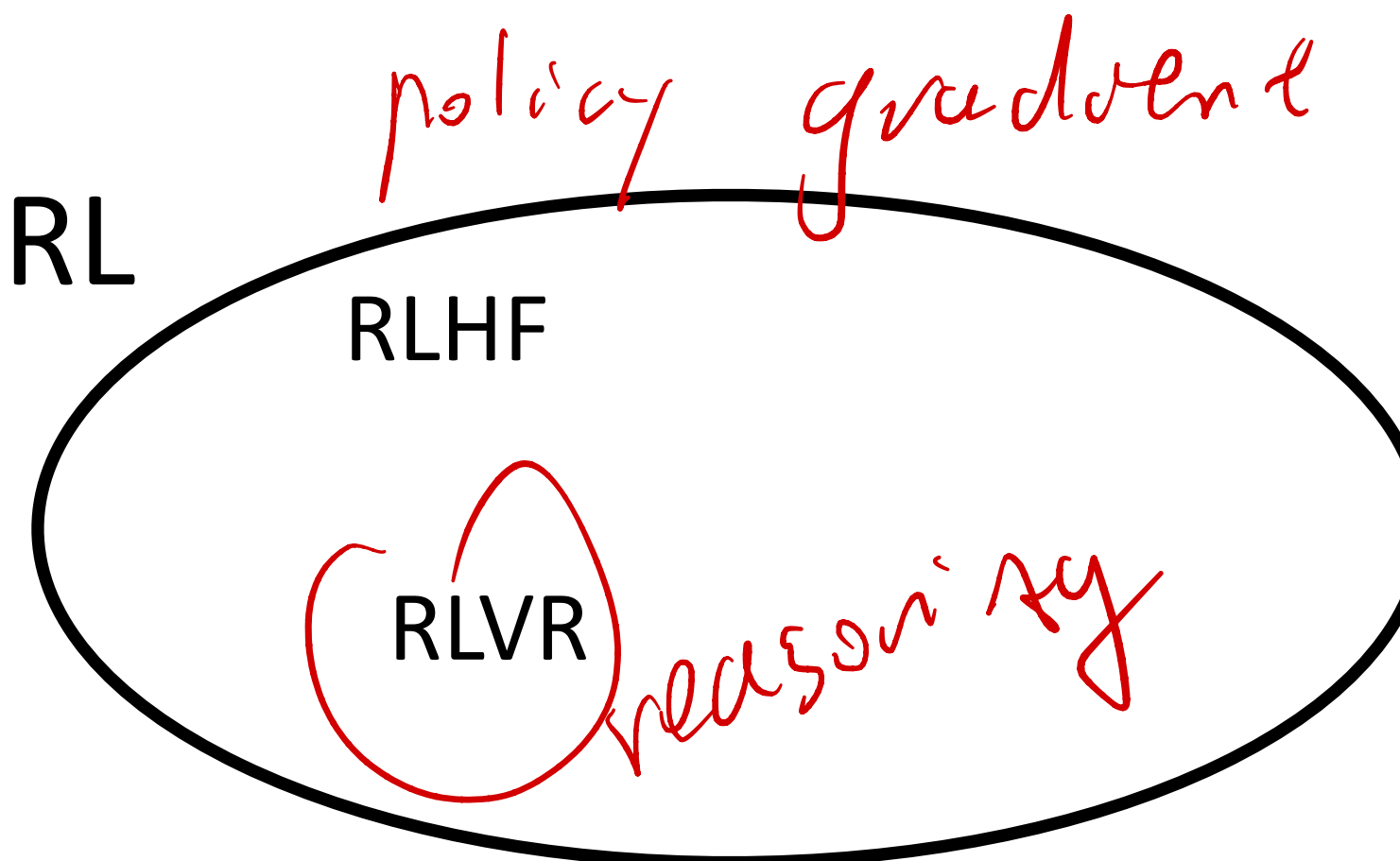
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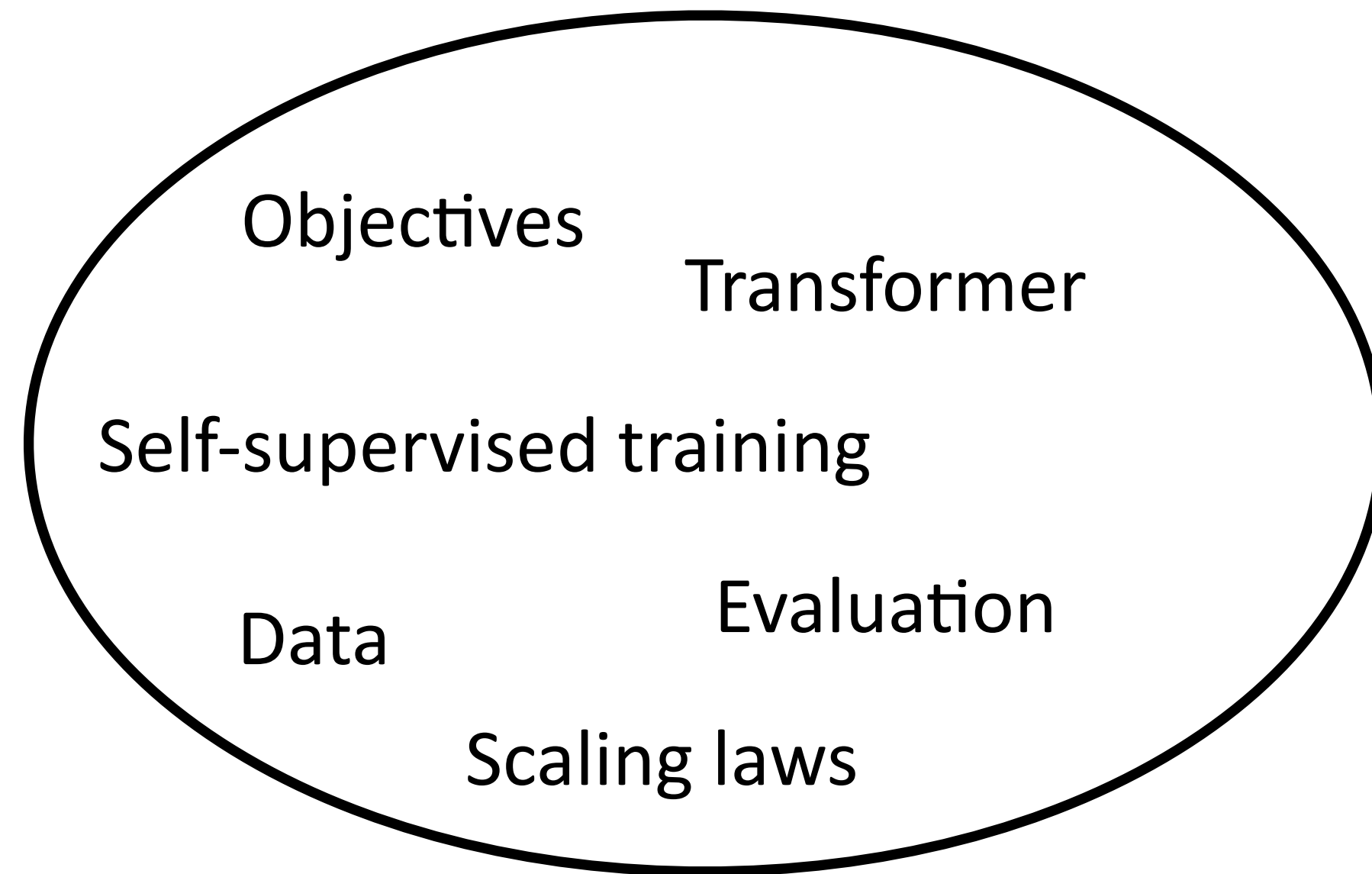


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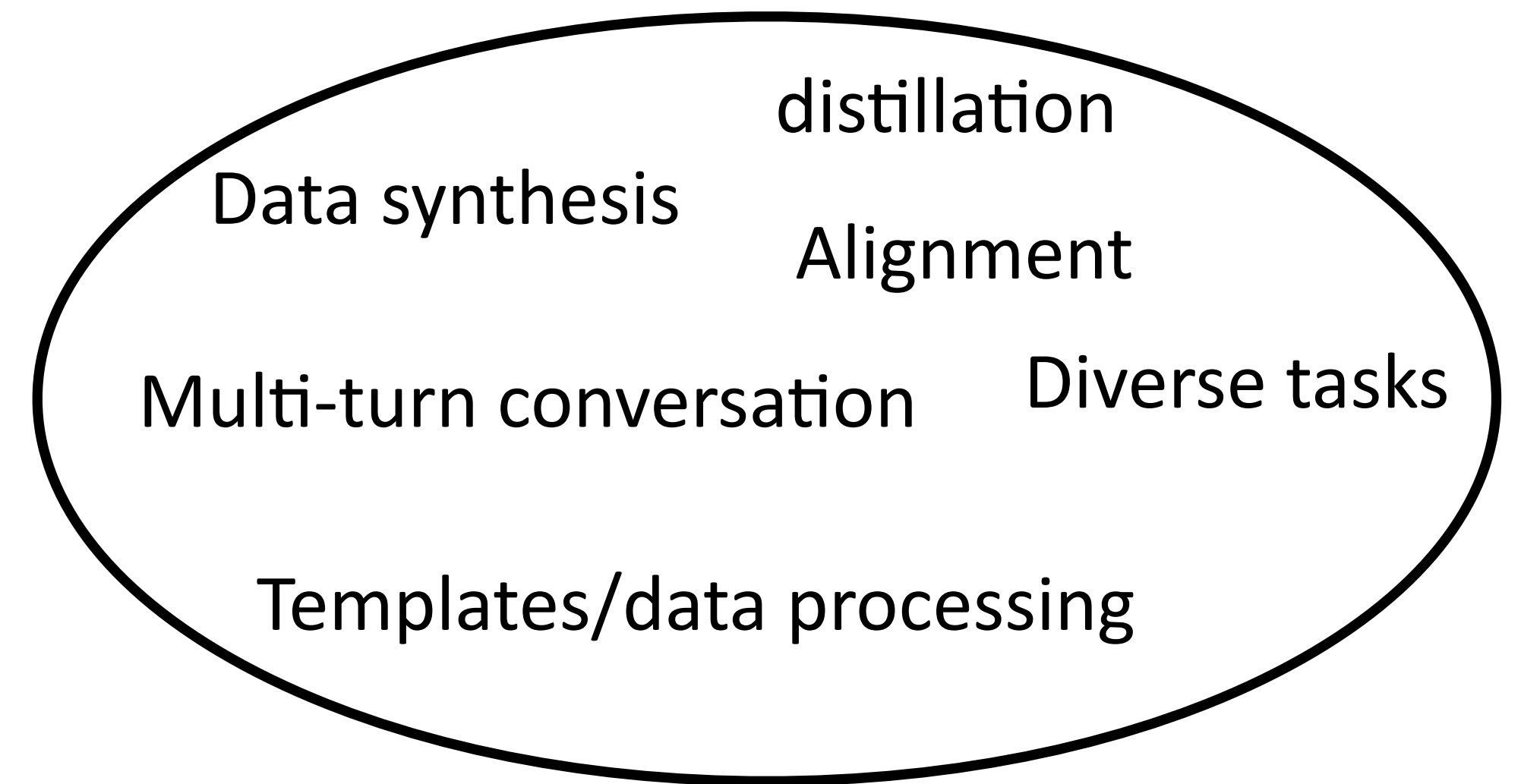


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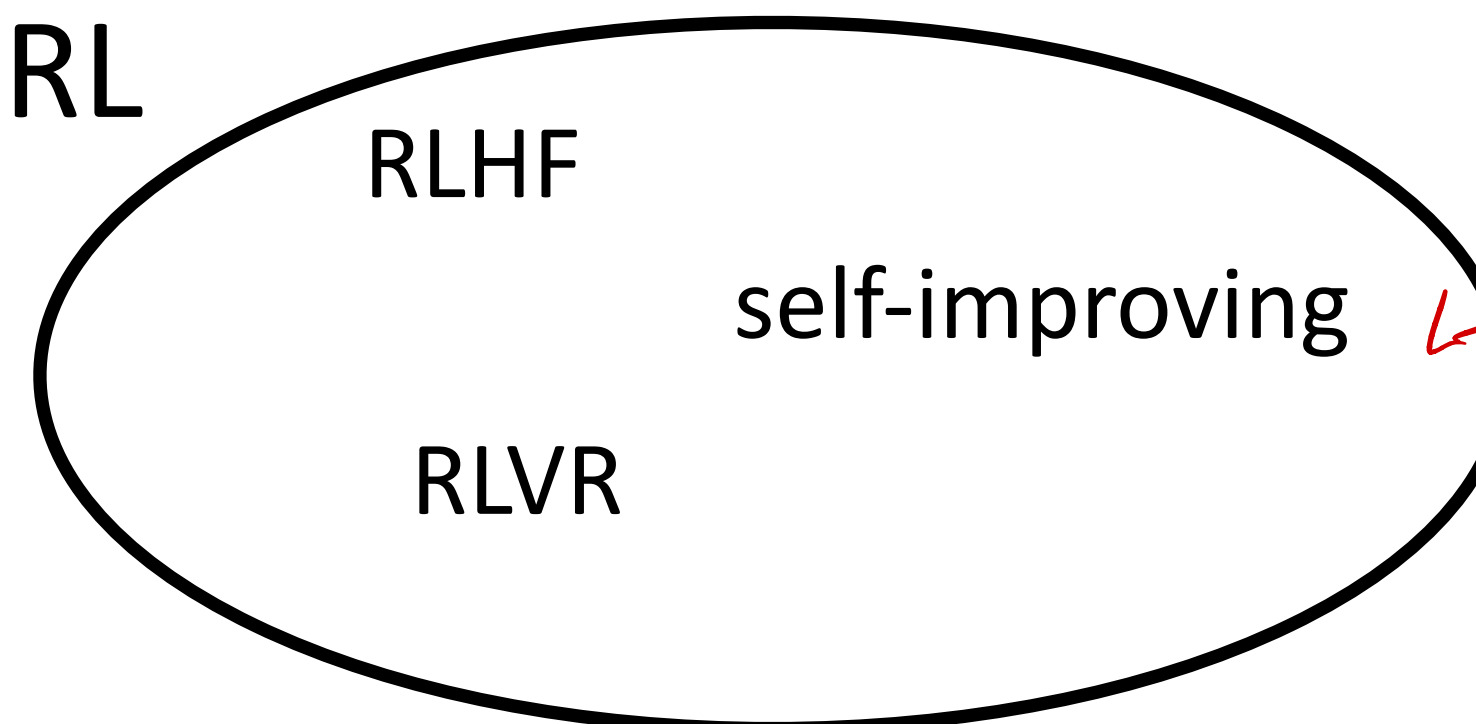
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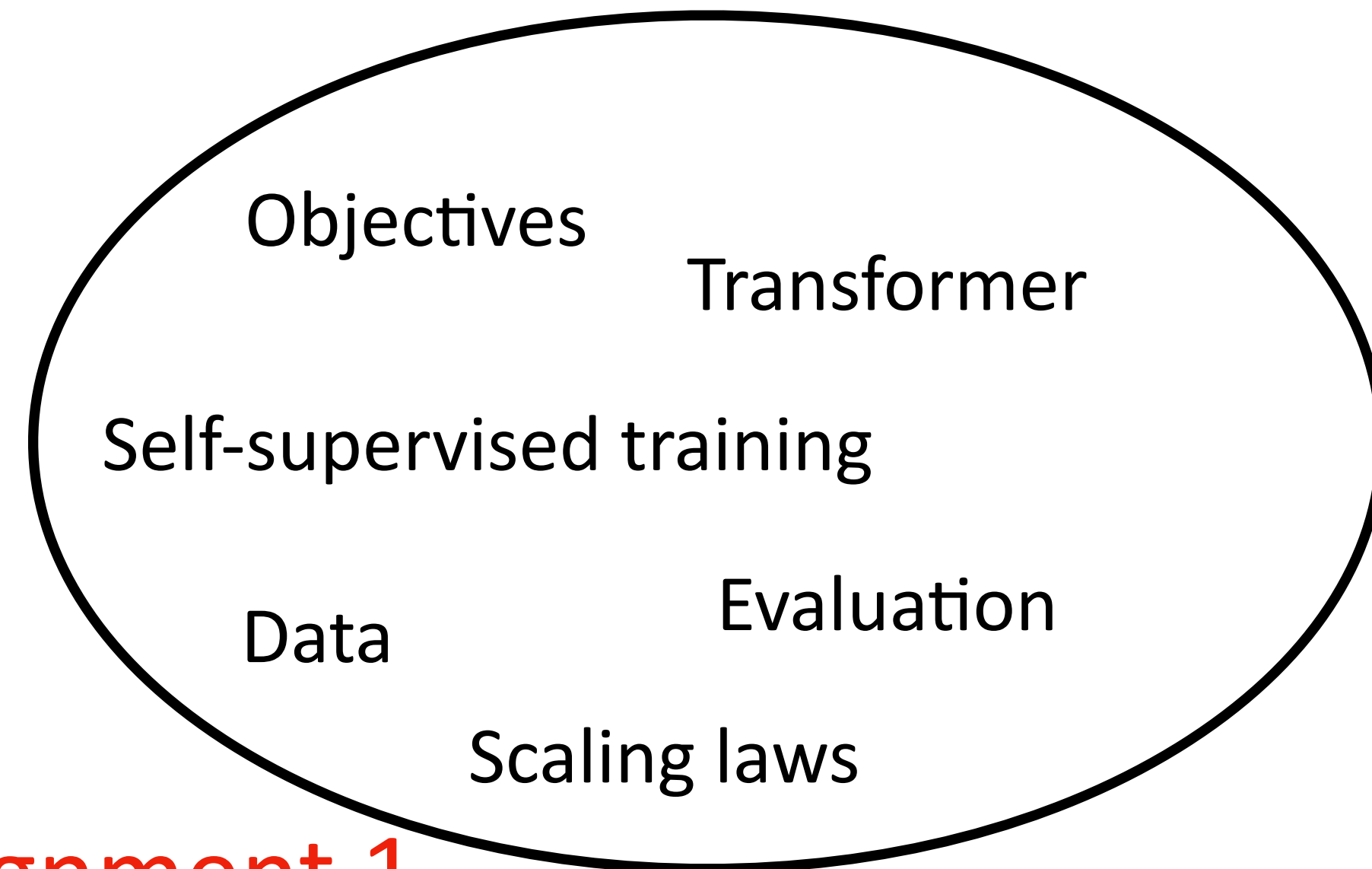
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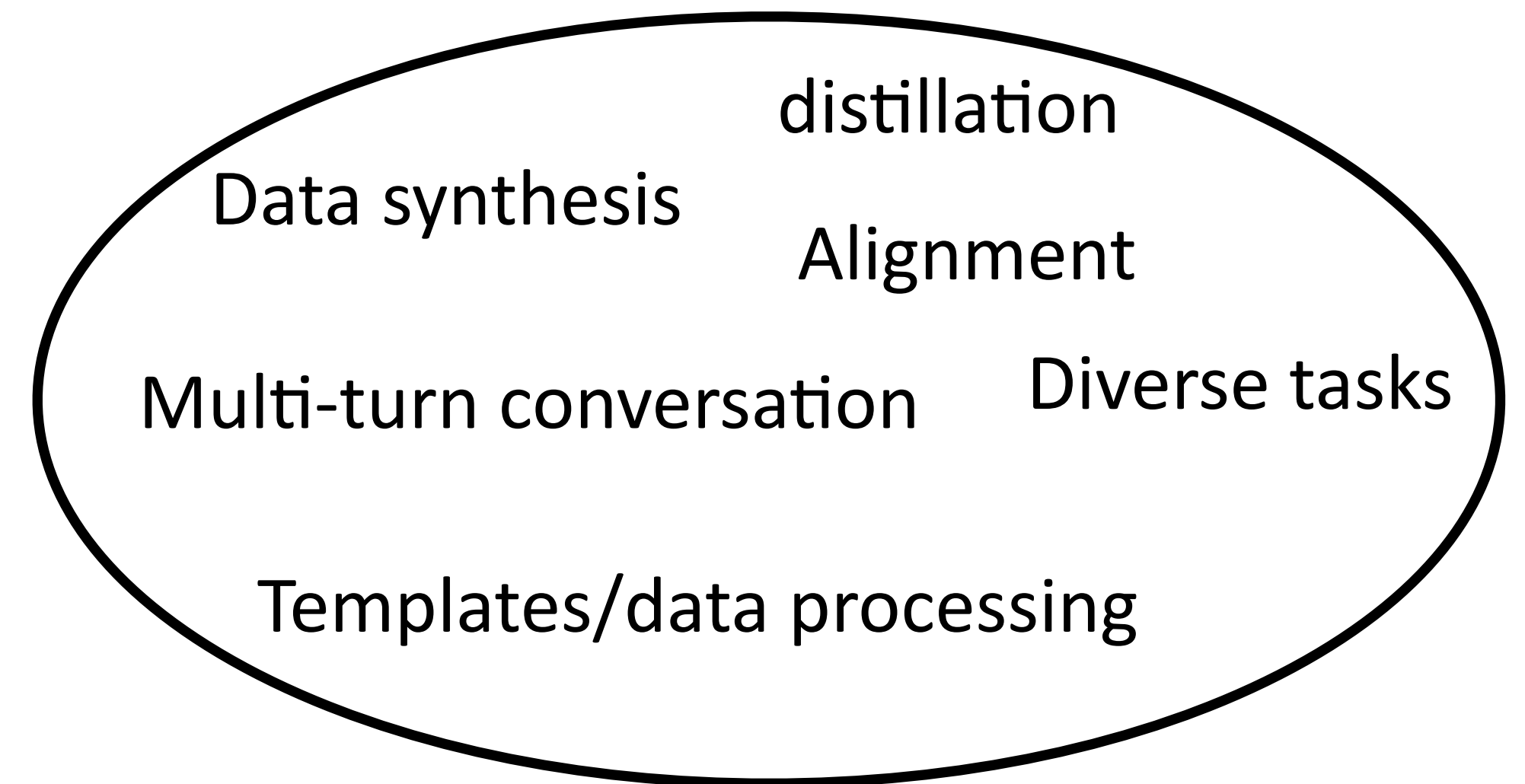
no imitation learning

Review — Method

Pretraining

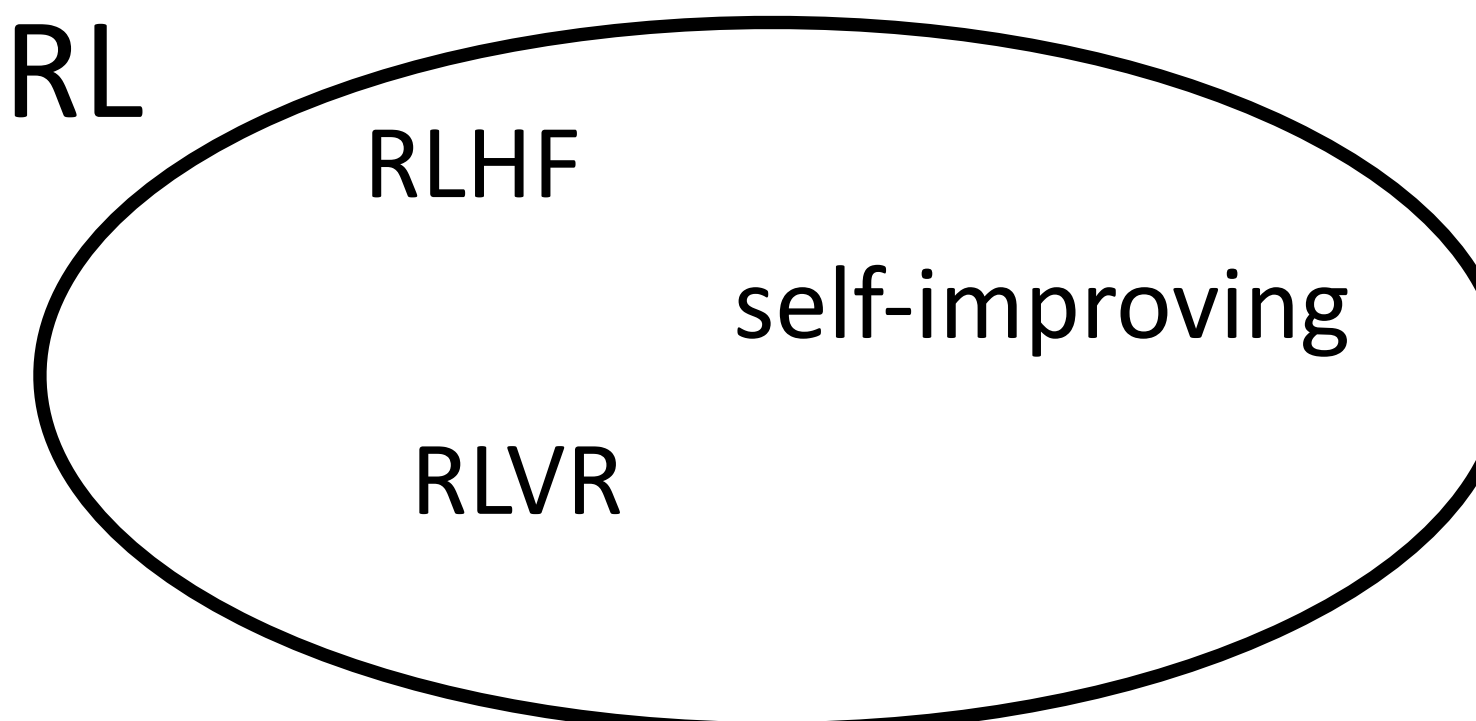


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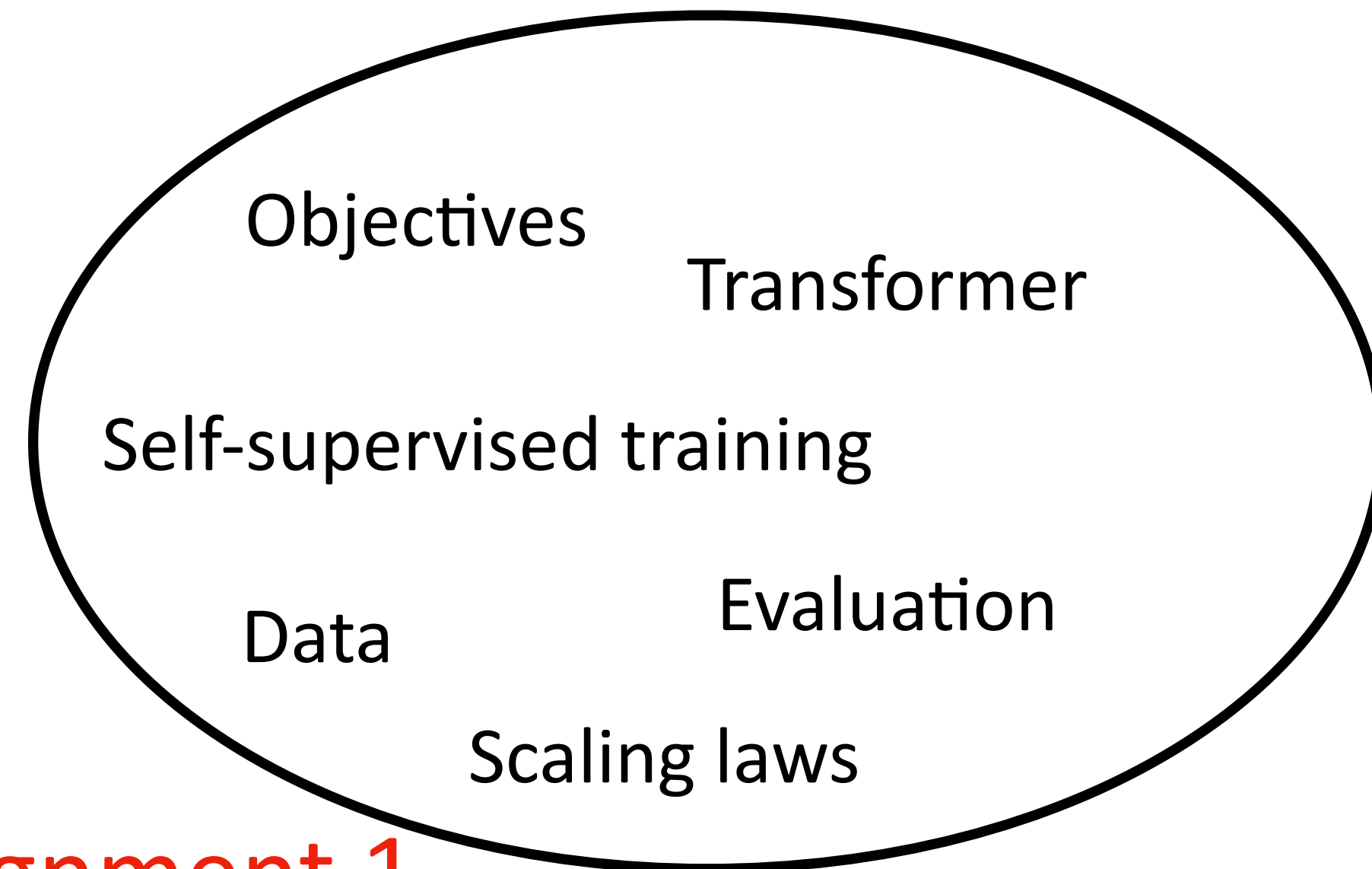
Assignment 1

RL



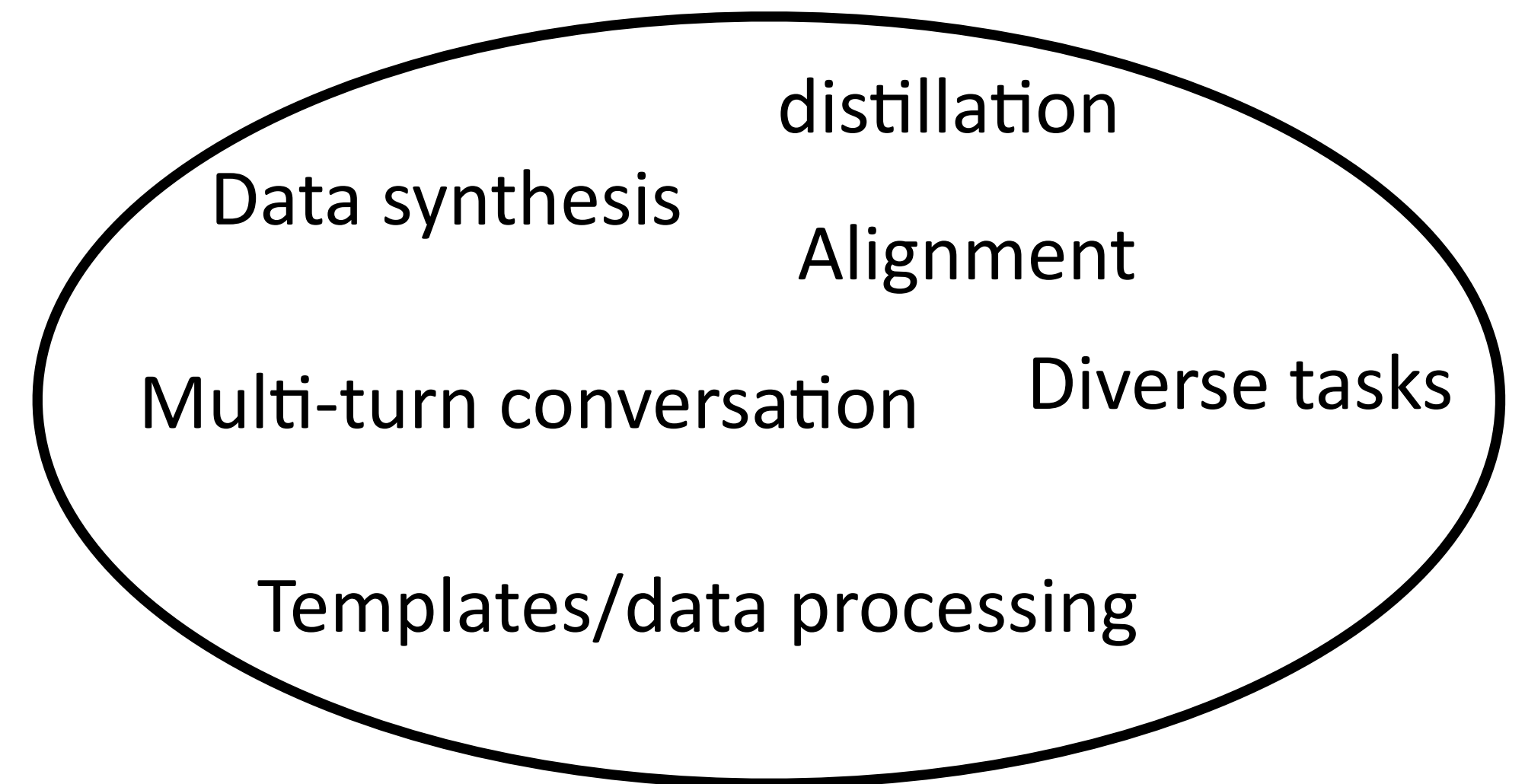
Review — Method

Pretraining



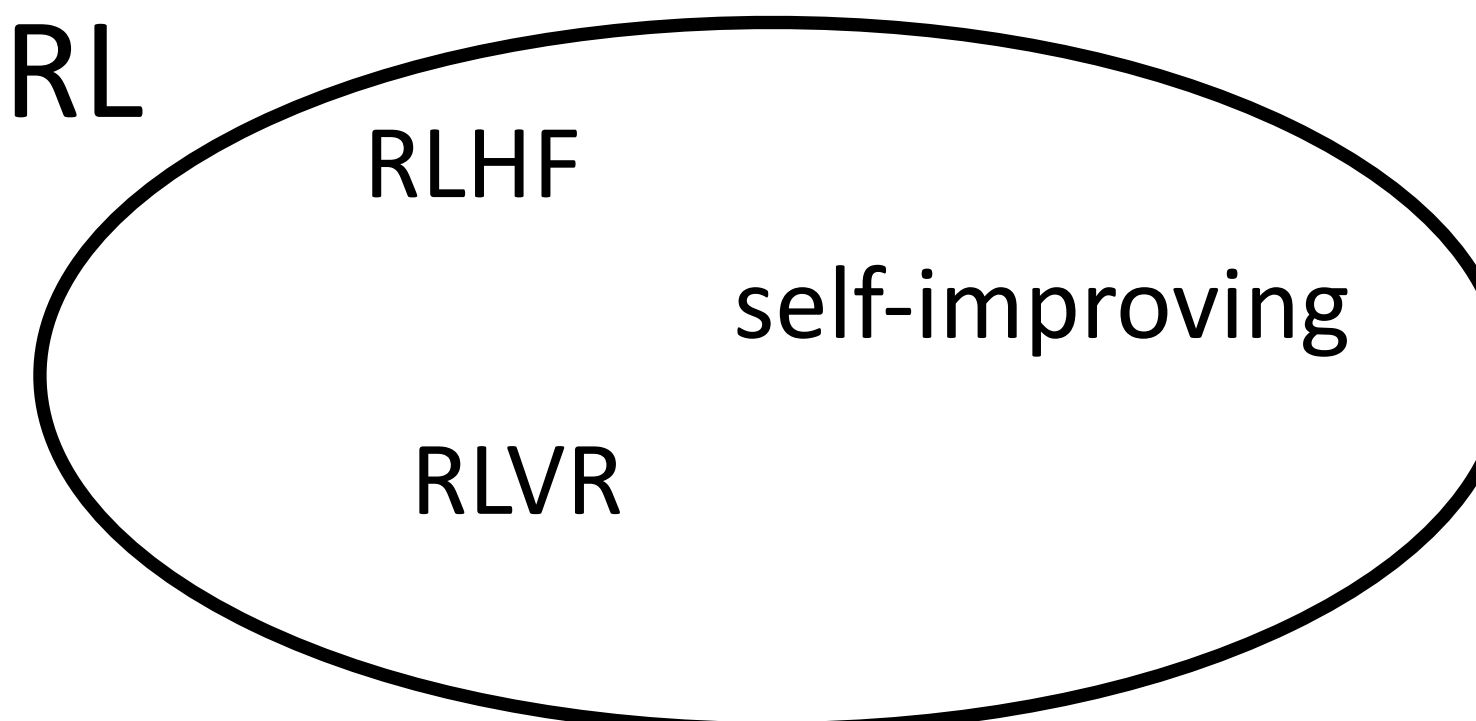
Assignment 1

SFT



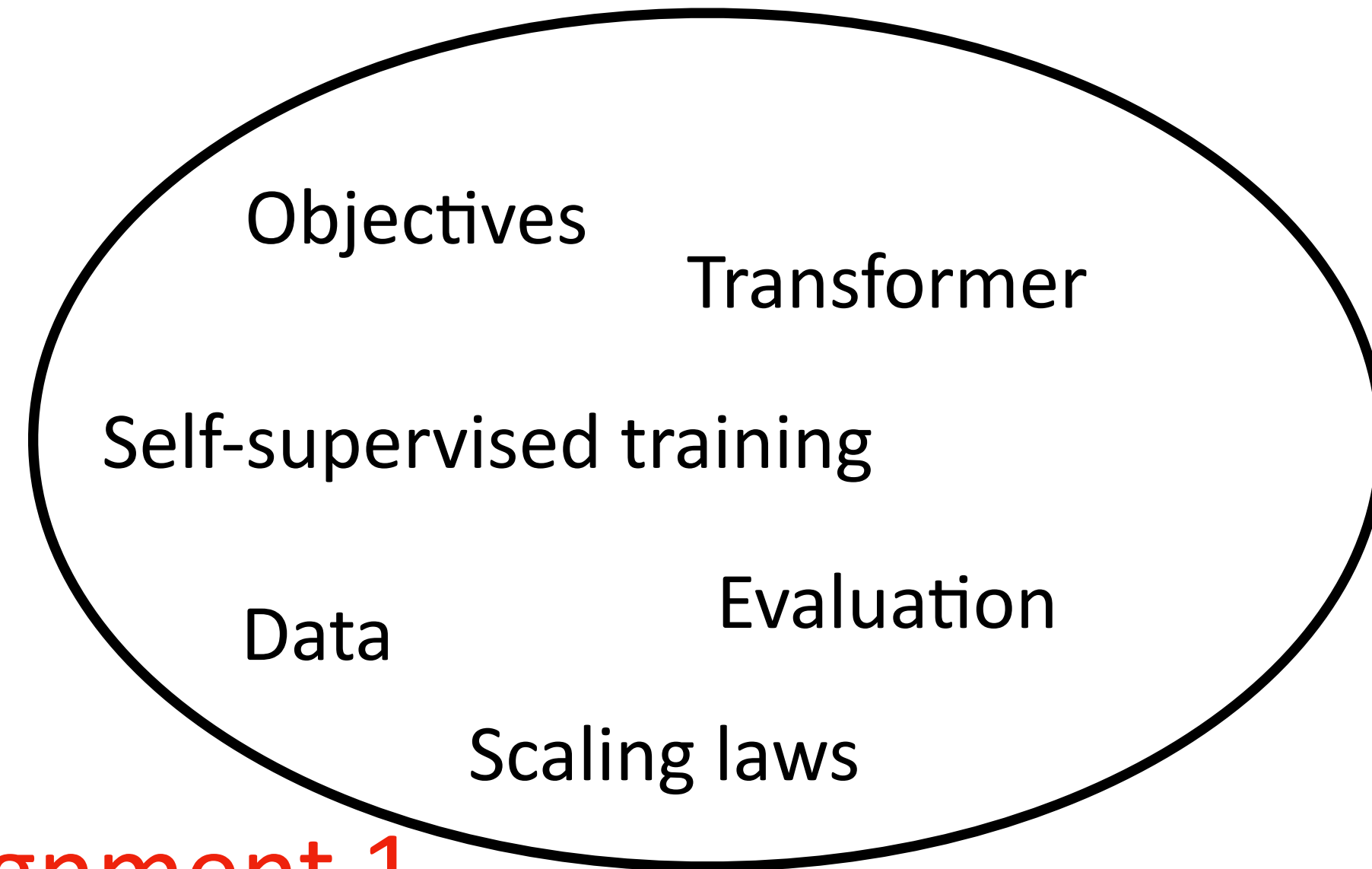
Assignment 2

RL



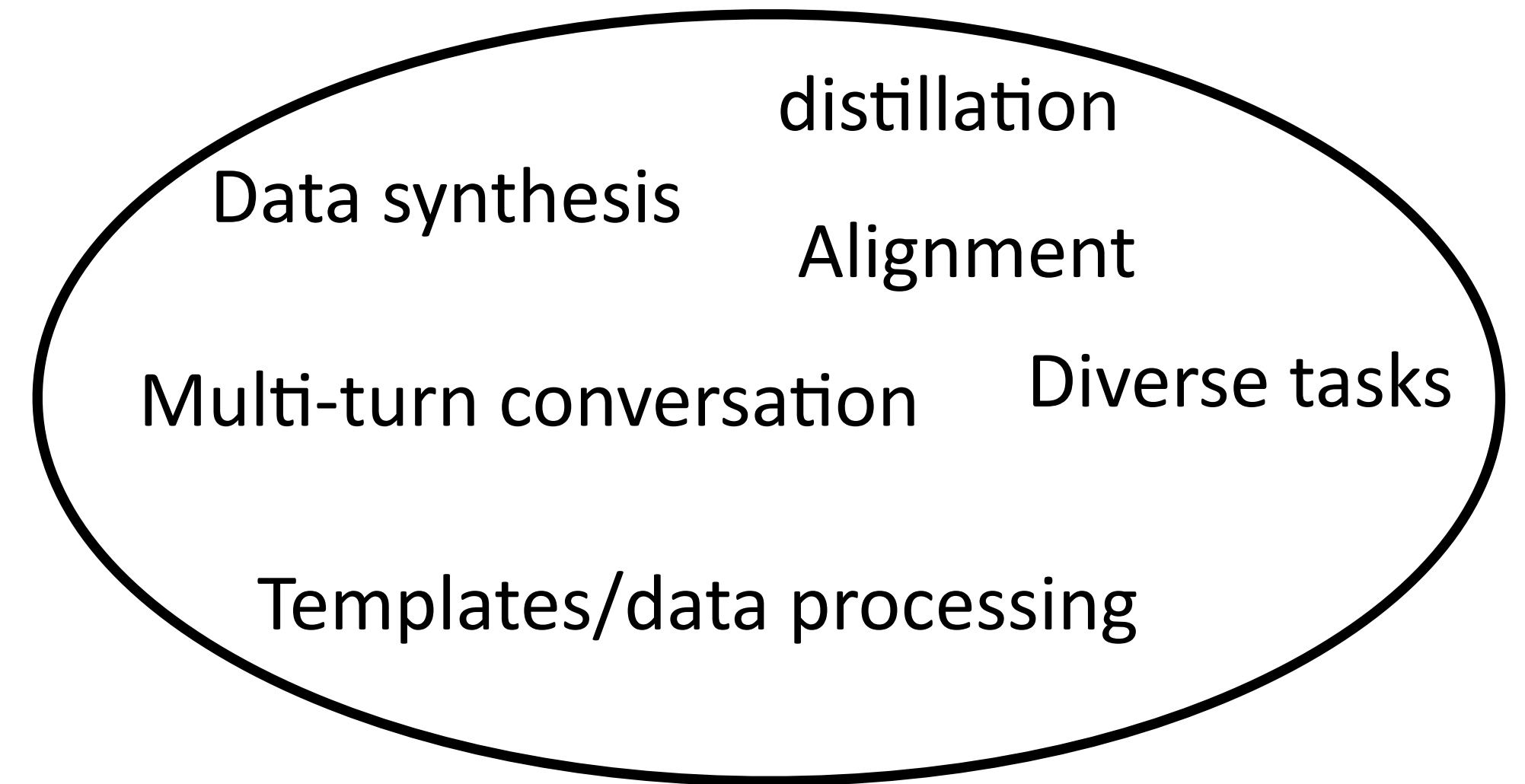
Review — Method

Pretraining



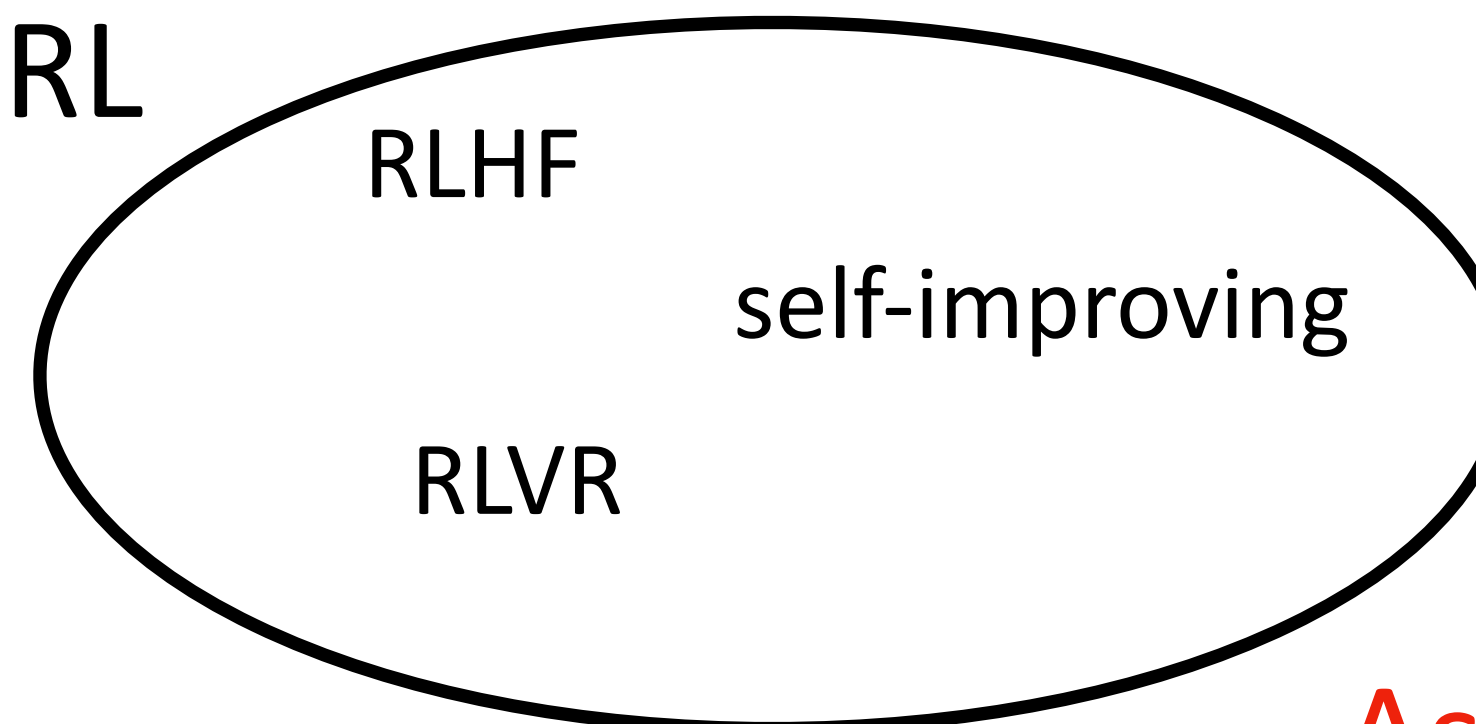
Assignment 1

SFT



Assignment 2

RL

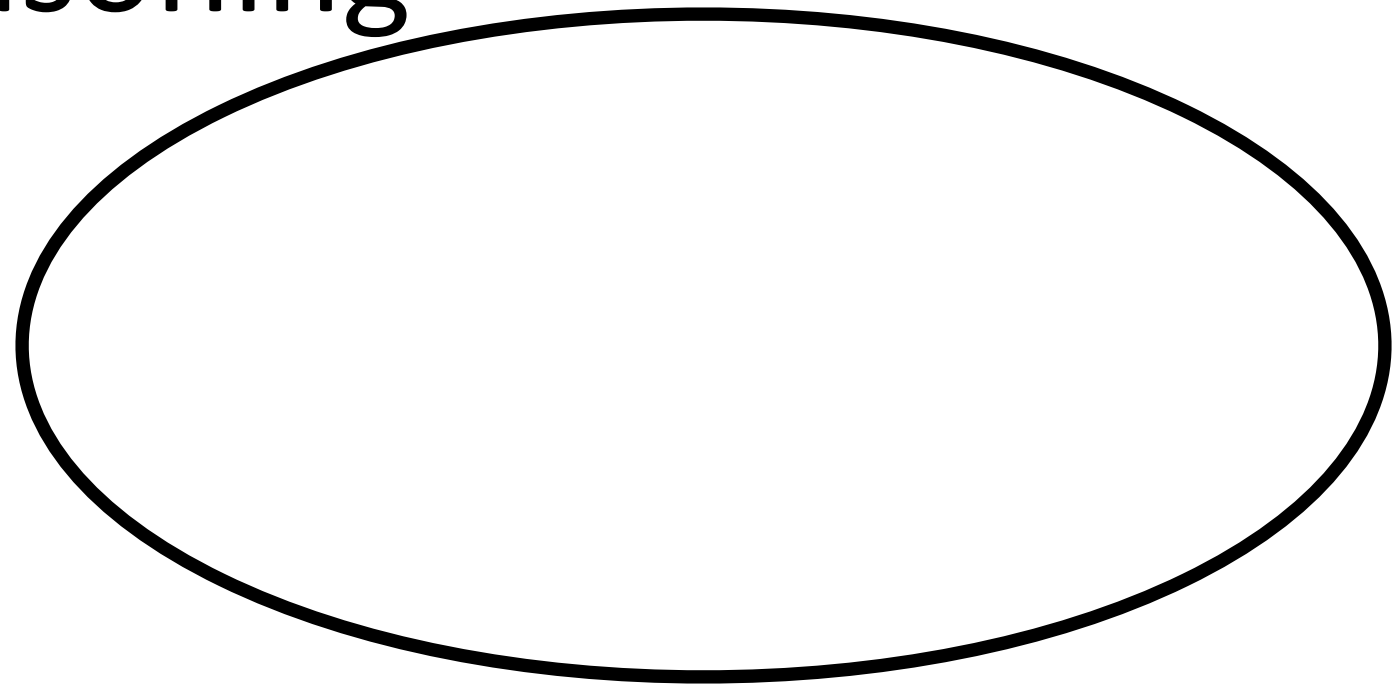


Assignment 3

Review — Applications

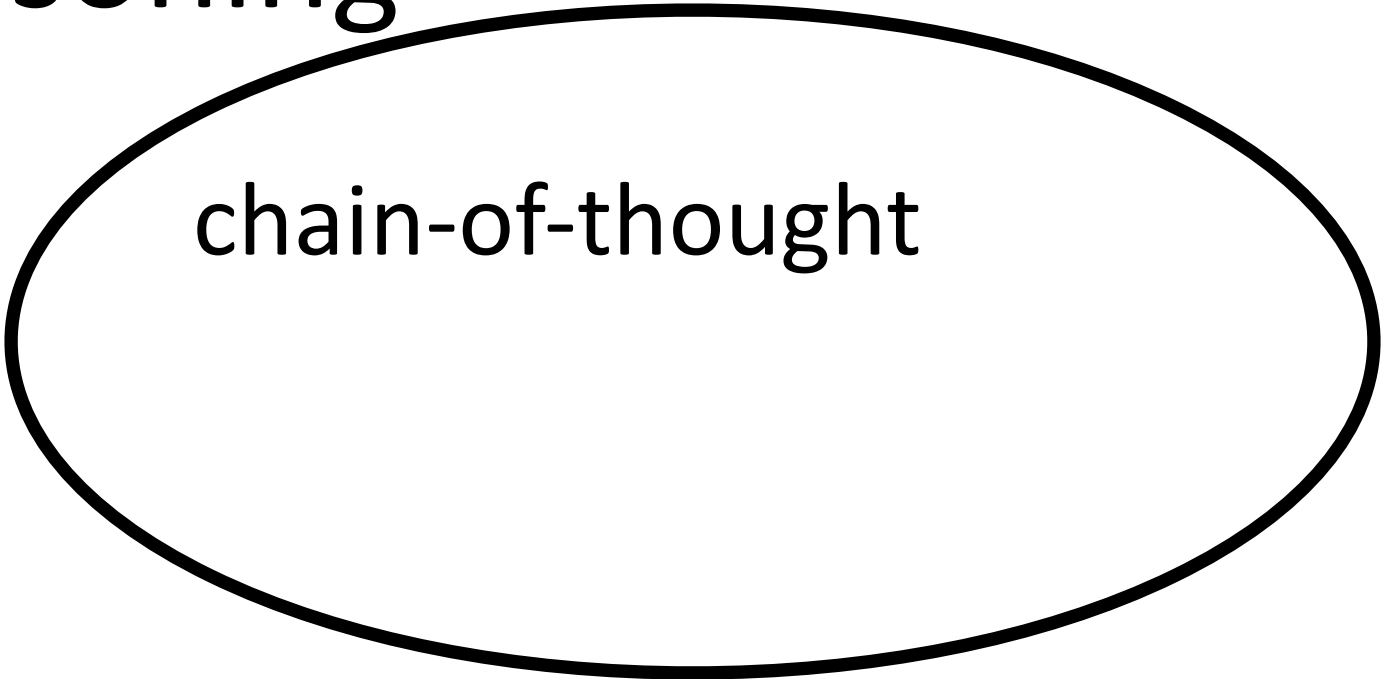
Review — Applications

Reasoning



Review — Applications

Reasoning



chain-of-thought

Review — Applications

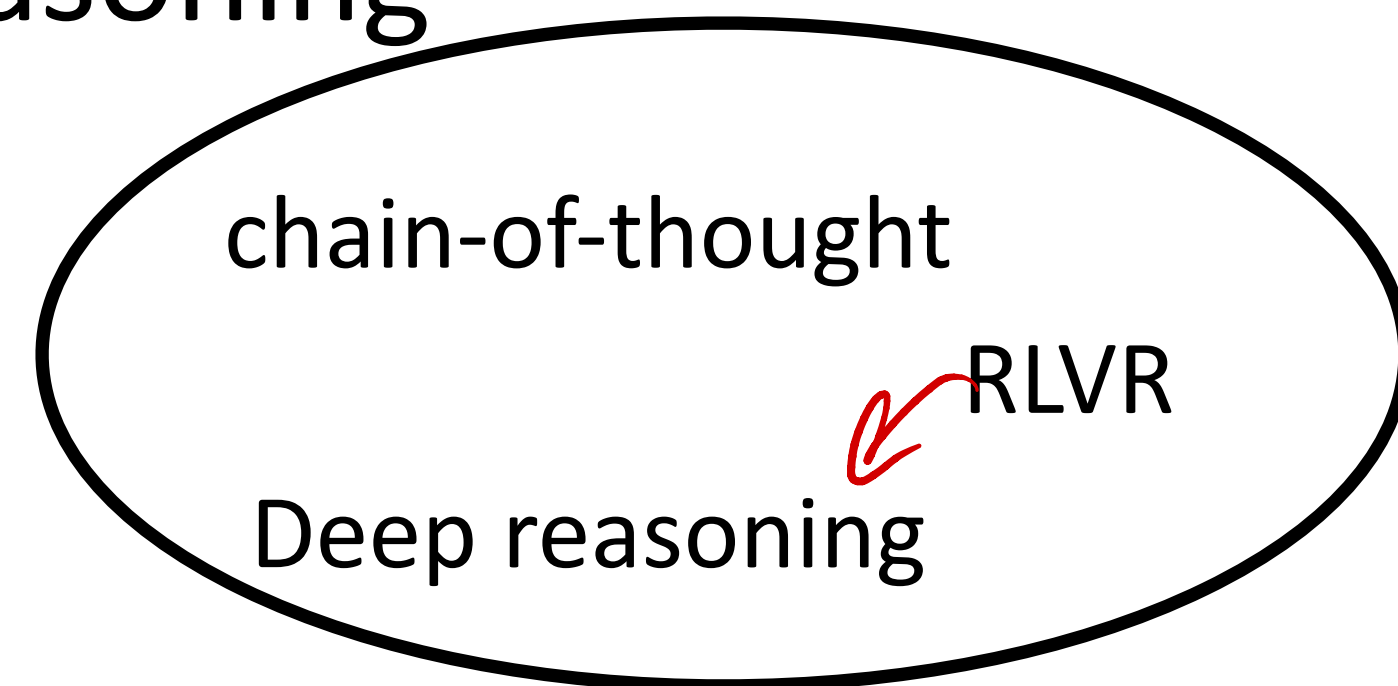
Reasoning

chain-of-thought

Deep reasoning

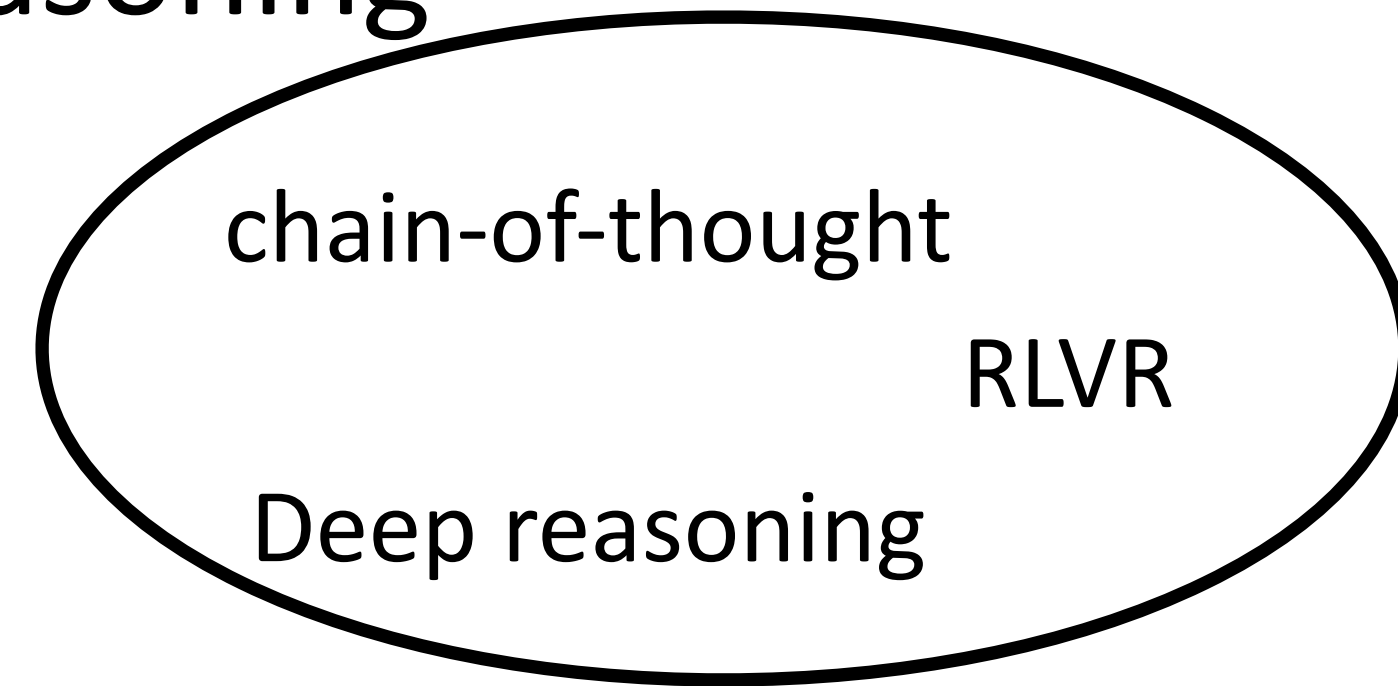
Review — Applications

Reasoning



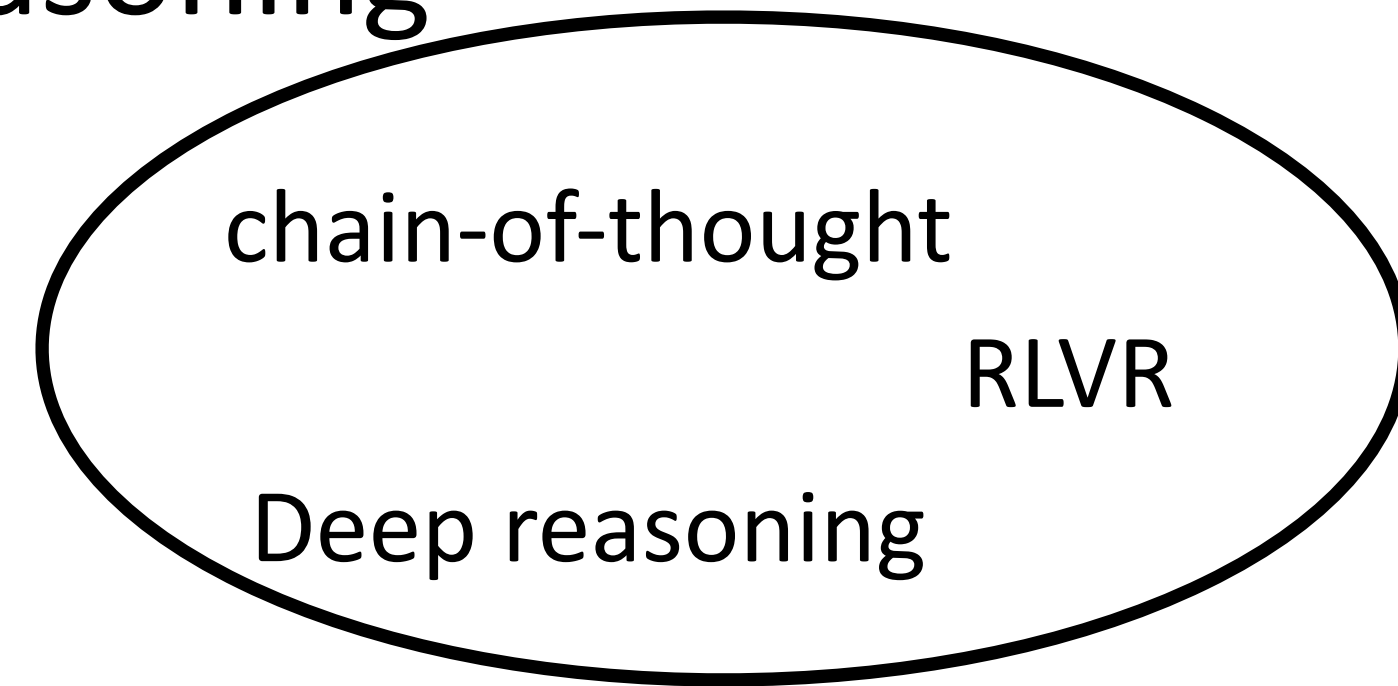
Review — Applications

Reasoning

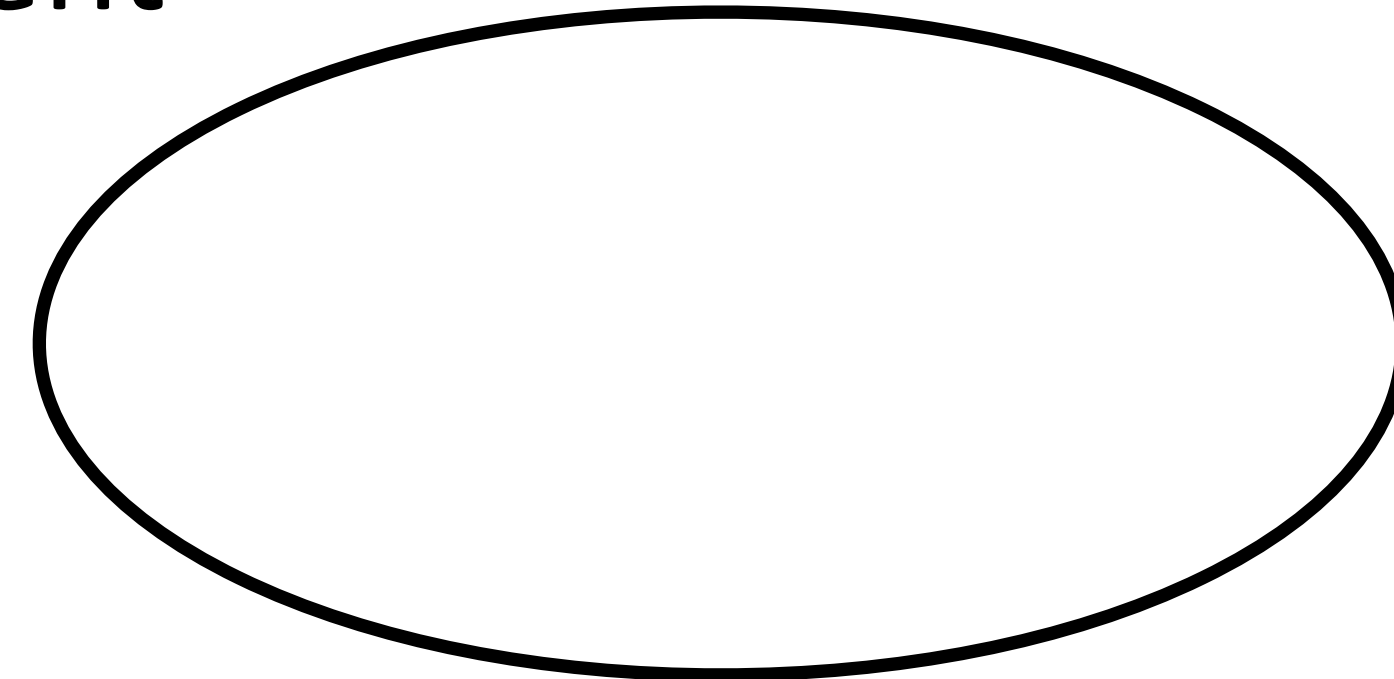


Review — Applications

Reasoning

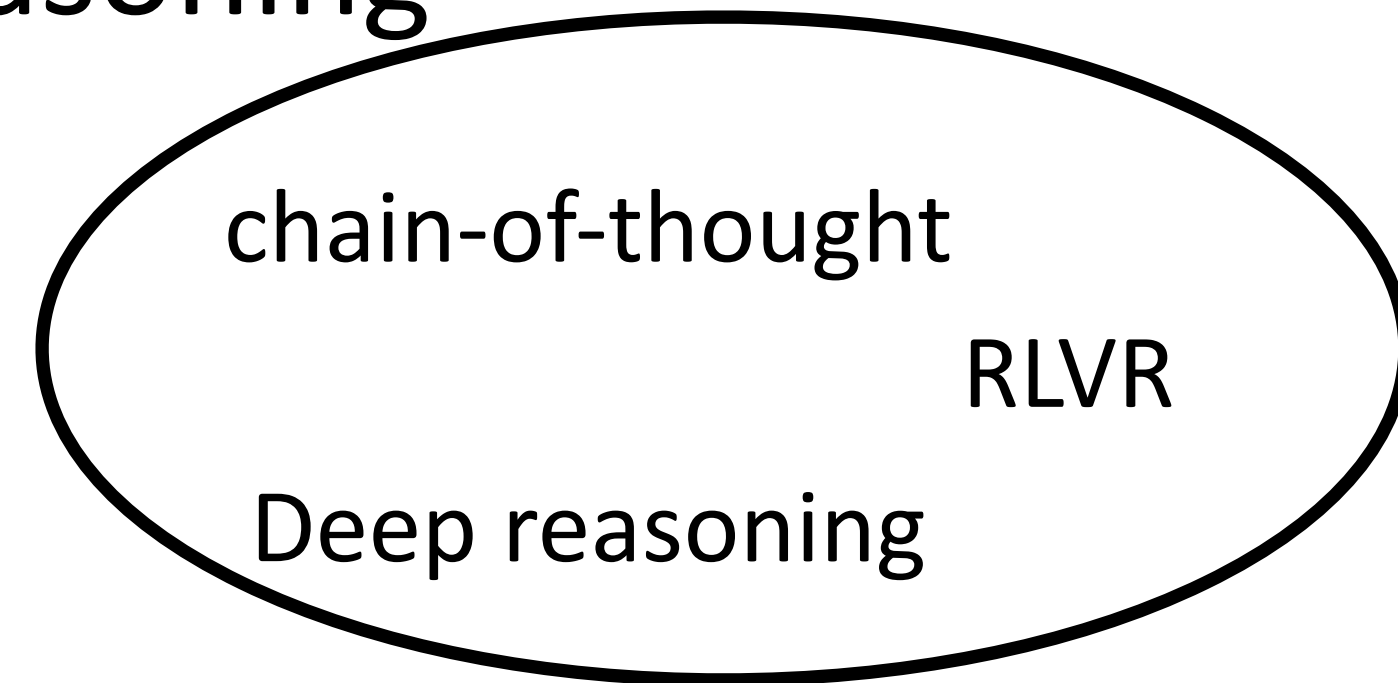


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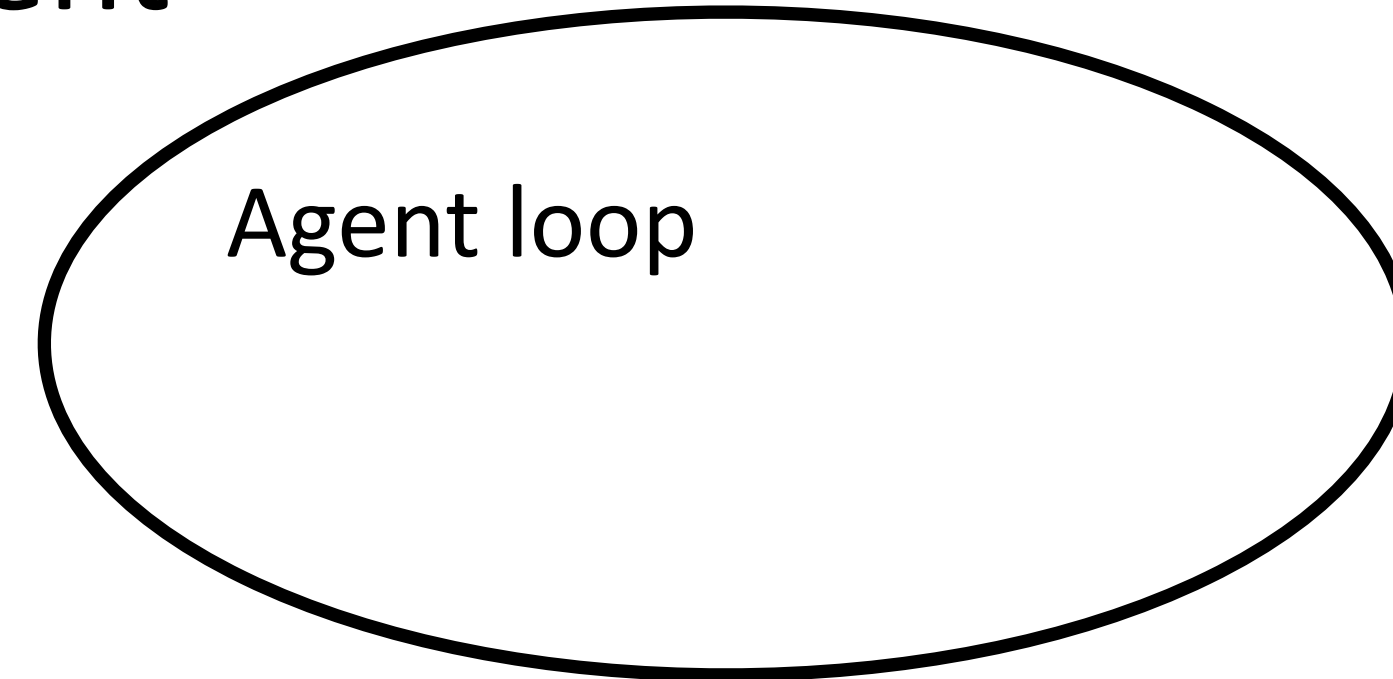


Review — Applications

Reasoning

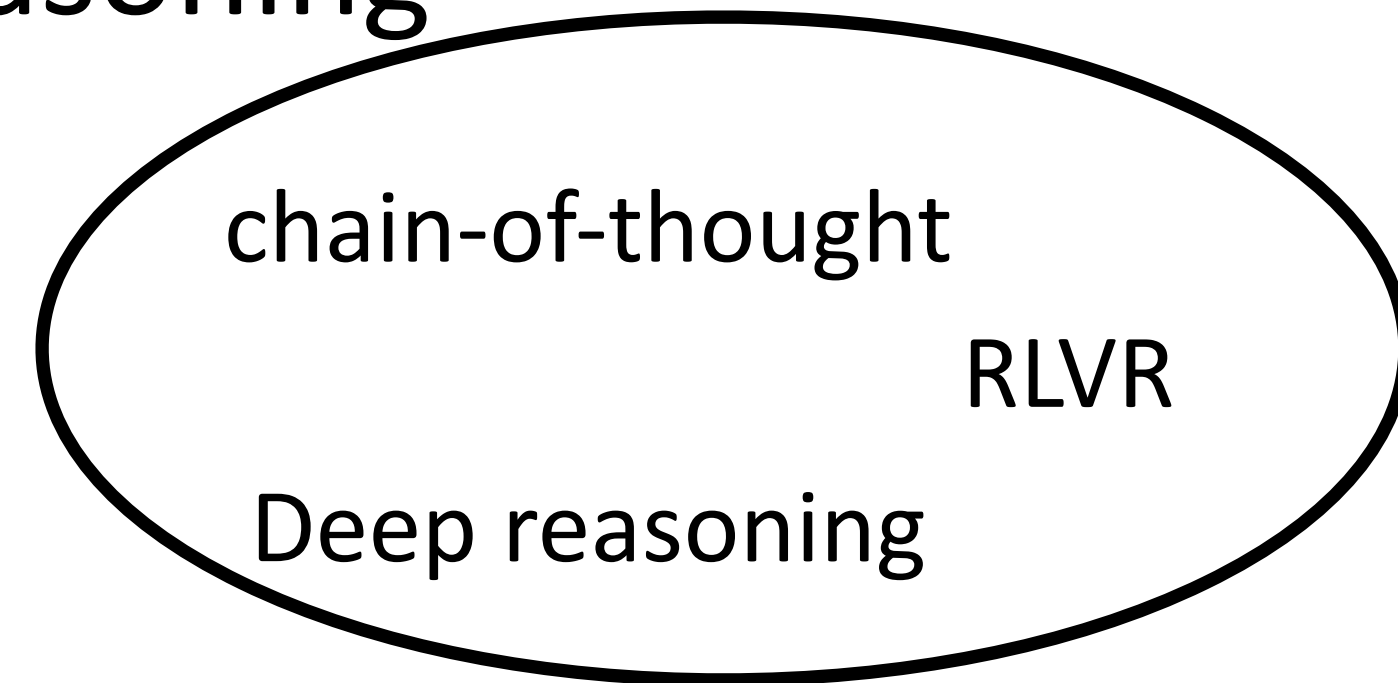


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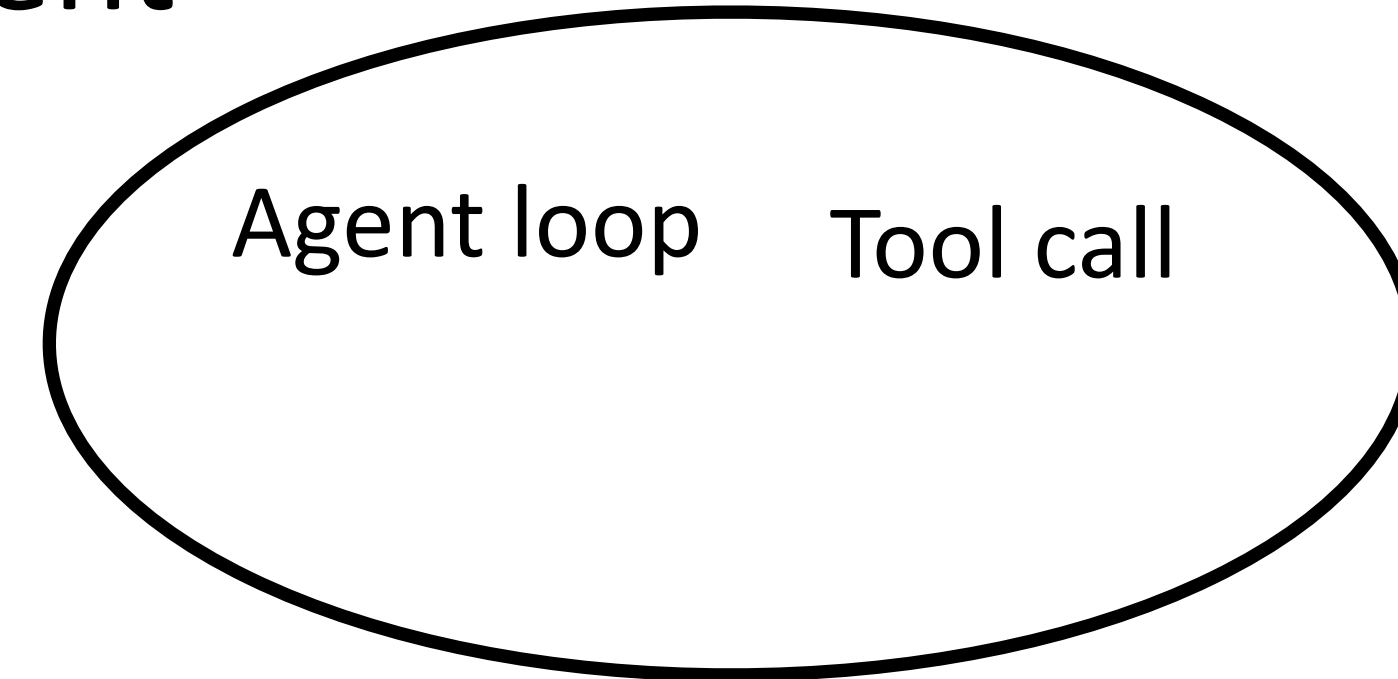


Review — Applications

Reasoning

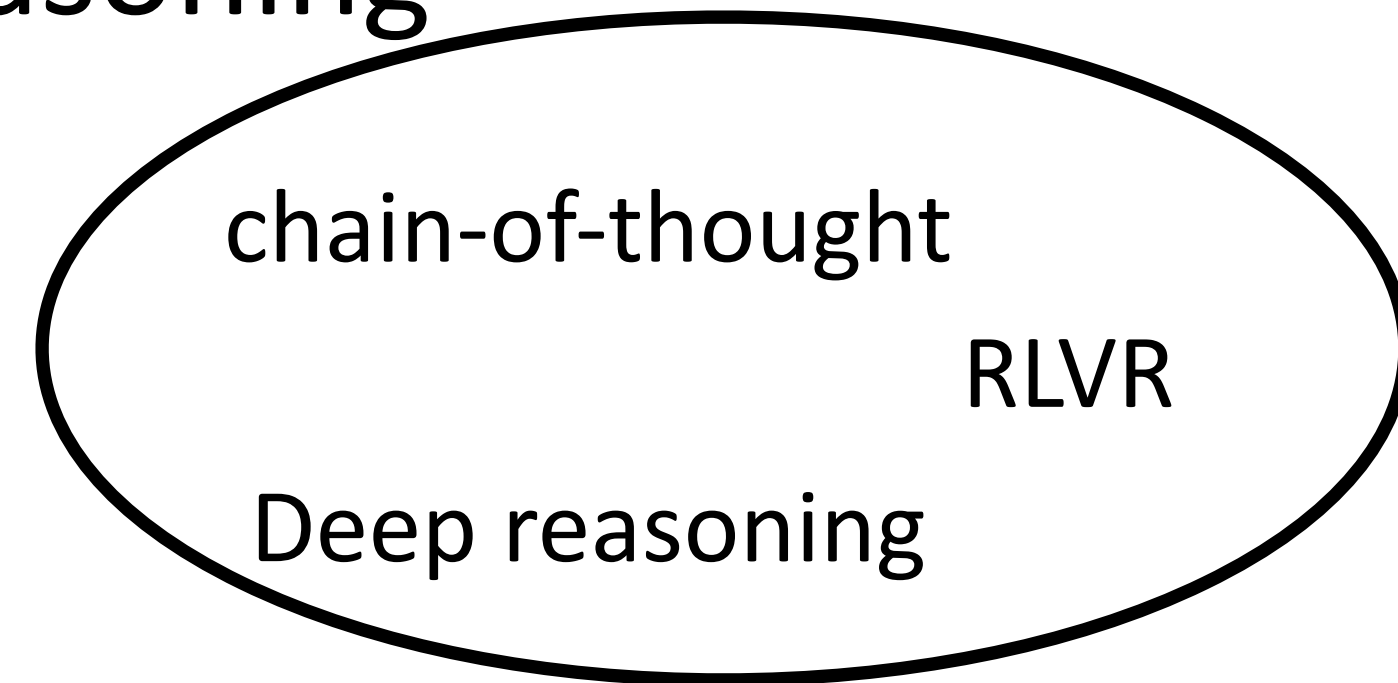


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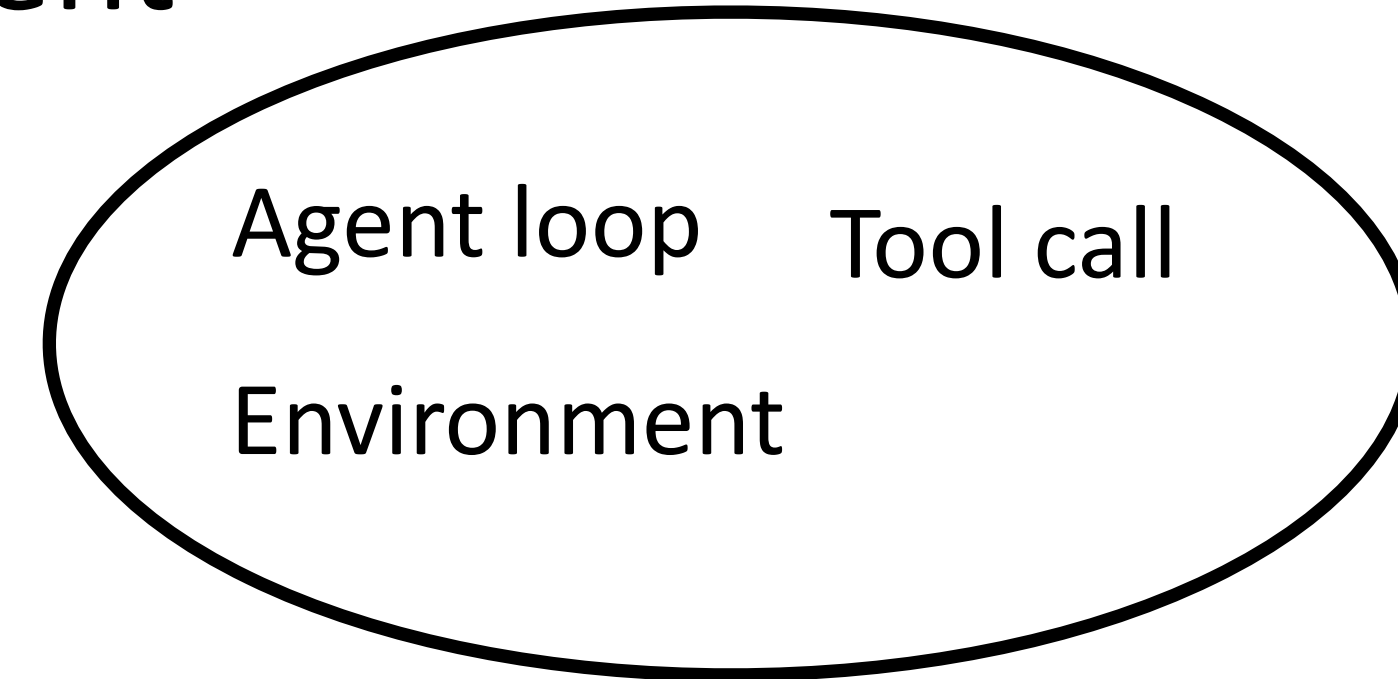


Review — Applications

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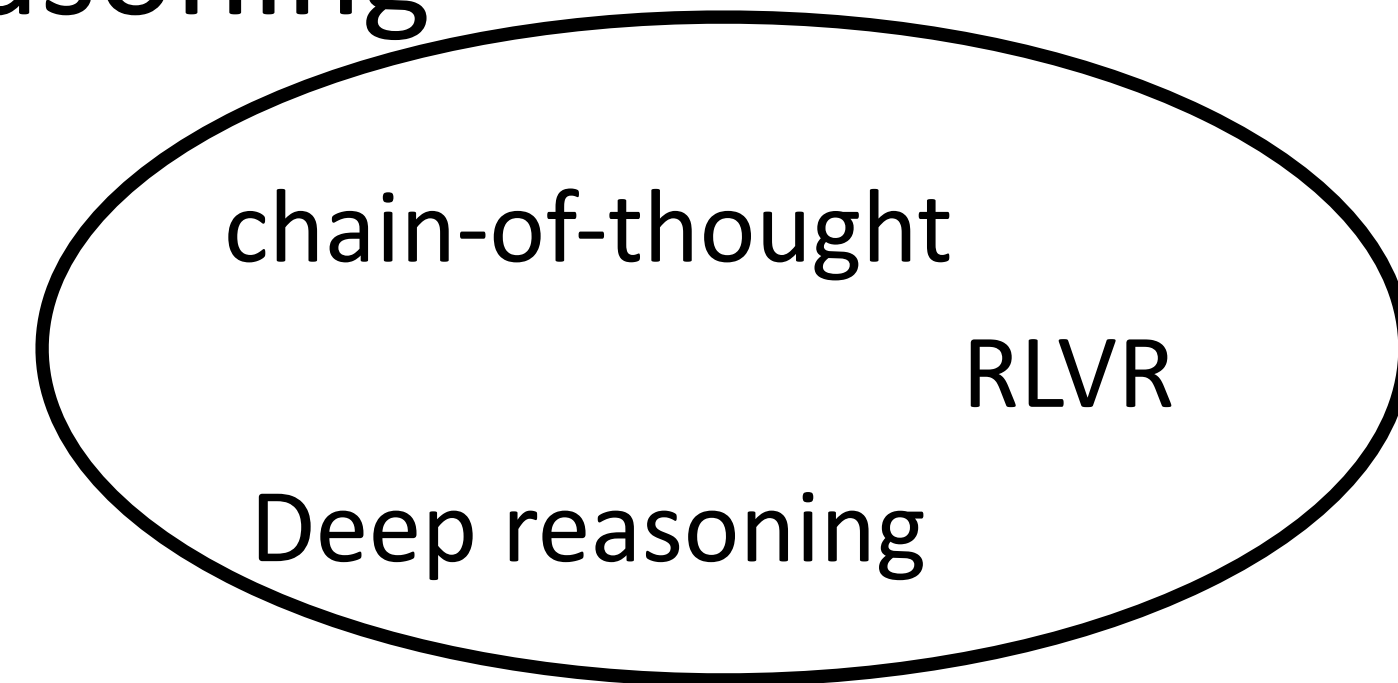


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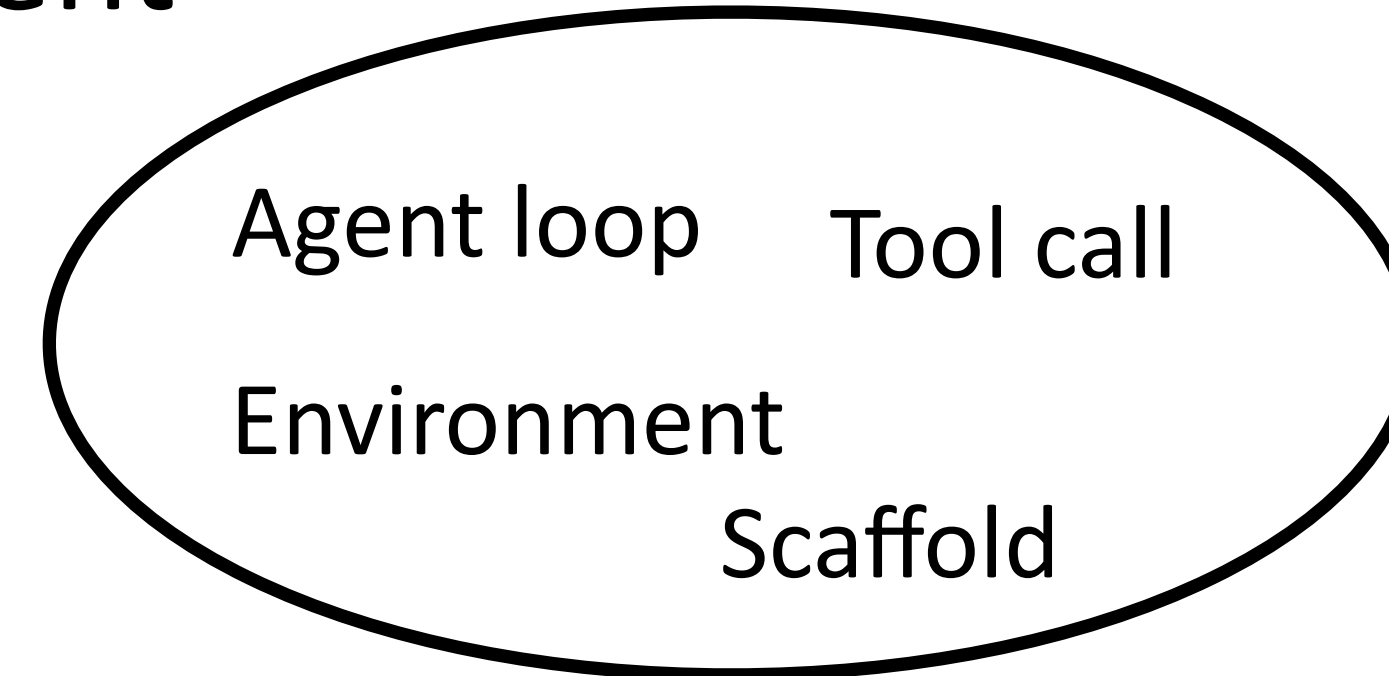


Review — Applications

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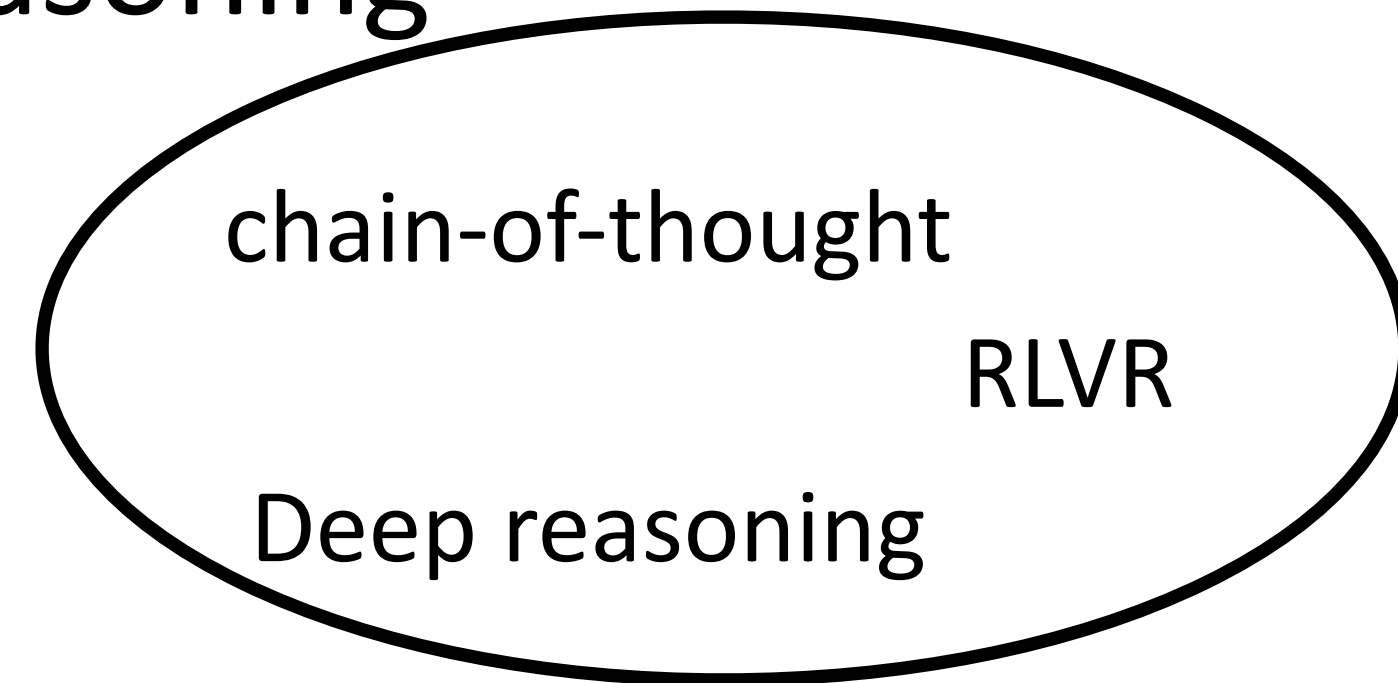


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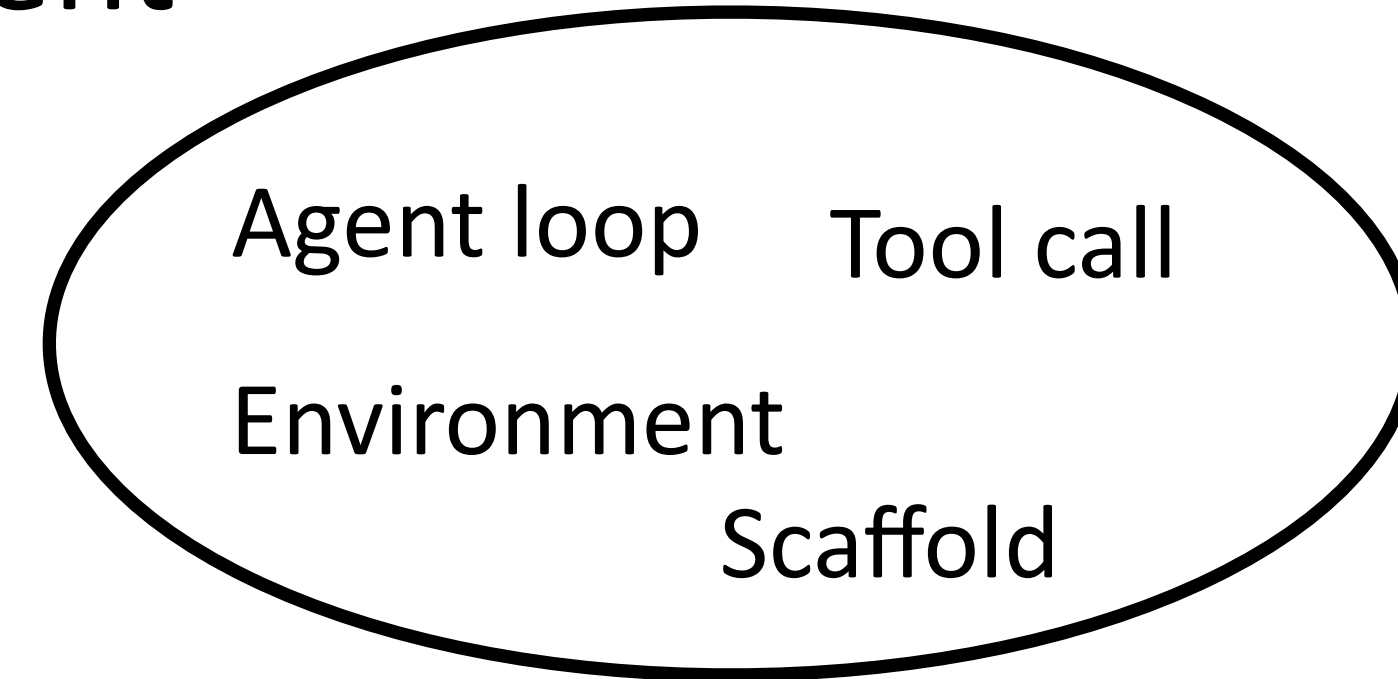


Review — Applications

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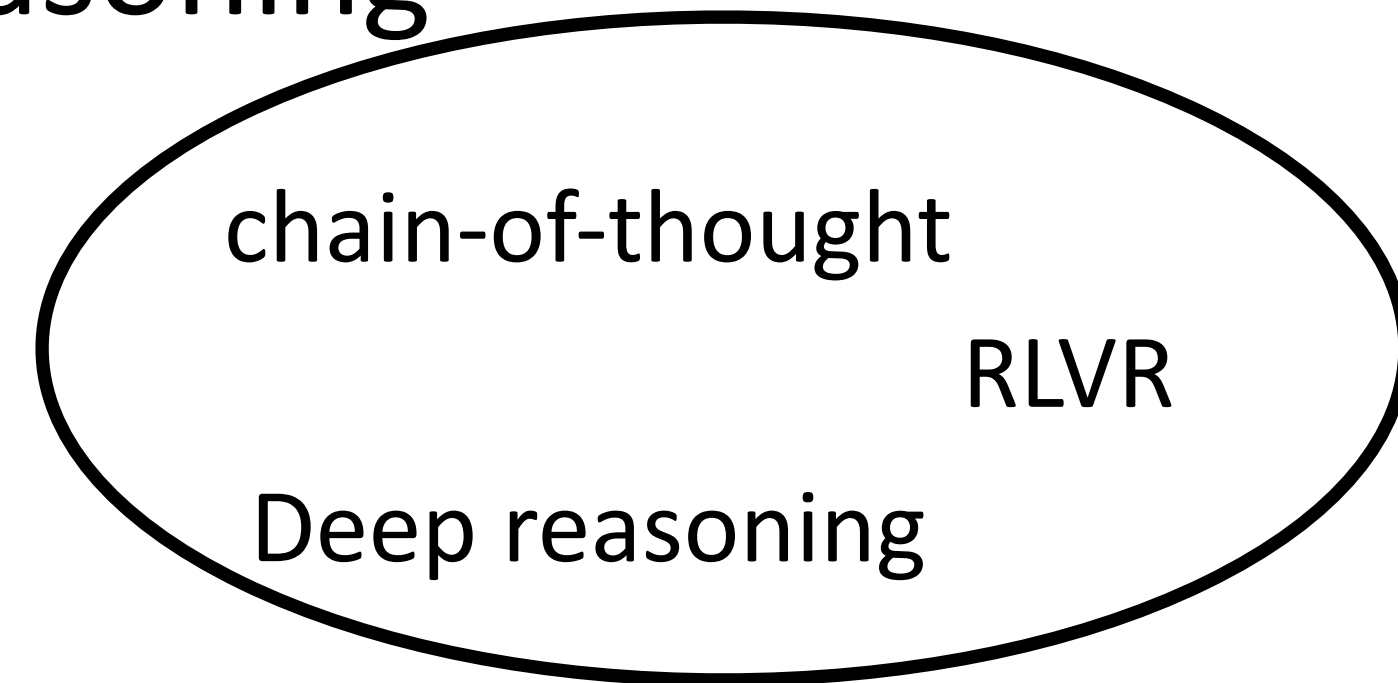


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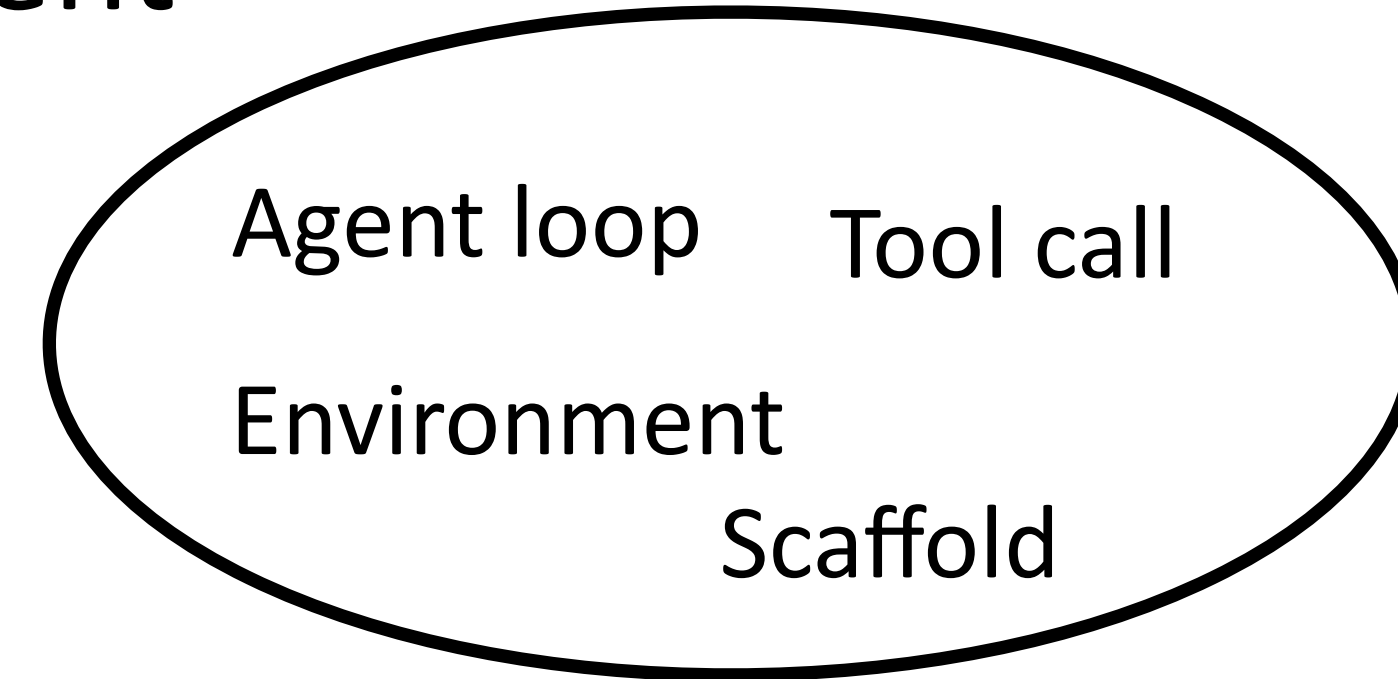


Review — Applications

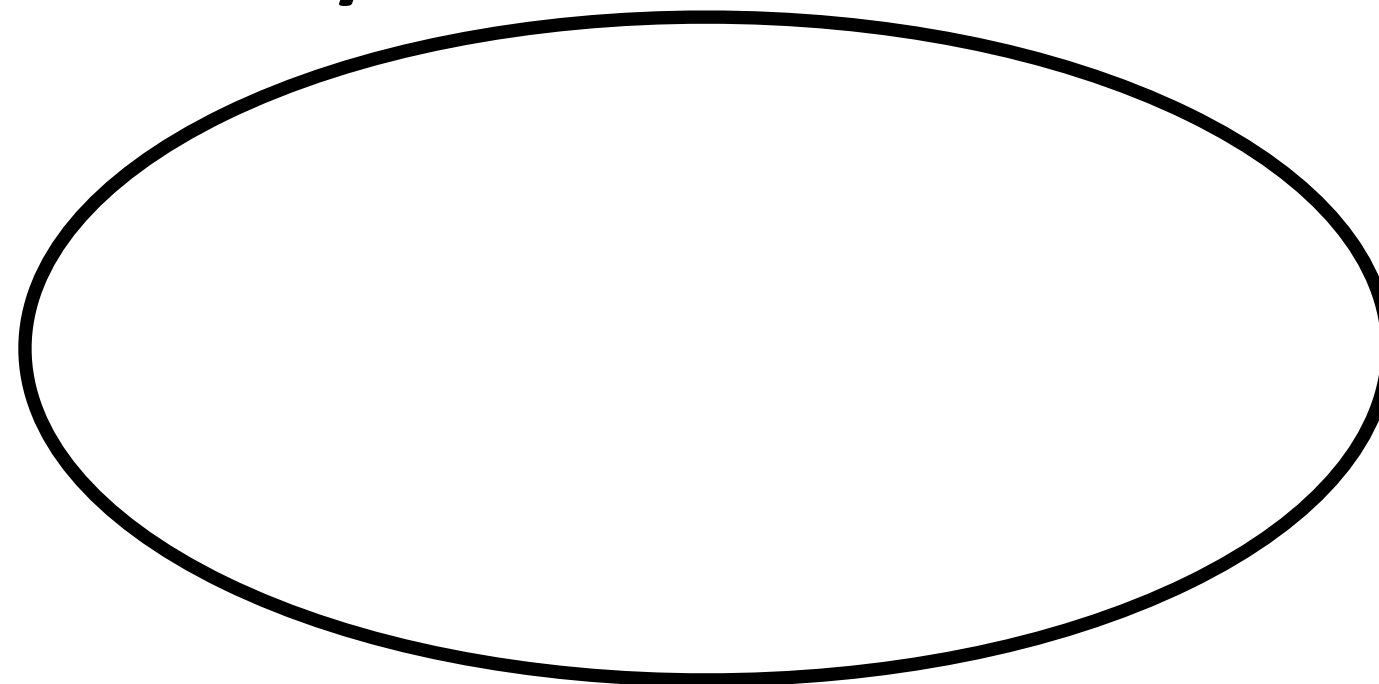
Reasoning



Agent

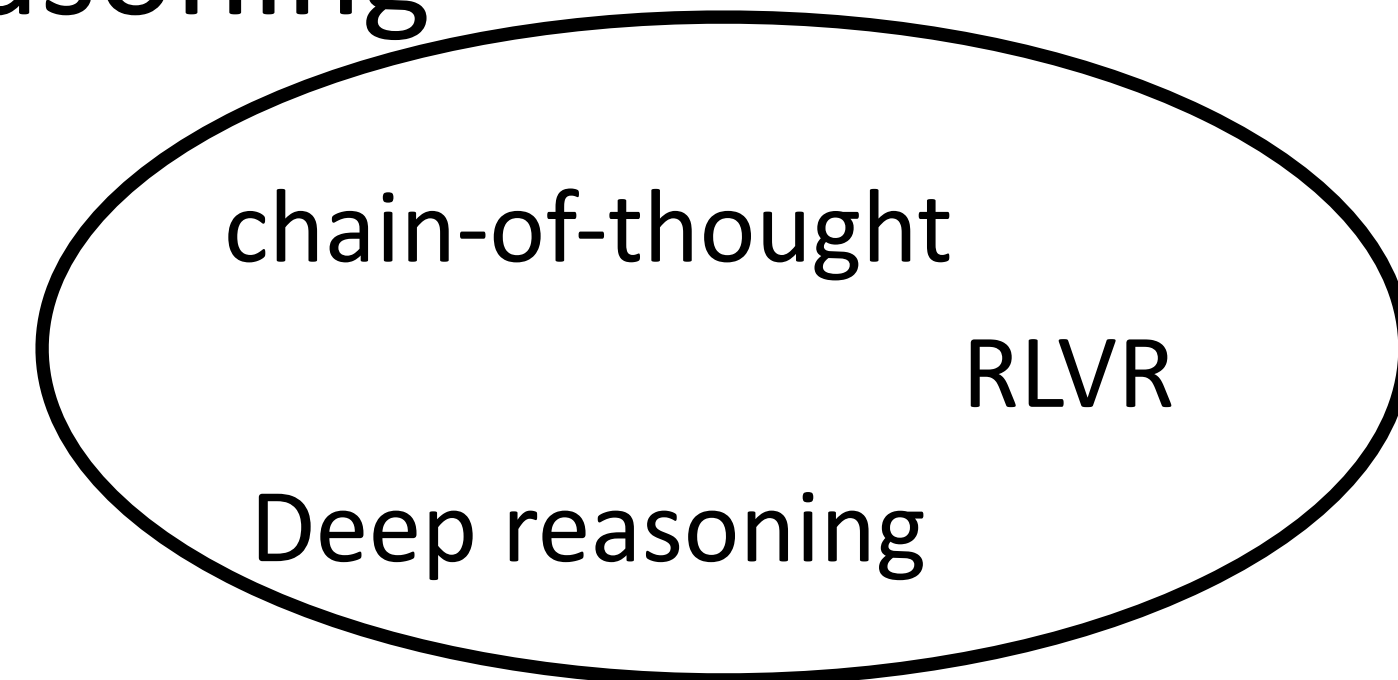


Ethics/Safety

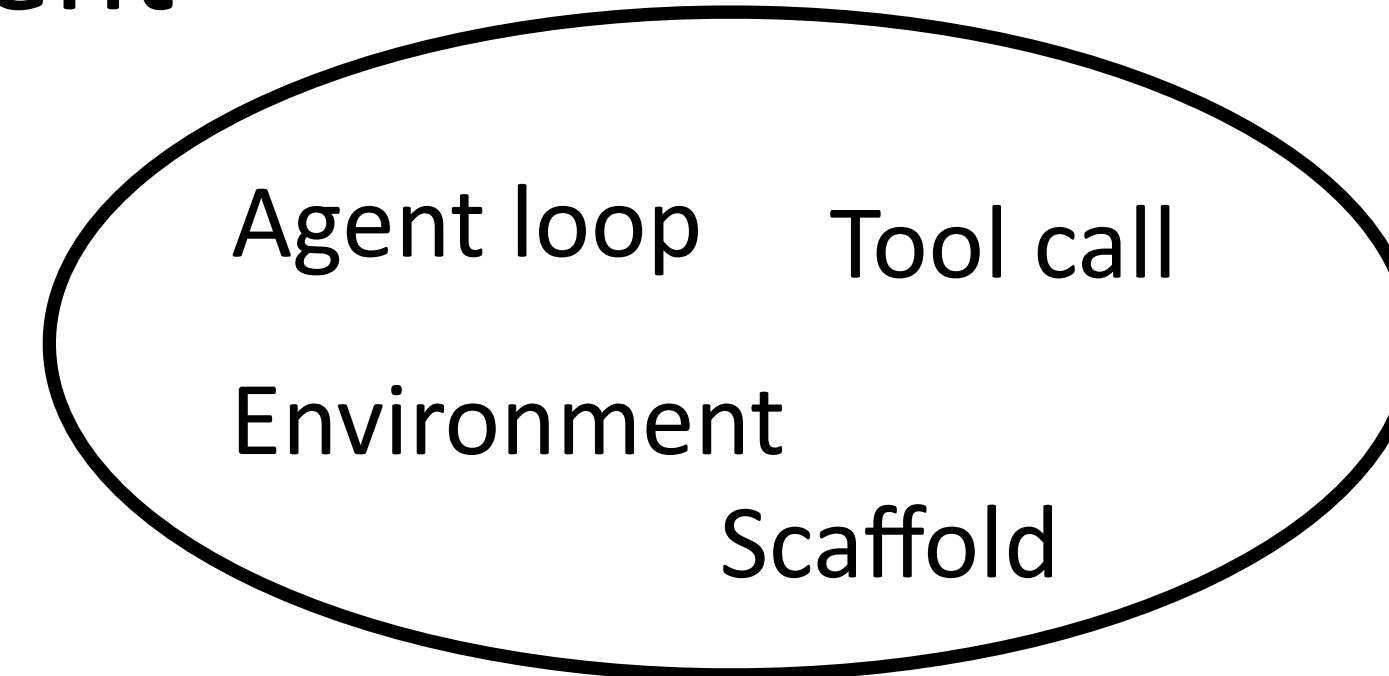


Review — Applications

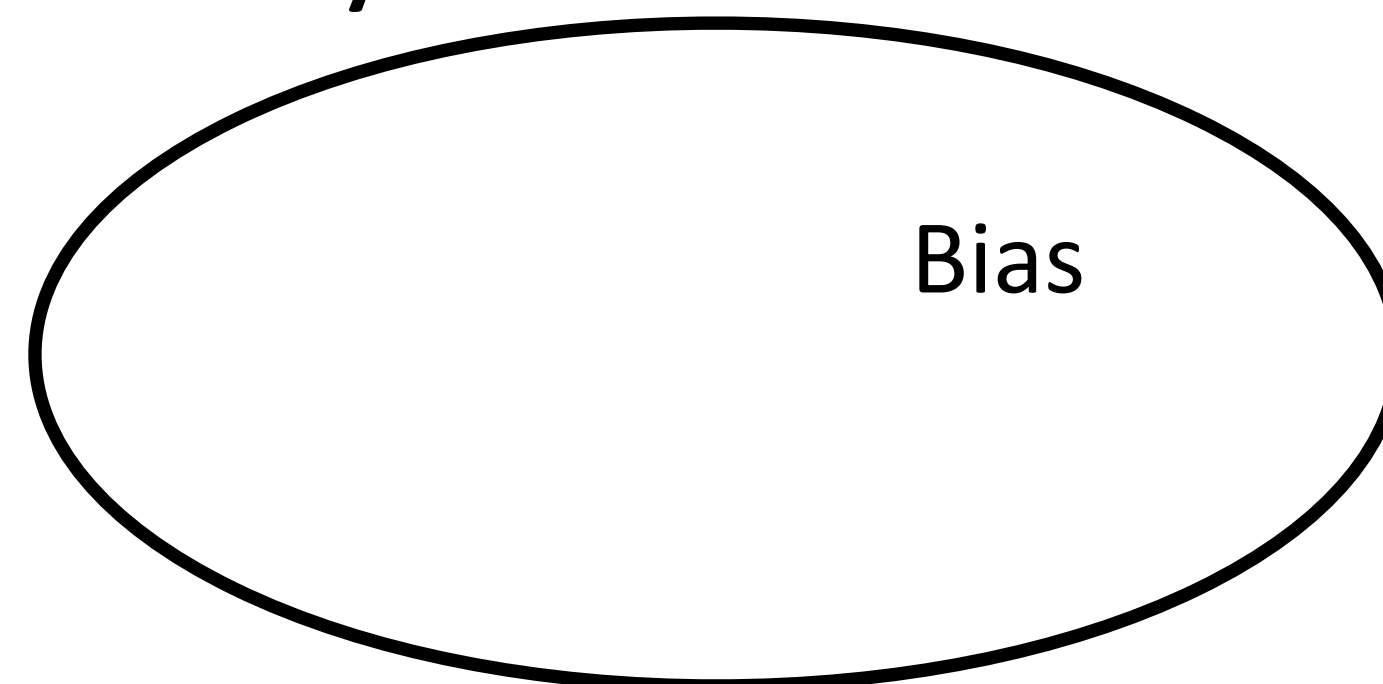
Reasoning



Agent

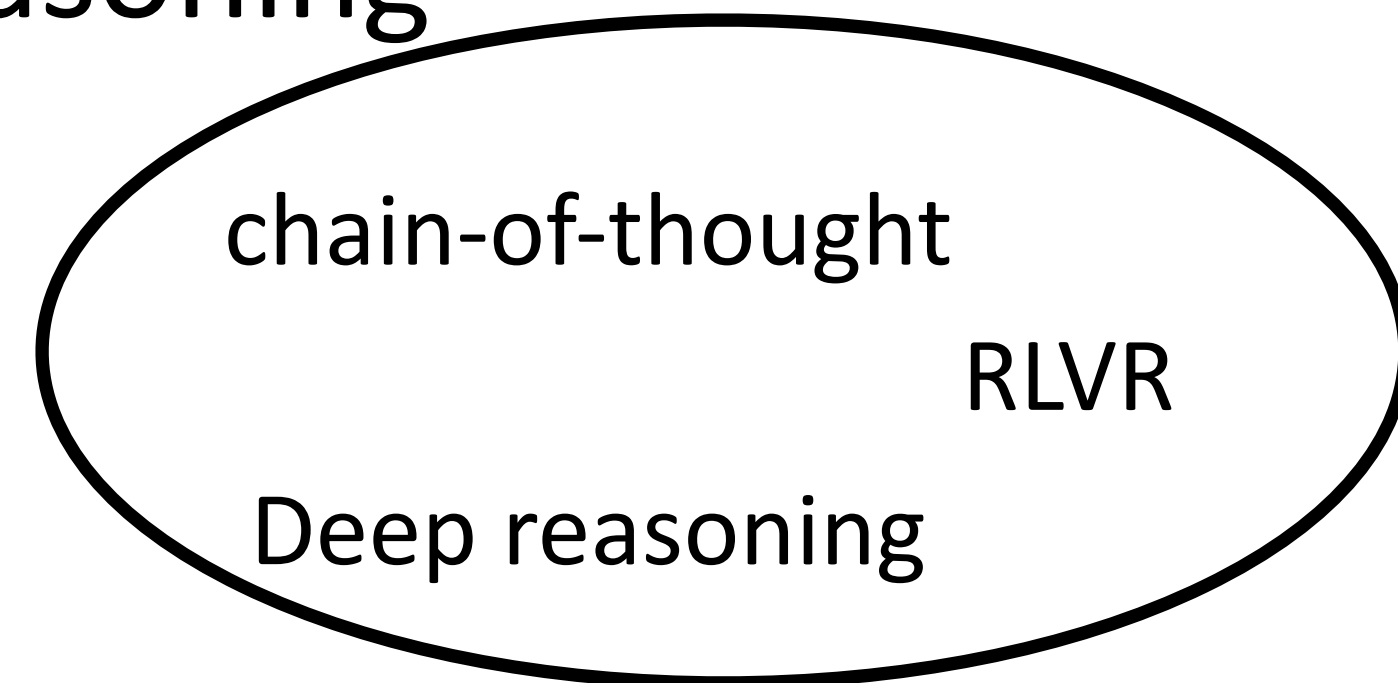


Ethics/Safety

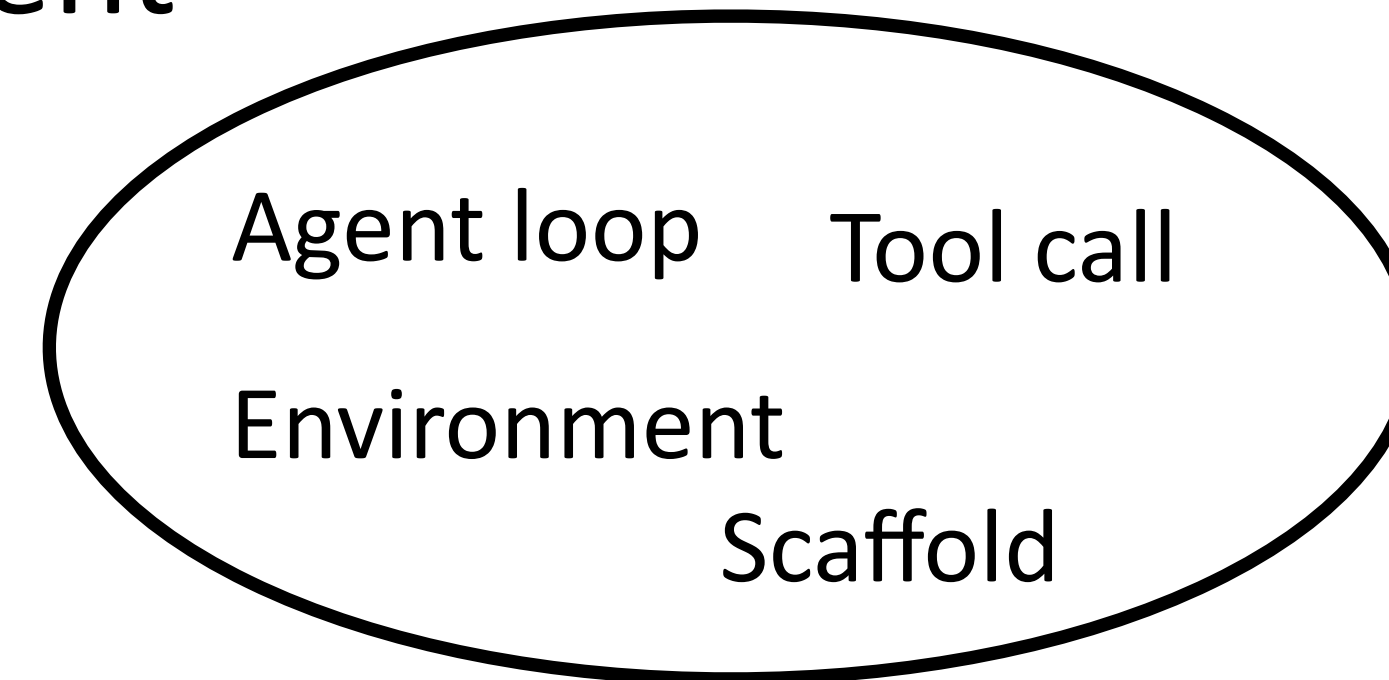


Review — Applications

Reasoning



Agent

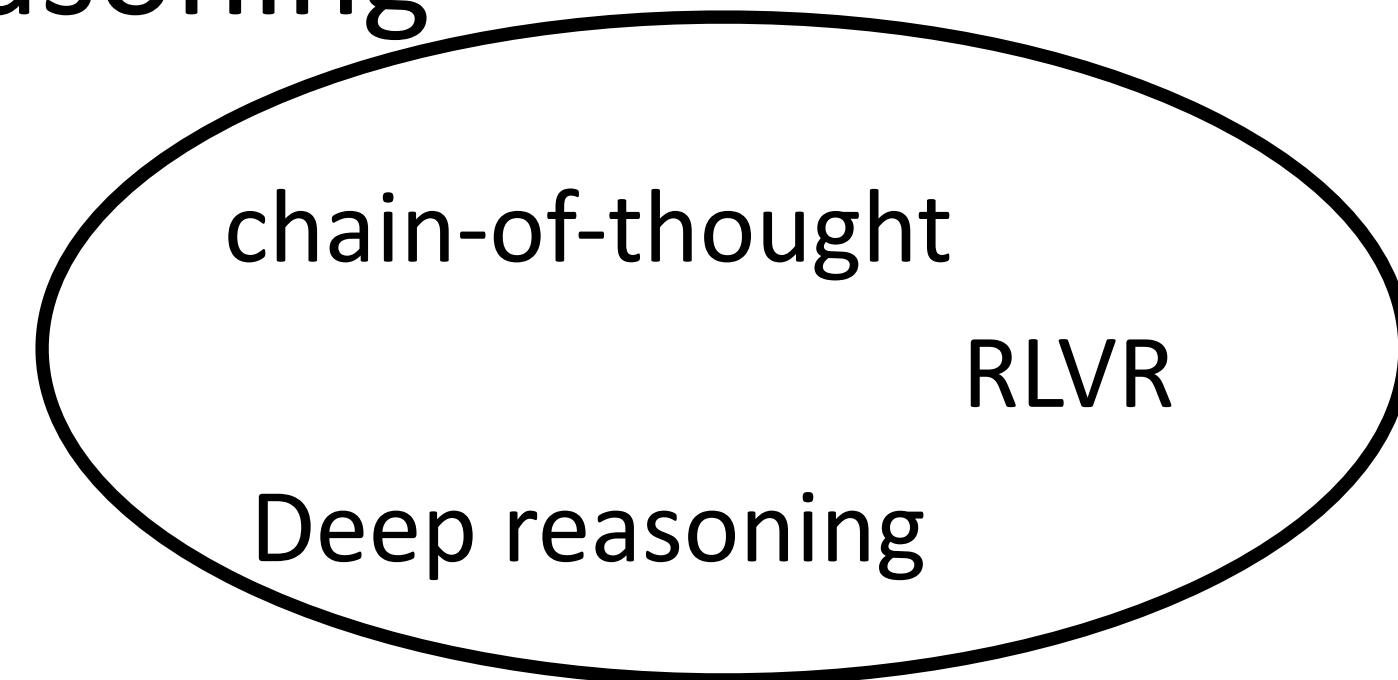


Ethics/Safety

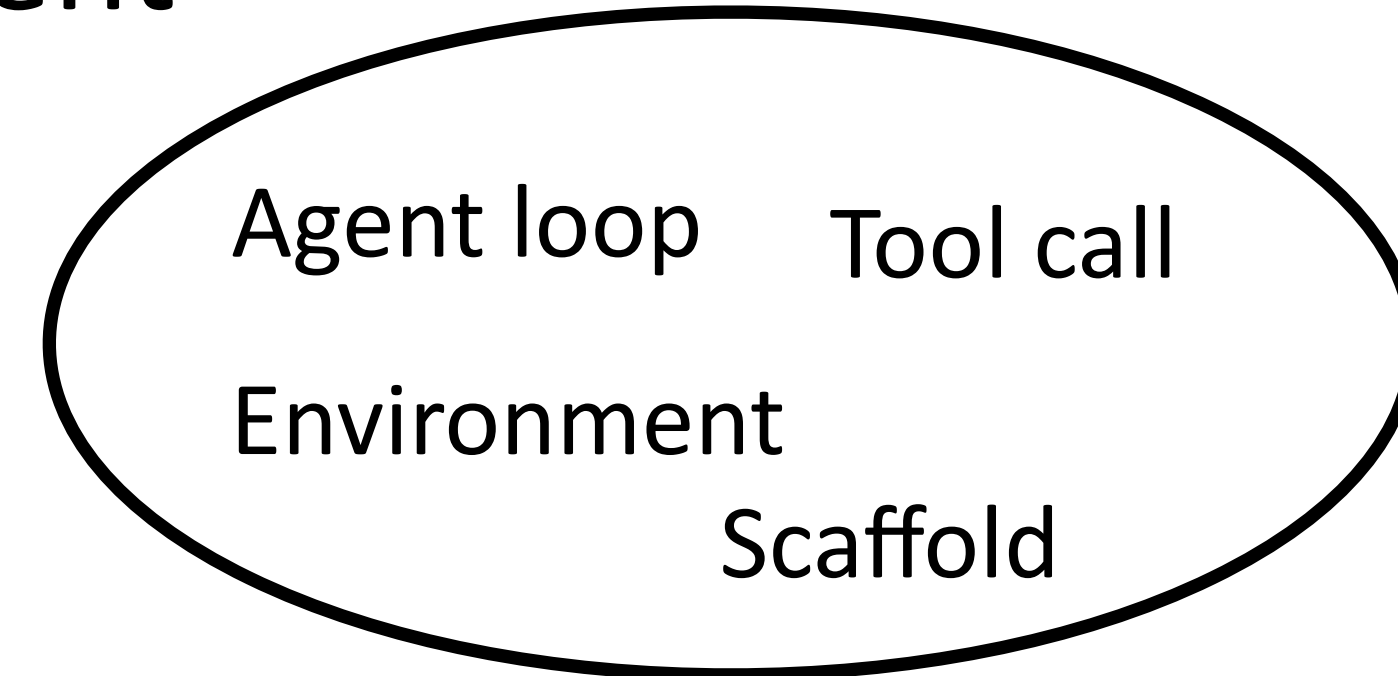


Review — Applications

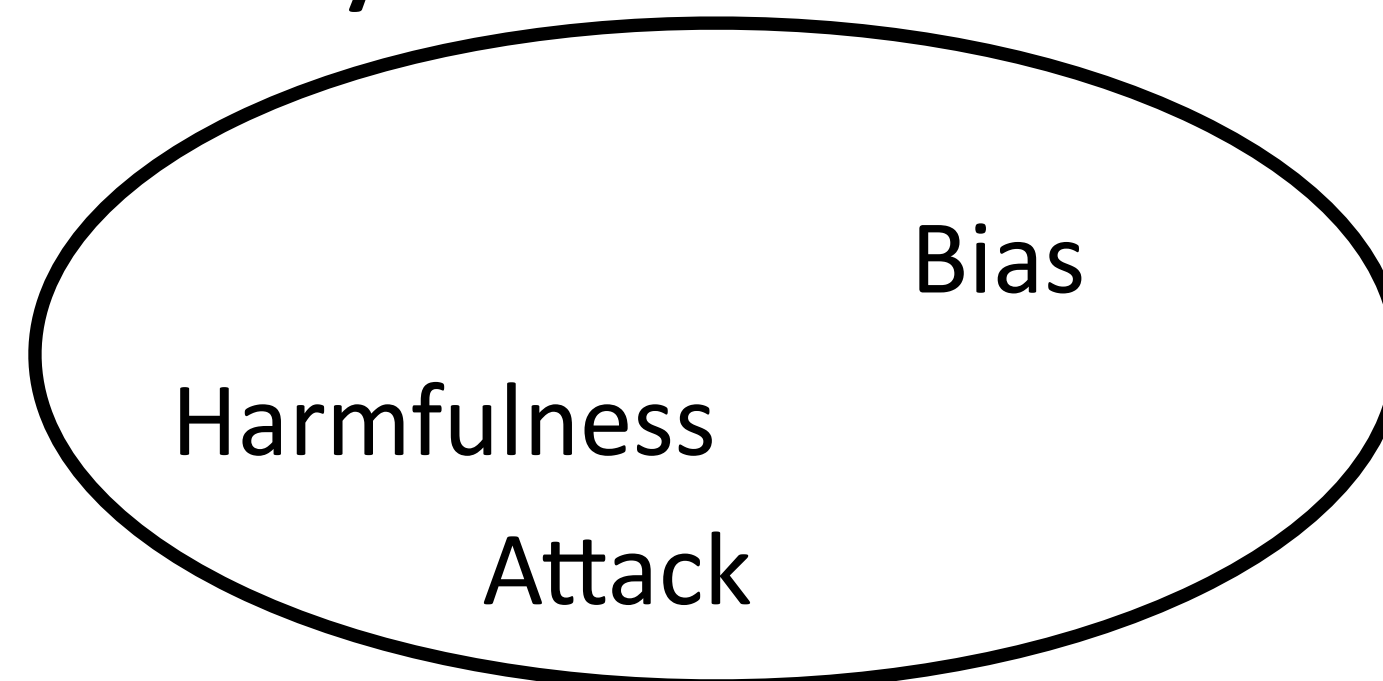
Reasoning



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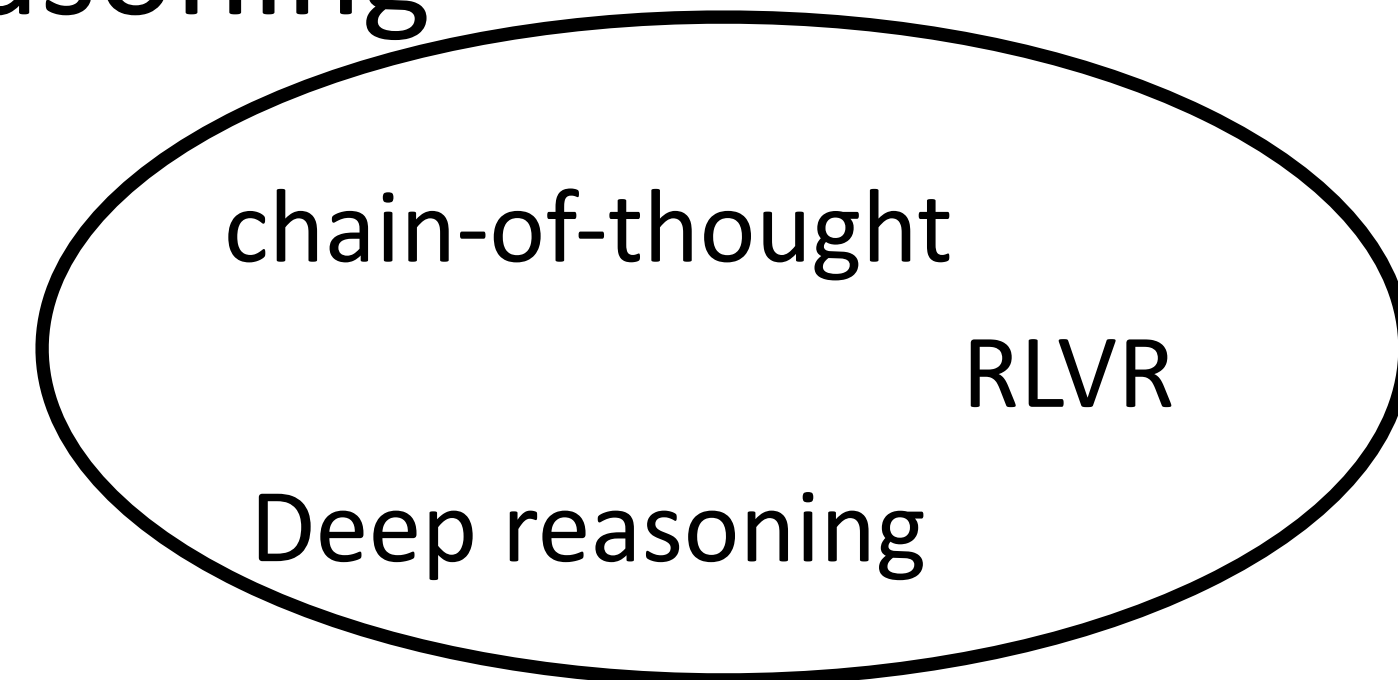


Ethics/Safety

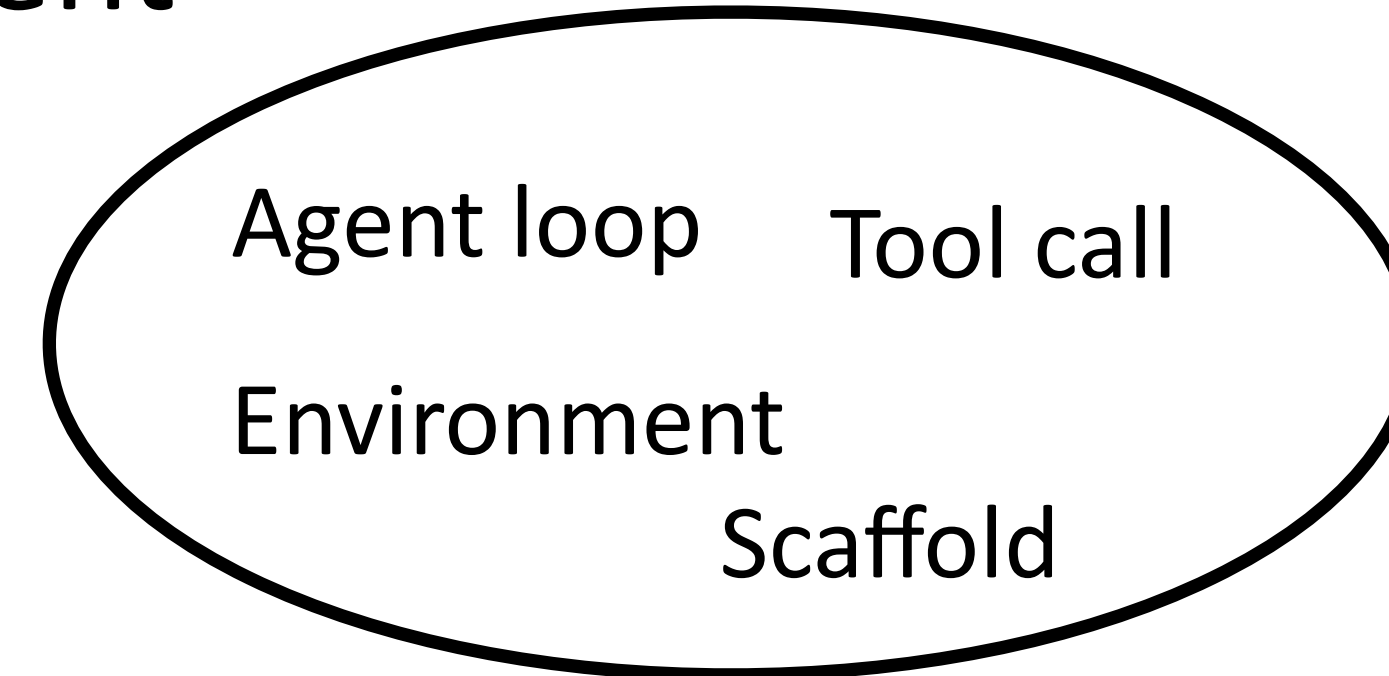


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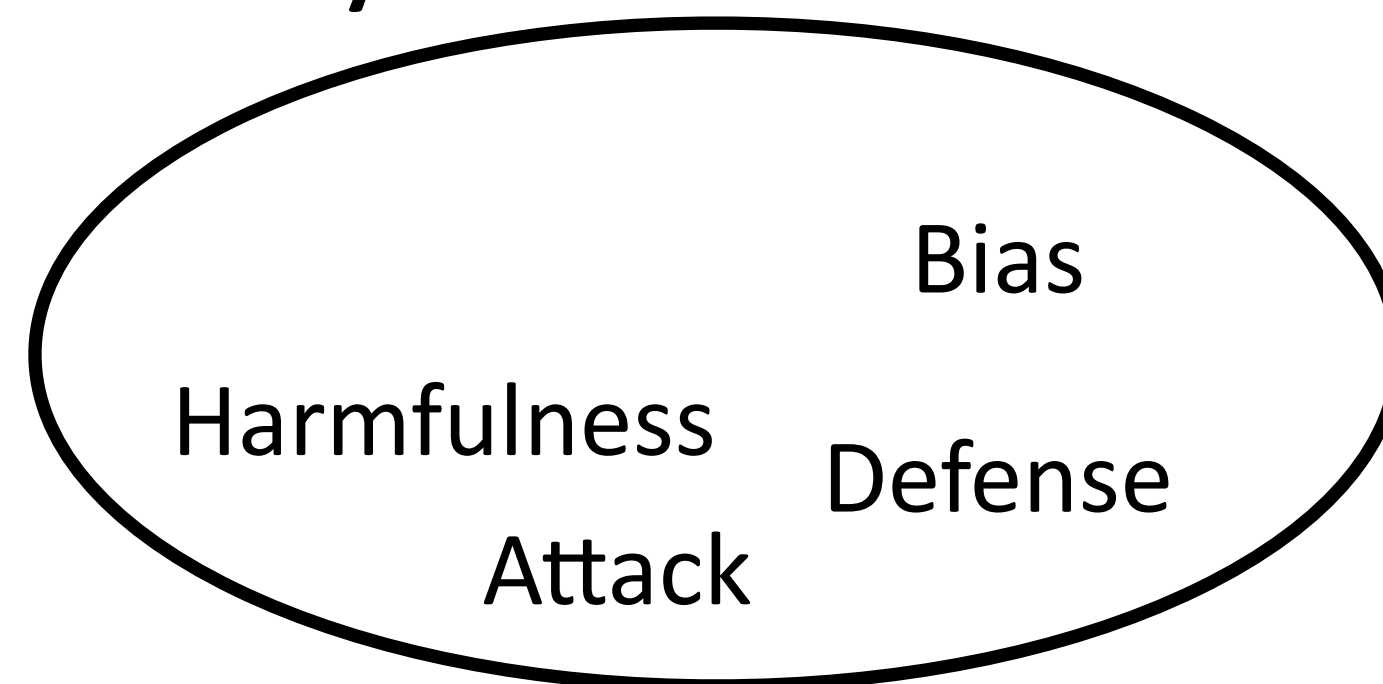
Reasoning



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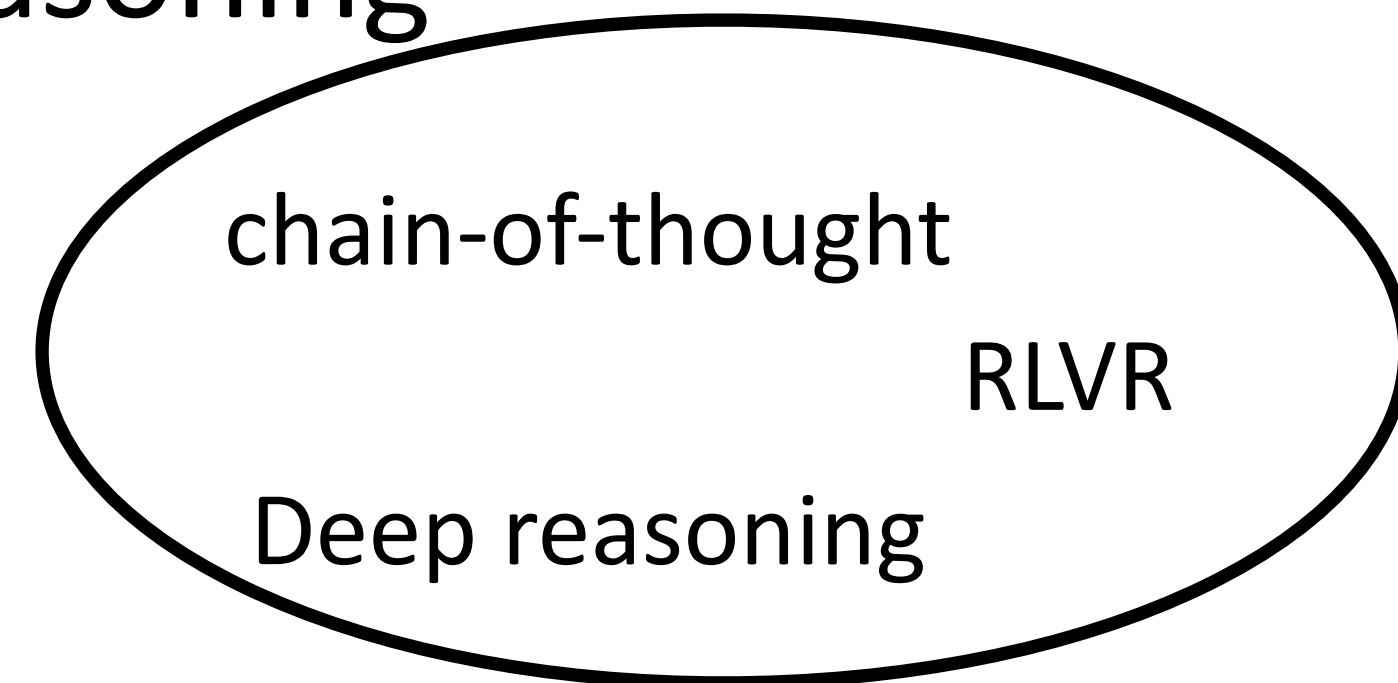


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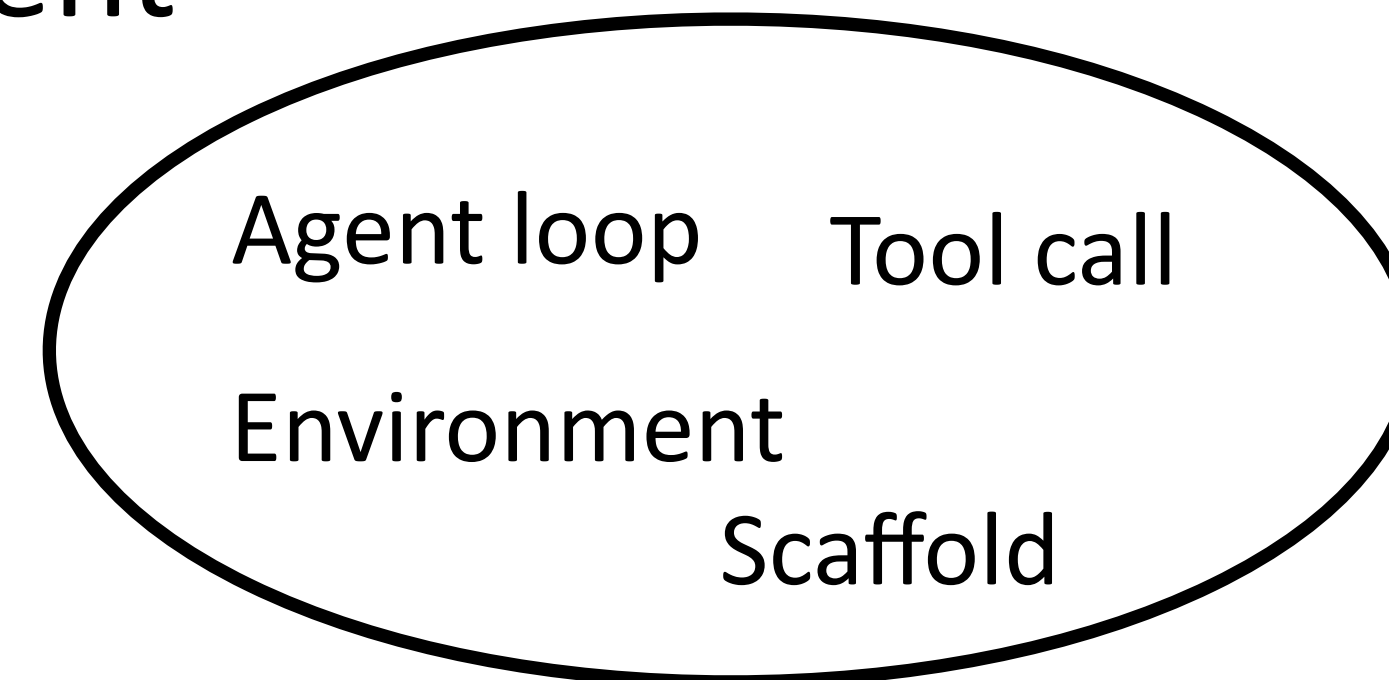


Review — Applications

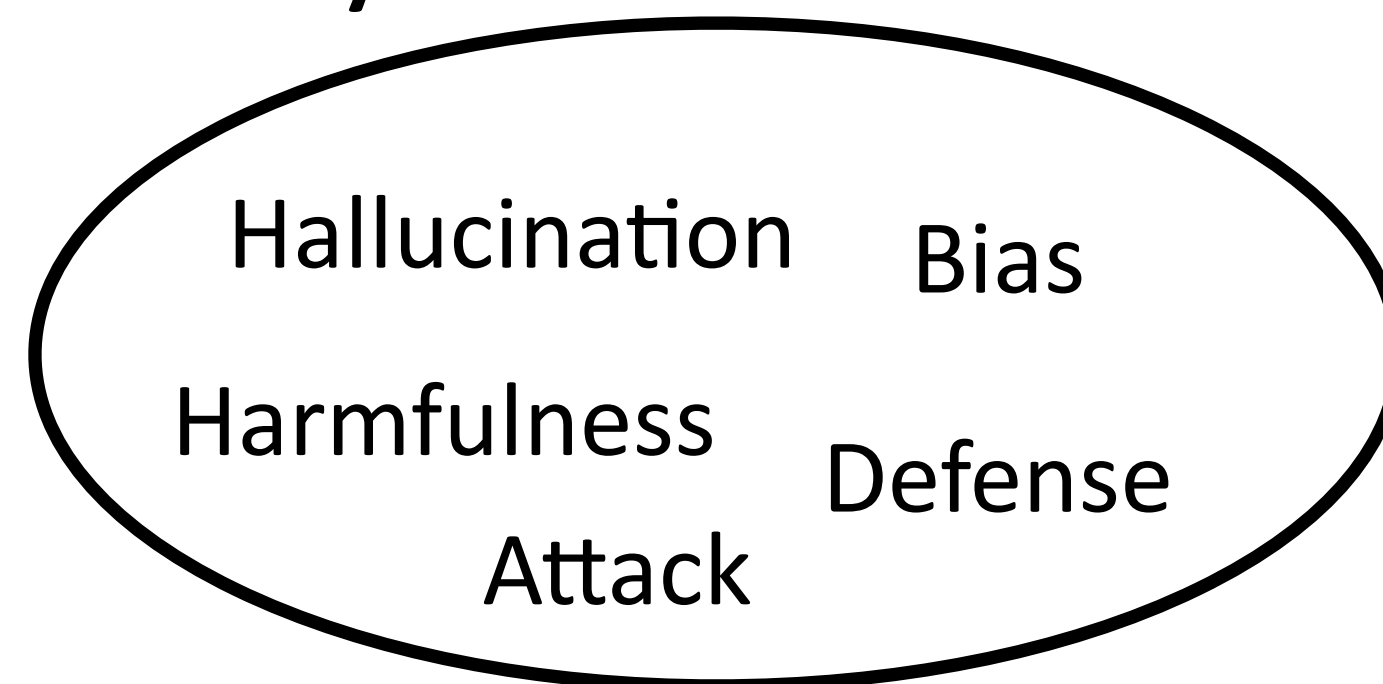
Reasoning



Agent

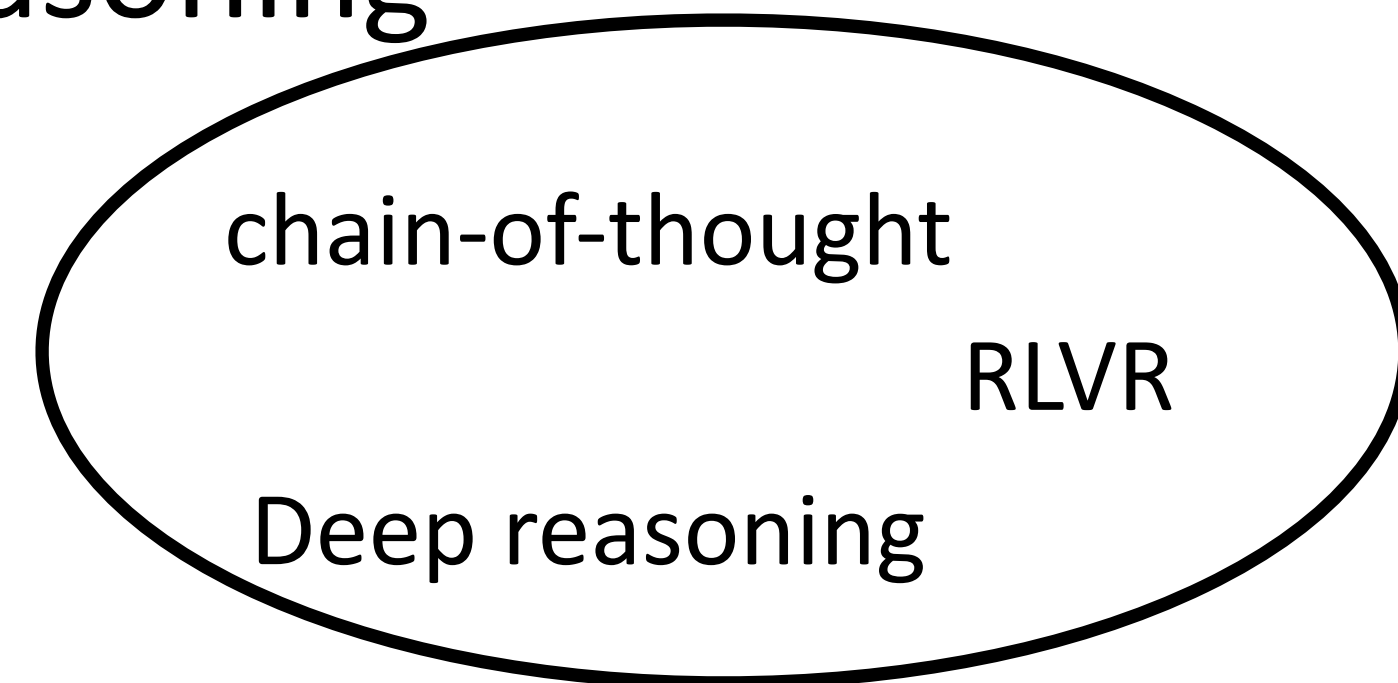


Ethics/Safety

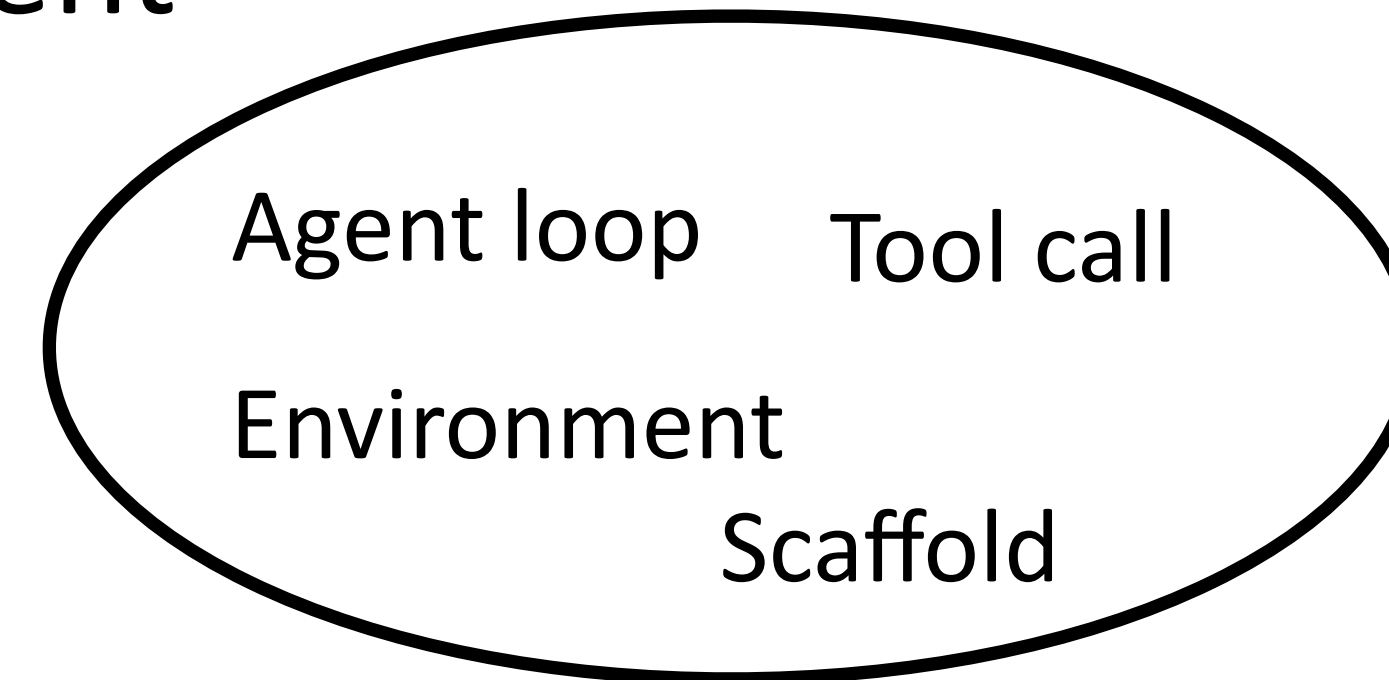


Review — Applications

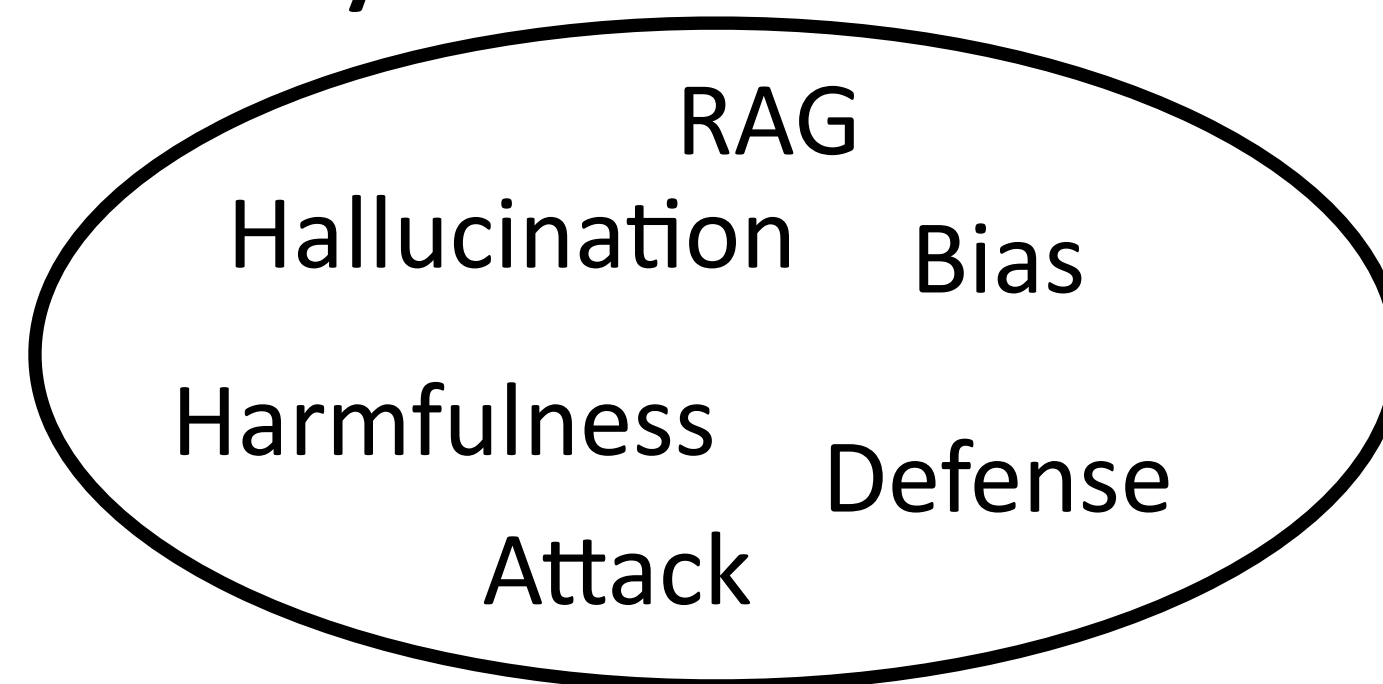
Reasoning



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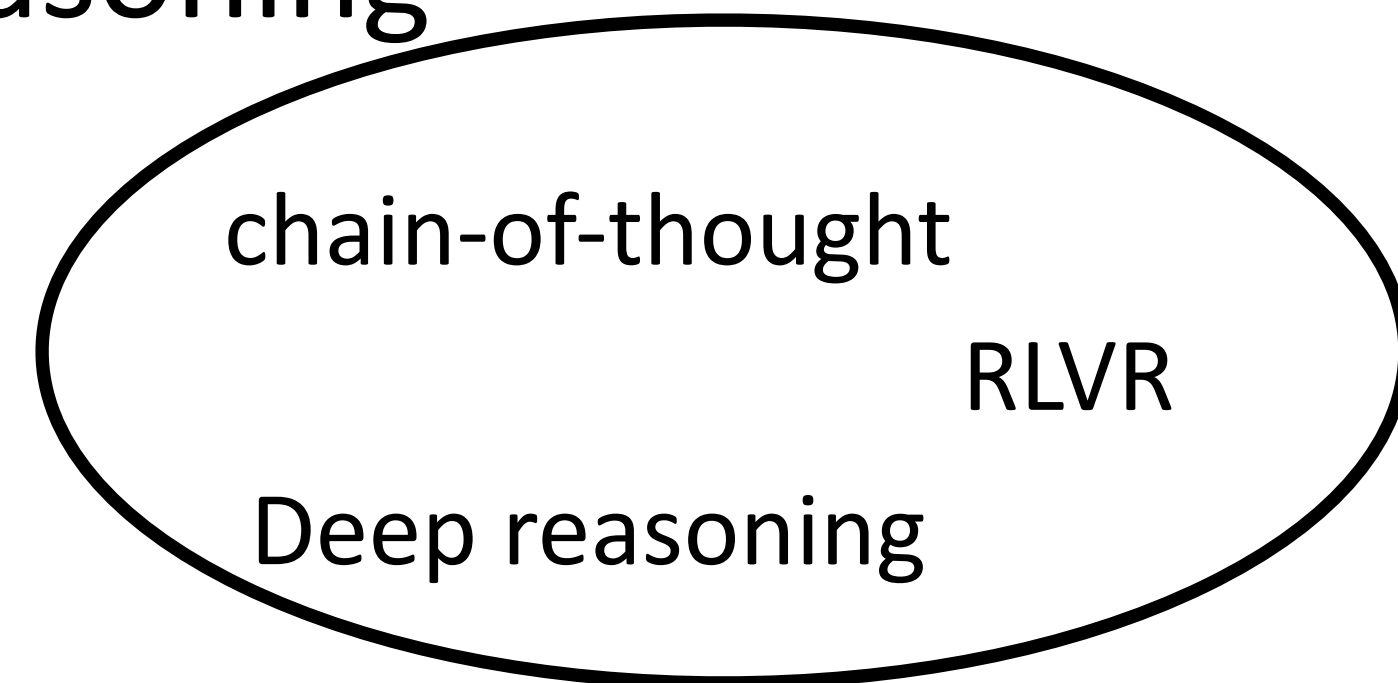


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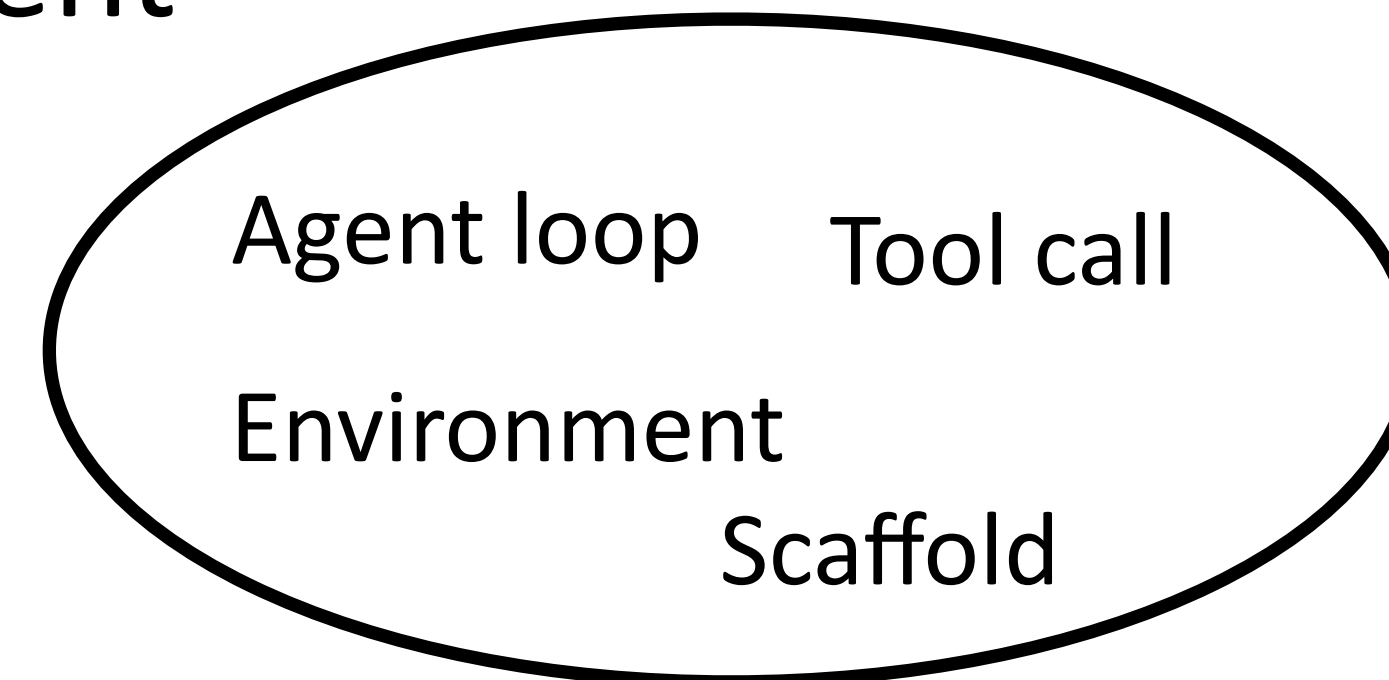


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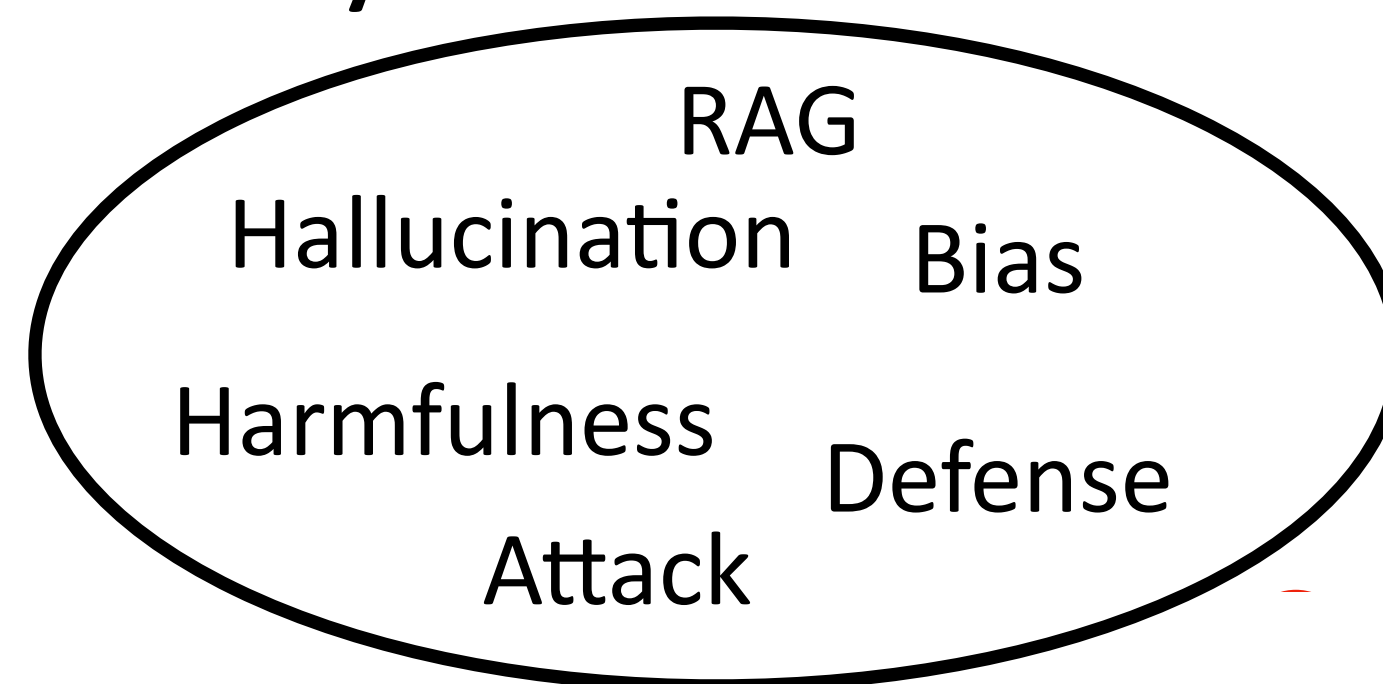
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Ethics/Safety



About Large Language Models

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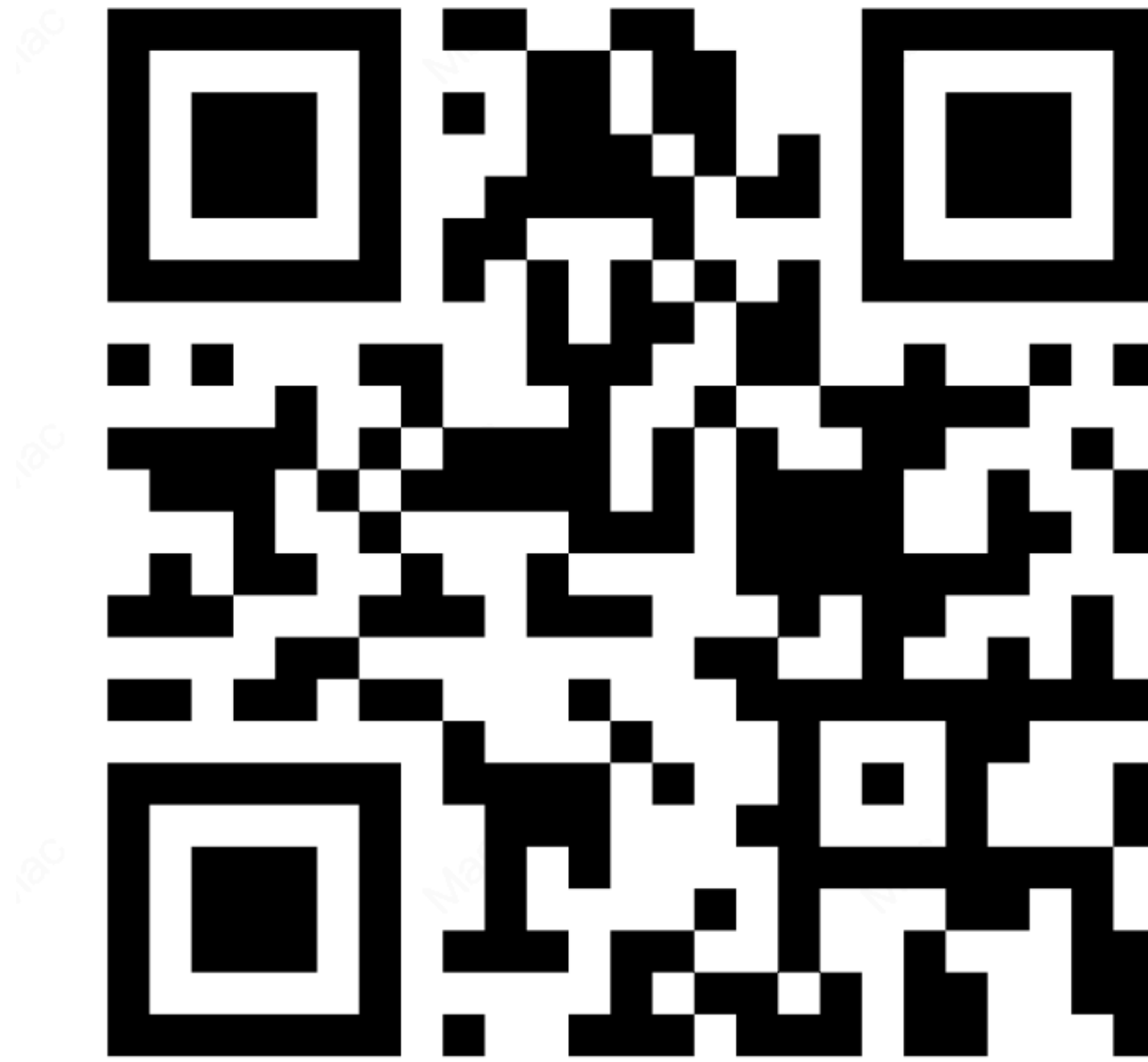
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3. Always embrace new technologies / applications
4. LLMs allow you to build amazing stuff quickly that can make actual impacts

Student Feedback Questionnaire (SFQ)

Anonymous and not mandatory, ddl tomorrow



<https://sfq-survey.ust.hk>

Thank You for taking the course!